

ELECTRONICS

Electronics

The physics behind the performance of signal processing devices is known as electronics.

Active Components

The active electronic components are those which has an ability to amplify the input signal, *e.g.*, transistor (BJT), VJT, FET etc.

Passive Components

The passive electronic components cannot amplify the signal but some of those may change the phase of an input signal, *e.g.*, resistor, capacitor and inductor etc.

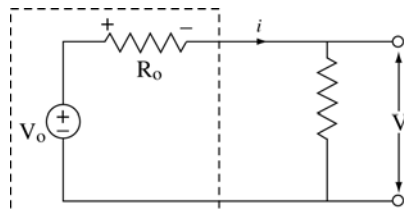
Kirchoff's Laws of Circuit Analysis

1. **Kirchoff's Current Law (KCL)**—At any instant of time the algebraic sum of current entering a junction is always equal to zero.

2. **Kirchoff's Voltage Law (KVL)**—At any instant of time the algebraic sum of the voltage drops in a loop to zero.

Model of a Battery

We can approximately represent the behaviour of a battery as follows :



A battery consists of an ideal voltage source and an internal resistance R_o then the voltage at its output terminal will be

$$V = V_o - iR_o$$

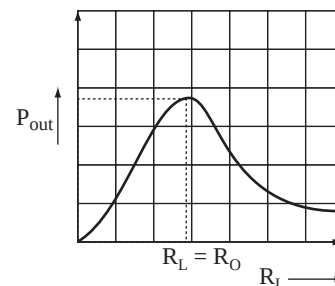
Matching

In an electric circuit, energy is transferred from the source to various branches in the circuit. The rate of this energy transfer is known as power delivered. The power across the load resistance R_L can be given by

$$P = \frac{V_o^2/R_L}{\left[1 + \left(\frac{R_o}{R_L}\right)\right]^2}$$

$$R_L \rightarrow 0, P \rightarrow 0$$

$$R_L \rightarrow \infty, P \rightarrow 0$$



For maximum power transfer $\left(\frac{dP}{dR_L}\right) = 0$

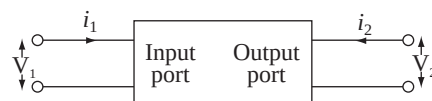
or

$$R_o = R_L$$

The maximum power transfer takes place when the load resistance is equal to the source resistance. This condition is called 'Matching'. Under matched condition the source will deliver the maximum power

$$P_{\max} = \frac{V_o^2}{4R_L} \quad (\text{for } R_L = R)$$

Hybrid parameters



In a two port network an ideal current and an ideal voltage source are connected at input and output respectively. So input current and output voltage are independent variables, whereas input voltage and output current are dependent variables. Then the equations relating the variables will be

$$\begin{aligned} V_1 &= h_{11} i_1 + h_{12} V_2 \\ i_2 &= h_{21} i_1 + h_{22} V_2 \\ \text{for } V_2 &= 0 \\ h_{11} &= \frac{V_1}{i_1} \end{aligned}$$

Short circuit input resistance parameter is

$$h_{21} = \frac{i_2}{i_1}$$

Short circuit forward current transfer ratio is

$$h_{12} = \frac{V_1}{V_2}$$

Open circuit reverse voltage ratio is

$$h_{22} = \frac{i_1}{V_2}$$

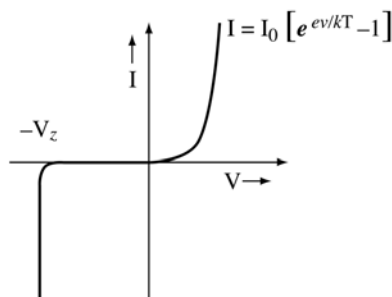
Since all the parameters have different dimensions, these are known as hybrid parameters.

Special Diodes

(a) **Zener diode**—The reverse bias breakdown voltage of a junction can be varied by choice of junction doping concentration. This breakdown mechanism is called zener effect in which the covalent bonds in the potential junction breaks abruptly. A zener is operated under reverse bias conditions. In a well fabricated zener the reverse current became independent of reverse voltage after zener breakdown. That is why a zener is called breakdown diode sometimes.

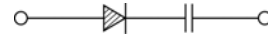


Zener diode symbol



Zener V-I characteristics

(b) **Varactor or varicaps diode**—The varactor is a acronym for voltage variable capacitor diode. The junction capacitance of a diode is not constant but it a function of applied voltage and the design of junction. This variable capacitance property of diode is useful in tuning of a radio receiver.



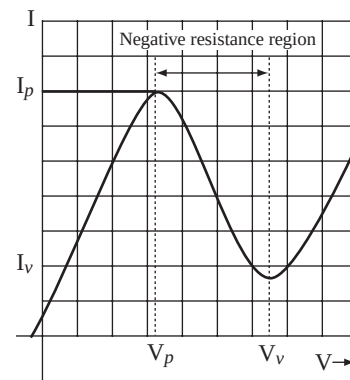
A varactor equivalent

If the p - n junction is abrupt, the capacitance varies as the square root of the reverse bias. In a graded junction, the capacitance varies as

$$C \propto \frac{1}{V_r^n} \text{ for } V_r \gg V_0 \quad (\text{for hyper abrupt junction } n = 2)$$

(c) **The tunnel diode**—This type of diode is controlled by a quantum mechanical phenomena of tunneling between the two sides of a diode.

An interesting feature of this type of diode is its negative resistance region in V-I curve [See figure].



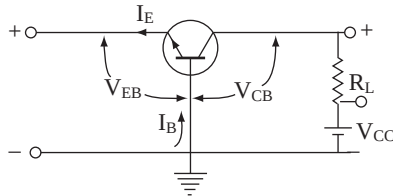
I-V characteristics of a tunnel diode

For negative resistance region the dynamic resistance $\frac{dV}{dI}$ is negative. This region is the direct result of quantum mechanical tunneling between a degenerate p -type material and a degenerate n -type material.

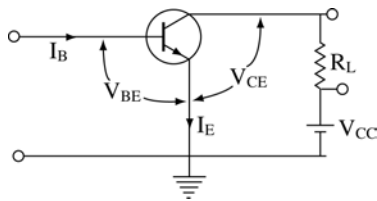
Transistors Operating Modes

Depending upon the common terminal between the input and output circuits of a transistor it may be operated in three modes.

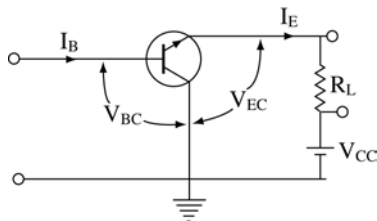
(a) **CB mode**—Common base or grounded base configuration is shown in the figure in which an n - p - n transistor is connected in CB mode.



(b) **CE mode**—In this mode emitter is the common terminal between input and output mode. An amplifier when operated in such a mode provides a phase change of π .



(c) **CC mode**—In common collector configuration the collector is the common terminal as shown in figure.



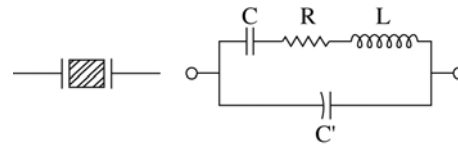
Comparative performance of CB, CC and CE amplifiers—

Parameter	CB	CE	CC
Current gain	α	β	Highest ($1 + \beta$)
Voltage gain	High	Highest	Nearly unity
Power gain	Moderate	Highest	Lowest
Input impedance	Lowest	Moderate	Highest
Output impedance	Highest	Moderate	Lowest
Phase reversal	No	Yes	No

Crystal Oscillators

The quartz crystal shows an interesting property when an alternating electric potential is applied on its opposite surface. The surface expands and contracts and resonance takes place if the natural frequency of the crystal equals the applied electrical frequency. The crystal frequency depends on its dimensions. This property of quartz crystal is used to construct the quartz oscillator which has precise frequency stability.

A crystal oscillator and its electrical equivalent is shown in the figure alongside. In electric equivalent circuit the inductance corresponds to inertia, resistance corresponds to friction and capacitor corresponds to reciprocal of damping constant for a mechanical system.

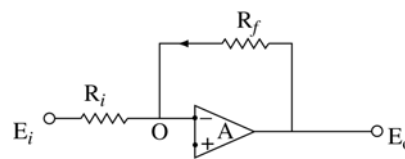


C' represents the electrostatic capacitance between electrodes with the crystal as a dielectric. $C' > C$

Operational Amplifiers

The operational amplifiers are high gain negative-feedback amplifiers which are used to perform mathematical operations and highly useful in digital computation.

A schematic diagram of an operational amplifier is shown below. The common grounded connection is forbidden.



The negative and positive terminals are inverting and non-inverting terminals. The amplification and feedback are made very large for extreme stability.

The current in the resistors R_i and R_f will be $\frac{E_i}{R_i}$ and $\frac{E_o}{R_f}$ respectively according to Ohm's law.

By Kirchoff's first law (KCL) the sum of these two currents at junction 'o' will be

$$\frac{E_i}{R_i} + \frac{E_o}{R_f} = 0$$

$$\text{The output voltage } E_o = \frac{-R_f}{R_i} E_i$$

The polarity of output voltage is reversed and magnitude is multiplied by ratio of feedback resistance to input resistance, i.e., R_f/R_i

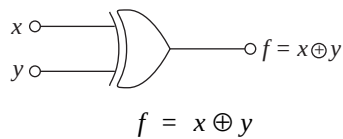
The gain can be given by

$$\frac{E_o}{E_i} = -\frac{R_f}{R_i} \quad (A \gg 1)$$

The feedback amplification is independent of open loop gain if the open loop gain is too high.

Exclusive or Function

The exclusive-OR or XOR is a logic function which provides an output only when its both inputs are at different values. Its symbol and truth table is given below :



x	y	f
1	0	1
0	1	1
0	0	0
1	1	0

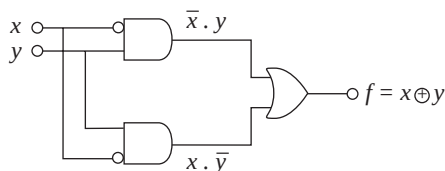
The logic operations are

$$x \oplus y = y \oplus x$$

$$1 \oplus x = \bar{x}$$

and

$$0 \oplus x = x$$



De-Morgan's Theorems

For the simplification of Boolean expressions De-Morgan explained the equivalence of OR and AND functions with the help of two laws.

1. For the AND of n inputs named A, B, C, ... n , $A \cdot B \cdot C \dots = \bar{A} + \bar{B} + \bar{C} + \dots + \bar{n}$

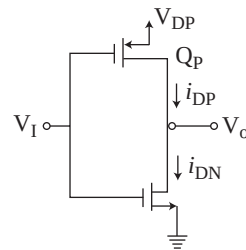
and its dual relation

2. For the OR of n inputs named A, B, C ... n
 $A + B + C + \dots + n = \bar{A} \cdot \bar{B} \cdot \bar{C} \dots \bar{n}$

These laws help in the construction of all the gates from NAND and NOR gates only. That is why NAND and NOR gates are known as universal gates.

CMOS Digital Logic Inverter

The basic circuit element for digital circuit design is the logic inverter. This element is used to build logic gates and more complicated digital circuits. The basic CMOS logic inverter is we will assume the body and source terminals are connected together, so there's no body effect to consider. We'll also assume that the MOSFETs are matched transistors.



The operation of this circuit can be summarized as—

When V_i is 'low', Q_N is 'off', Q_P is 'on'

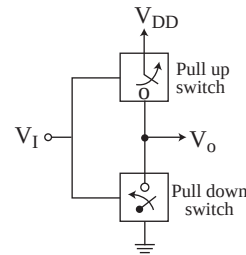
$\Rightarrow V_o$ is 'high'

when V_i is 'high', Q_N is 'on', Q_P is 'off'

$\Rightarrow V_o$ is 'low'.

Conceptually, this CMOS circuit is intended to junction as "complementary switches".

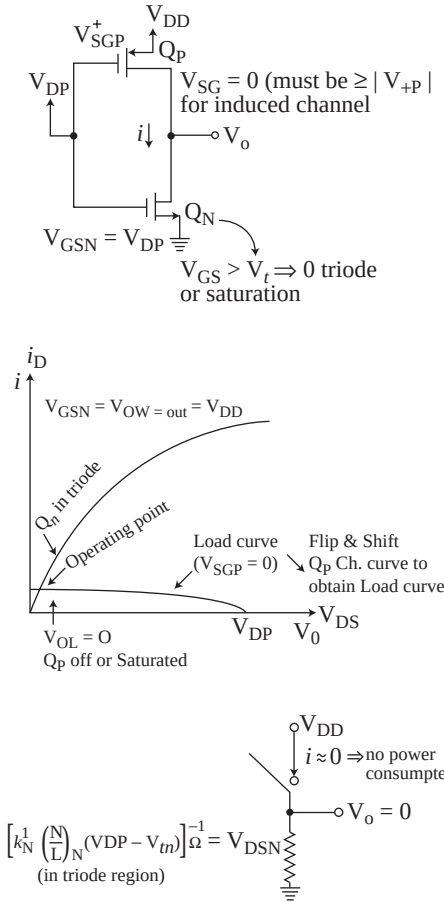
The directions of the arrow indicate the complementary nature of the two switches. When one is closed the other is open's and **vice-versa**.



Graphical Analysis of the CMOS Logic Inverter

Here Q_P is treated as an active load for Q_N through the converse would produce identical results.

Graphical solution when V_I is “high”



We'll consider the two extremes of the input :

$$V_I = 0 \text{ ("low")}$$

$$V_I = V_{DD} \text{ ("high")}$$

With Q_N considered the driving transistor and Q_P the active load, then for a graphical solution, we'll be plotting characteristic and load curve in the $i_{DN} - v_{DSM}$ plane.

For this digital inverter circuit in figure we see that

$$i_{DP} = i_{DN} \text{ [Fig. (1)]}$$

The v_{DS} values are different for the two transistors. By KVL

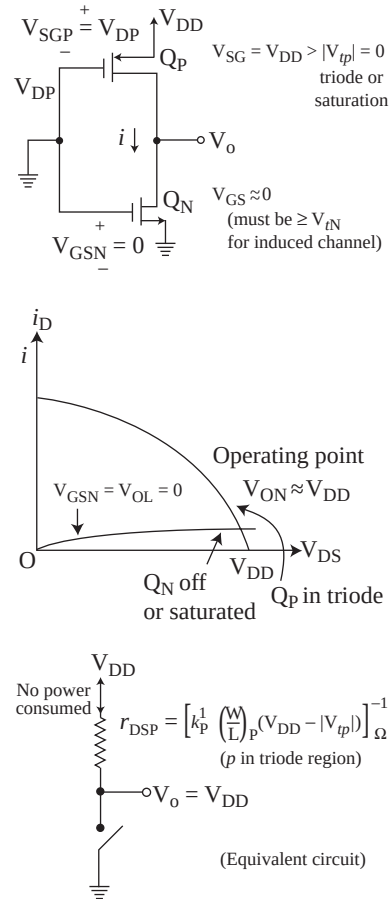
$$V_{DD} - v_{SDP} = v_{DSN} \quad \dots(1)$$

How we interpret this equation can give us the answer on how to plot the Q_P ch. curve in the $i_{DN} - v_{DSN}$ plane.

The minus sign in (1) tells up to flip the Q_P characteristic curve about the vertical axis.

The V_{DD} term tells us to shift this flipped curve V_{DD} units to the right.

Graphical solution when v_I is “low”—



Summary of the CMOS Digital Logic Inverter

1. The output voltage levels are ~ 10 mV and $\sim V_{DD} - 10$ mV, thus the output signal swing is nearly the maximum possible (Gives rise to the so called noise margins).

2. The static power dissipation is nearly zero (a fraction of a μW) in both states. The dynamic power dissipation is not zero as the gates are changing states.
3. How output resistance in either state : low resistance to ground in the “low” output state, and low resistance to V_{DD} in the “high” state.
4. Input resistance of the inverter is very large (ideally infinite). The inverter can derive a very large number of similar inverters with little loss in signal level.
5. The gate input capacitance is not negligible, so it will take time to “charge” up. Low output resistance helps to keep the time constant $\tau = R_{\text{out}} C_{\text{in}}$ small when driving similar inverter.

Characteristic Curve for the CMOS Logic Inverter

Rather using a graphical solution, it is fairly straight forward to numerically, construct the characteristics curve v_0 versus V_I . To obtain an equation to plot for the inverter characteristic curves we will use the equation for the MOSFET current in the triodes and saturation region of operation.

The $i_D - v_{DS}$ characteristic curve for Q_N is given by

$$i_{DN} = k_n' \left(\frac{W}{L} \right)_n \left[(v_I - v_{tn}) v_0 - \frac{1}{2} v_0^2 \right] \quad \dots(2)$$

$$v_I = v_{GS}$$

$$v_0 = v_{DS}$$

$$v_0 \leq v_I - v_{tn}$$

$$i_{DN} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_n (v_I - v_{tn})^2 \quad \dots(3)$$

$$v_0 \geq v_I - v_{tn}$$

While the $i_D - v_{DS}$ Ch. curve for Q_p is given by

$$i_{DP} = \frac{1}{2} k_p' \left(\frac{W}{L} \right)_p (V_{DD} - v_I - |V_{tp}|)^2 \quad \dots(4)$$

$$v_0 \leq v_I + |V_{tp}| \quad \left\{ \begin{array}{l} \text{Where } v_{GS} = V_{DD} - v_I \\ \text{And } v_{SD} = V_{DD} - v_0 \end{array} \right.$$

For the digital logic inverter circuit, these two currents must be equal—

$$i_{DN} = i_{DP} \quad \dots(5)$$

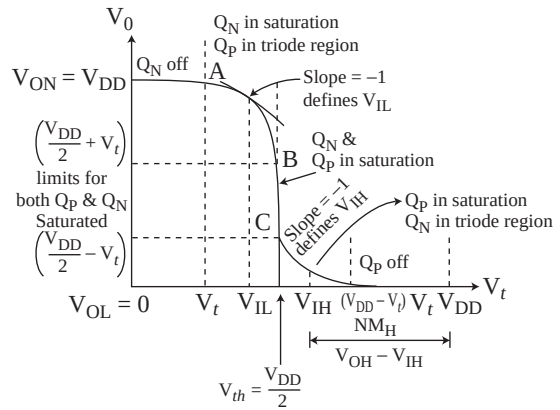
To construct the complete characteristic curve for this logic inverter, we solve (5) for v_0 using (2) through (5) as v_I varies from 0 to V_{DD} .

In the case of symm MOSFETs in which

$$V_{tn} = |V_{tp}|$$

$$\text{and } k_n' \left(\frac{W}{L} \right)_n = k_p' \left(\frac{W}{L} \right)_p,$$

the voltage transfer characteristic curve will be symmetrical as—



(The voltage transfer Ch. curve)

$$NM_H = V_{OH} - V_{IH}$$

$$NM_H = \frac{1}{8} (3V_{DD} + 2V_t)$$

$$NM_L = NM_H$$

Noise Margin

From this characteristic curve we can observe that there is a range of input values (from 0 to V_{IL}) that produces a high output voltages and a range of input values (from V_{IH} to V_{DD}) that produces a low output voltages. This is one advantage that digital circuits have over analog ones.

These ranges of input voltages are called **noise margins**. In particular, the noise margin for high input NM_H is defined as—

$$NM_H = V_{OH} - V_{IH} \quad \dots(7)$$

And the noise margin for low input NM_L is defined as

$$NM_L = V_{IL} - V_{OL} \quad \dots(8)$$

Refer to figure above the voltages V_{IH} and V_{IL} are defined where the slope of the curve is -1 . Solving equation (5) at these two points, we get

$$V_{IH} = \frac{1}{8}(5V_{DD} - 2V_t) \quad \dots(9)$$

$$\text{and } V_{IL} = \frac{1}{8}(3V_{DD} + 2V_t) \quad \dots(10)$$

Using (9) and (10) in (7) and (8) we can determine express for two Nm

$$NM_H = \frac{1}{8}(3V_{DD} + 2V_t)$$

$$\begin{aligned} NM_L &= NM_H \\ &= \frac{1}{8}(3V_{DD} + 2V_t) \end{aligned}$$

These two NM are equal as we assumed the two MOSFETs in the inverter circuit to be symmetrical.

Summary for FETS

1. FET are unipolar devices.
2. The JFET operates with a reverse biased pn junction (gate to source).
3. The high input resistance of a JFET is due to the reverse biased gate-source junction.
4. Reverse bias of a JFET produces a depletion region within the channel, thus increasing channel resistance.
5. For an n -channel JFET, V_{GS} can vary from zero negatively to cut-off, $V_{GS(off)}$. For a p -channel JFET, V_{GS} can vary from positively to $V_{GS(off)}$.

6. I_{DSS} is the constant drain current when $V_{GS} = 0$.

(True for both JFETs and D-MOSFETs).

7. The devices are square law device.
8. Unlike JFETs and D-MOSFETs, the E-MOSFET can operate with $V_{GS} = 0V$.
9. Mid-point bias for a JFET is $I_D = \frac{I_{DSS}}{2}$, obtained by setting $V_{GS} = V_{GS(off)}/3.4$.
10. MOSFETs differ from JFETs in that the gate of a MOSFET is insulated from the channel by SiO_2 layer, whereas the gate and channel in a JFET are separated by a PN junction.
11. A depletion MOSFET (D-MOSFET) can operate with a zero, positive or negative gate to source voltage.
12. An E-MOSFET has no I_{DSS} parameter

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 \quad \text{JFET/D MOSFET}$$

transfer characteristic

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)$$

$$g_{ms} = \frac{2I_{DSS}}{|V_{GS(off)}|}$$

$$I_D = K(V_{GS} - V_{GS(th)})^2$$

$$k = k_n \frac{W}{L} \text{ or } k_p \frac{W}{L}$$

Digital Electronics

Binary number system—Base = 2; There are two no's : 0 and 1.

- Single binary number is called a **Binary digit** or **bit**.
- Binary number groups called **words**.
- Words are subdivided into 8-bit groups called **bytes**.
- One-half a byte is sometimes referred to as a **nibble** (a term not often used any more).

$$\begin{array}{ll} \mathbf{O}_{16} = \mathbf{O}_{10} = 0000_2 & \mathbf{9}_{16} = \mathbf{9}_{10} = 1001_2 \\ \mathbf{9}_{16} = \mathbf{1}_{10} = 0001_2 & \mathbf{A}_{16} = \mathbf{10}_{10} = 1010_2 \\ & \mathbf{B}_{16} = \mathbf{11}_{10} = 1011_2 \\ & \mathbf{C}_{16} = \mathbf{12}_{10} = 1100_2 \\ & \mathbf{D}_{16} = \mathbf{13}_{10} = 1101_2 \\ & \mathbf{E}_{16} = \mathbf{14}_{10} = 1110_2 \\ & \mathbf{F}_{16} = \mathbf{15}_{10} = 1111_2 \end{array}$$

Note—One nibble = One hex digit.

- Since $2^4 = 16$, each hex digit effectively represents the same numeric count as four binary digits.
- Another way to say this is that one column in a hex number is the same as four columns of a binary no.

[illegible]

Hexadecimal-Binary Conversion

Most computers process 32 or 64 bits at a time.

Example—
0111100010100101101011110111110

- Separate into 4-bit groups, starting at the right.

0111	1000	1010	0101	1010	1111		
7	8	A	5	A	F		
						1011	1110
						B	E

- **Another example—**

Grouping—

$$\begin{aligned} & 1001011100 \cdot 11110011_2 \\ &= \underline{10} \ \underline{0101} \ \underline{1100} \cdot \underline{1111} \ \underline{0011} \\ &= \frac{0010}{2} \ \frac{0101}{5} \ \frac{1100}{C} \cdot \frac{1111}{F} \ \frac{0011}{3} \end{aligned}$$

Binary Hex and Hex-Binary Examples

Convert—More binary to hex

101110100010; 101101110·01010011;

1111111101·10000111

- **To Convert hex-binary**

2375]₁₆ =; CD·89; 37AC·6; 3·DCAB.

- Note that leading zeros are added or removed as appropriate in the conversion processes.

- ## ● Integer Binary/Decimal Conversions

Number to be converted :

1 0 1 0 1 1 0 1

— Power of 2 of column :

7 6 5 4 3 2 1 0

128 64 32 16 8 4 2 1

$$128 + 0 + 32 + 0 + 8 + 4 + 0 + 1 = 173$$

Fractional Binary/Decimal Conversions—

- For numbers right of the binary point, we use the same approach, remembering that the powers of 2 are negative

No. to be converted : 0·1 0 1 1

Power of 2 :	-1	-2	-3	-4
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No. represented : .5 .25 .125 0.0625

$$\cdot 5 + 0 + \cdot 125 + 0 \cdot 0625 = 0 \cdot 6875$$

Second example—0.01011

$$= 0.34375$$

Decimal → Binary Integer Conversions

Simplify divide the decimal no by 2 :

- The quotient becomes the new no. to divide again.
- The remainder, **which is always be 1 or 0**, becomes one bit of the binary no., **least significant digit first**. The last quotient will always be 0, the last remainder always 1.

- Div. continues until the quotient is 0 (with a last remainder of 1). Thus, converting 26 to binary.

$$\begin{array}{rcl}
 26 \div 2 = 13, & \text{remainder } 0 & \longrightarrow \\
 13 \div 2 = 6, & \text{remainder } 1 & \longrightarrow \\
 6 \div 2 = 3, & \text{remainder } 0 & \longrightarrow \\
 3 \div 2 = 1, & \text{remainder } 1 & \longrightarrow \\
 1 \div 2 = 0, & \text{remainder } 1 & \longrightarrow
 \end{array}
 \begin{array}{c}
 0 \\
 1 \\
 0 \\
 1 \\
 1
 \end{array}
 \quad 11010_2 = 26_{10}$$

- Convert 117_{11} to binary :

Ans. 1110101_2

Converting Decimal Fractions to Binary fractions—

- The method of **successive multiplication** :

Example 0.25 to binary :

$$\begin{array}{lcl}
 0.25 \times 2 = 0.5 & \downarrow & : \text{ note 0 to left of binary points} \\
 0.5 \times 2 = 1.0 & \downarrow & : \text{ note 1 to left of binary points}
 \end{array}$$

- There is no part of the decimal fraction left, thus we are done.
- Read the binary fraction as the numbers to **left** of binary point in the two results, the remainder from the **first** multiplication first.
- So $0.25_{10} = 0.01_2$

$$\begin{array}{lcl}
 \text{Eq. } 0.375 : & 0.375 \times 2 = 0.75 & \downarrow \\
 & 0.75 \times 2 = 1.5 & \downarrow \\
 & 0.5 \times 2 = 1.0 & \downarrow \\
 & 0.375_{10} = 0.011_2 &
 \end{array}$$

Multiply decimal fraction by 2. Record the number left of the binary point. This will always be a 1 or 0.

Multiply the remaining fraction by 2, repeating above action.

$$\begin{array}{lcl}
 0.427 : & 0.427 \times 2 = 0.854 & \downarrow \\
 & 0.854 \times 2 = 1.708 & \downarrow \\
 & 0.708 \times 2 = 1.416 & \downarrow \\
 & 0.416 \times 2 = 0.832 & \downarrow \\
 & 0.832 \times 2 = 1.664 & \downarrow \\
 & 0.664 \times 2 = 1.328 & \downarrow \\
 & 0.328 \times 2 = 0.656 & \downarrow \\
 & 0.656 \times 2 = 0.312 & \downarrow
 \end{array}
 \quad
 \begin{array}{c}
 0.427_{10} \\
 01101101_2
 \end{array}$$

Continue until the fraction is eliminated.

Thus we must change the **last rule**.

Continue until the decimal fraction is eliminated or the binary fraction has **at last** twice as many places as the decimal fraction.

Converting mixed decimal numbers—

Read $\leftarrow$ for integers

$\rightarrow$ for fractions

$$36.125 = 100100.001_2; 19.375_{10} = 10011.011_2$$

$$7.33_{10} \approx 111.0101_2$$

$$10.17_{10} \approx 1010.00101_2$$

Signed Arithmetic—In real ‘world’ of mathematics, computers must represent both positive and negative binary numbers.

e.g., $124 - 237 = -113$

Thus needs to be a consistent method of representing negative numbers in binary Arithmetic operations.

There are various approaches, by they all involve using one of the digits of the binary no to represent the sign of the number.

Two methods are the sign/magnitude representation and the one’s complement method of representation.

Binary Sign Representation—

Sign-magnitude—The left bit is the sign (0 for +ve numbers and 1 for – numbers) all are the number magnitude.

1.

left bit is the sign bit

Advantages—

Simple to implement.

Useful for floating point representation.

Disadvantages—

Sign bit independent of magnitude; can be both + 0 or –0.

One Complement—

The negative numbers of the same magnitude as any given positive numbers is its one’s complement.

If $m = 01001100$, then $\bar{m} = 10110011$

The most significant bit is the sign is 0 for +ve binary no and – for negative number.

Note the problem : If $m = 00000000$, then $\bar{m} = 1111 1111$; there are two zeros in the method.

2’s Complement Negative Binary Nos.

m is the 2’s complement of n , then : $m = \bar{n} + 1$

Eq. $n = 01010100$, then $m = 10101011 + 10 = 1010 1100$

$n = 010111 11$, then $m = 10100000 + 1 = 1010 0001$

$n = 0111 1111$, then $m = 1000 0000 + 1 = 1000 0001$

For integer 2s complement representation addition and subtraction of both + and – no’s always work out correctly.

Furthermore left most bit is always 1 for negative no. **always 0 for a +ve number.**

This means that an n bit binary number in two’s complement can represent a magnitude range of $\pm 2^{n-1} - 1$.

Eq. Let $n = 0000 0000$

Then $m = \bar{n} + 1$

$$= 1111 1111 + 1$$

$$= (1) 0000 0000$$

(1) discard, 0000 0000.

The 2’s complement 0 is 0.

Example : $-50 : 50 = 0011 0010$;

$$1's C = 11001101;$$

$$2's C = 11001110$$

$-127 :$ $-127 = 0111 1111$;

$$1's C = 1000 0000;$$

$$2's C = 1000 0001$$

-1 $1 = 0000 0001$;

$$1's C = 1111 1110;$$

$$2's C = 1111 1111$$

But +ve decimal no’s are converted simply to positive binary no’s as before (no 2’s complement).

Example : $+ 67 \rightarrow 0100 0011$

2’ complement Binary to decimal (2)

$$1111 1111 \rightarrow 0000 0000 + 1$$

$$= 0000 0001$$

$$= 1 \rightarrow -1$$

$$1010 0011 \rightarrow 0101 1100 + 1$$

$$= 0101 1101$$

$$= 93 \rightarrow -93$$

But for +ve binary no

$0000 0001 \rightarrow$ rest a negative no $\rightarrow 1$.

2's complement Binary Math

If we represent binary no's in 2's complement form, simple addition and subtraction is easy.

$$\begin{aligned}\text{To } a - b &= a + (2\text{'s complement of } b) \\ &= a + \bar{b} + 1\end{aligned}$$

Adding no's is the same as decimal addition

Subtract 0111 0101 from 0111 1100 :

2's complement of 0111 0101 is 1000 1010 + 1 = 1000 1011.

$$\begin{array}{r} \text{Add :} \quad \quad \quad 0111 \ 1100 \quad 124 \\ \quad \quad \quad 1000 \ 1011 \quad -117 \\ \hline \end{array}$$

$$\text{Thrown away } \leftarrow (1) \ 0000 \ 0111 \quad \underline{007}$$

$$\text{Add : } 1100 \ 0001 + 0110 \ 1110$$

Note that 2's complement of 1100 0001 is 0011 1111 so first no is -63

$$\begin{array}{r} \text{So} \quad \quad \quad 1100 \ 0001 \quad -63 \\ \quad \quad \quad 0110 \ 1110 \quad 110 \\ \hline \end{array}$$

$$\text{Thrown away } \leftarrow (1) \ 0010 \ 1111 \quad \underline{+47}$$

$$\text{Subtract : } 1101 \ 1101 \text{ from } 0101 \ 1100$$

(We are subtracting a negative no) :

$$\begin{array}{r} \quad \quad \quad 0101 \ 1100 \quad 92 \\ \quad \quad \quad + 0010 \ 0011 \quad -(-35) \\ \hline 2' \text{ complement} \quad \underline{0111 \ 1111} \quad +127 \end{array}$$

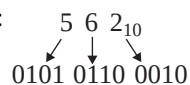
$$\begin{array}{r} \text{Add :} \quad 1000 \ 0001 \ + \ 0111 \ 0010 : \\ \quad \quad 1111 \ 0011 \rightarrow \text{negative number} \\ \quad \quad \quad 114 \\ \quad \quad \quad -127 \\ \hline \quad \quad \quad -13 \end{array}$$

$$\begin{aligned}\text{Check 2's C of } 1111 \ 0011 &\rightarrow 0000 \ 1100 + 1 \\ &= 0000 \ 1101 \\ &= 13\end{aligned}$$

So -13.

Binary codes—

BCD code (8421) :

**Excess -3 code**

To add in excess -3 add the binary no

$$\begin{array}{r} 8 \qquad \qquad 7 \\ +3 \qquad \quad +3 \\ \hline 11 \qquad \quad 10 \\ \downarrow \qquad \quad \downarrow \\ 1011 \qquad \quad 1010 \end{array}$$

ASC II code (2)

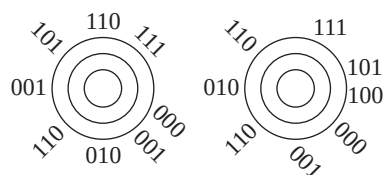
There are 128 basic ASC II characters 0 - 127₁₀, or 0 - 76₁₆ or (0000 0000 to 0111 1111 binary).

Gray-Code—Called min. change code t and is unweighted code. Because of this gray code is not suitable for arithmetic operation but finds application in input/output devices, some types of analog-to-digital converters; designation of rows and column in karnaugh map.

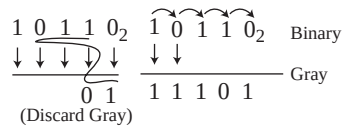
A three 3-bit Gray code may be obtained by merely reflecting the two bit code about an axis at the end of the code and assigning a third bit as 0 above axis and 1 below the axis,

	0 0	0 0 0	0 0 0 0
	0 1	0 0 1	0 0 0 1
	1 1	0 1 1	0 0 1 1
	1 0	0 1 0	0 0 1 0
(a)		1 1 0	0 1 1 0
		1 1 1	0 1 1 1
		1 0 1	0 1 0 1
		1 0 0	0 1 0 0
			1 1 0 0
			1
			1
			1

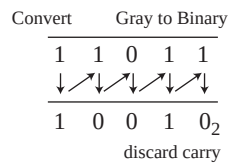
∴ Binary code.



From going 7(0111) to 8(1000) require four digits change but Gray code requires (0100 to 1100) only one bit change convert binary no's 10110₂ to Gray code



Convert Gray to Binary



Logic Gates and Boolean Algebra

AND, OR, NOT, NAND, NOR, XOR, XNOR gate etc.

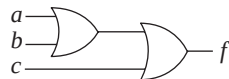
Creating Boolean Functions

By Combination of Boolean expression.

Boolean functions created in this way are called combinational logic and they are time-independent.

Let's now examine Boolean function → circuit design process.

Consider these function



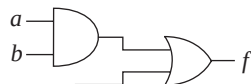
$$a + b + c = f$$



$$a + \bar{b} = f$$

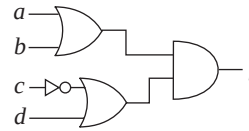


$$(a \cdot b) + c = f$$



Finally more complicated

$$(a + b) \cdot (\bar{c} + d) = f$$



Any Boolean function f maps a total of n inputs ($x_1, x_2, x_3, \dots, x_n$) into one output (0, 1), that is a Boolean function f of n variables produces a single output for each unique combination of inputs.

Principle

Any Combinational Logic Function may be represented by two levels of logic—One level of AND gates and another level of OR gates.

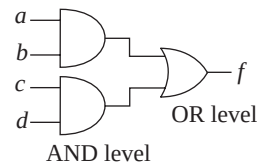
SOP and POS Boolean Representation—If we have AND function followed by an OR, then we have a “Sum Of Products” **SOP** representation.

If we have OR's followed by an AND, Product Of Sums (POS) representation.

A **sum of products** function looks like

$$f = (a \cdot b) + (c \cdot d)$$

$$f = (a \cdot b) + (c \cdot d)$$



Similarly POS :

$$f = (a + b) \cdot (c + d)$$

Using a Truth table to find Boolean expression—Making a truth table to represent the input/ output.

Defining a Boolean expression in SOP or POS form that satisfies the truth table.

Constructing a circuit that represents the “stuff in the middle” that performs the Boolean function on the inputs to get the desired output

Example SOP**Specification :**

- | | x | y | z | f | AND's |
|------------------------|-----|-----|-----|-----|---|
| | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 1 | 0 | |
| ● Output is true (= 1) | 0 | 1 | 0 | 0 | |
| For two Combination | 0 | 1 | 1 | 1 | $\bar{x}yz$ |
| Of the inputs | 1 | 0 | 1 | 1 | $x\bar{y}z$ |
| If we make the | 1 | 1 | 0 | 0 | |
| Boolean expression | | | | | |
| SOP, | 1 | 1 | 1 | 0 | |
| The first level will | | | | | |
| have only two | | | | | |
| So, | | | | | $f = (\bar{x} \cdot y \cdot z) + (x \cdot \bar{y} \cdot z)$ |

- We then define two AND functions that result in 1's being output function 1 for each x, y, z combination.

Boolean Notation; Minterms and Maxterms

- SOP is the more common form of notation. Since SOP expressions can contain a number of AND terms ORed together, the AND symbol and the parentheses are often omitted for simplicity.
- Thus, the SOP expression

$$f = (a \cdot b \cdot c) + (a \cdot c \cdot d) + (a \cdot b \cdot d) + (b \cdot c \cdot d)$$
could also be written as :

$$f = abc + acd + abd + bcd$$

- The AND symbols are **implied** in this representation and are still used when the expression might not be clear if they were left out.
- Since SOP is more compact, it is some times called the minterm form, and the AND components minterms. Similarly, POS is often referred to as the maxterm form, and the OR's maxterms, since POS terms are more unwildly.
- The names minterm and maxterm will be occasionally in future.

Product of Sums Boolean Form

A POS representation will contain a first level of OR's.

x	y	z	f
0	0	0	1
0	0	1	1

0	1	0	0	$x + \bar{y} + z$
0	1	1	1	
1	0	0	1	
1	0	1	1	
1	1	0	1	$\bar{x} + \bar{y} + z$
1	1	1	1	

In the SOP example, we looked for the output in F.

For POS, **we look for outputs that are 0.**

Here, f is 0 only for two combinations of x, y and z .

- We represent these conditions in OR form.
- Each OR will produce a 0 **only for the exact OR expression shown.**

$$f = (x + \bar{y} + z) \cdot (\bar{x} + \bar{y} + z)$$

- The Boolean expression is now. For

$$f = (x + \bar{y} + z) \cdot (\bar{x} + \bar{y} + z)$$

Note that $f = 0$ fully defined.

SOP Versus POS Representations

- **Any Boolean expression** can be represented in SOP or POS form.

a	b	f	SOP	POS
0	0	0		$a + b$
0	1	0		$a + \bar{b}$
1	0	1	$a\bar{b}$	
1	1	1	ab	

- Consider the truth table shown.

- If we consider f in SOP form,

$$\text{We have } f = a\bar{b} + ab$$

- And in POS form

$$f = (a + b) \cdot (a + \bar{b})$$

- **Either representation is correct.** In general, **we pick the representation that is simplest.**

- **For example**

x	y	z	f	OR's	AND's
0	0	0	0	$x + y + z$	
0	0	1	0	$x + y + \bar{z}$	
0	1	0	0	$x + \bar{y} + z$	
0	1	1	1		$\bar{x} \cdot y \cdot z$
1	0	0	0	$x + y + z$	
1	0	1	1		$x \cdot \bar{y} \cdot z$
1	1	0	0	$\bar{x} + \bar{y} + z$	
1	1	1	0	$\bar{x} + \bar{y} + \bar{z}$	

POS :

$$f = (x + y + z) \cdot (x + y + \bar{z}) \\ \cdot (x + \bar{y} + z) \cdot (\bar{x} + y + z) \\ \cdot (\bar{x} + \bar{y} + z) \cdot (\bar{x} + \bar{y} + \bar{z})$$

SOP :

$$f = \bar{x}yz + x\bar{y}z$$

- **Finding the truth table, given SOP Expression**

Boolean expression-truth table.

- Consider the SOP Boolean exp.—

$$f = xyz + \bar{x}yz + x\bar{y}z + xy\bar{z}$$

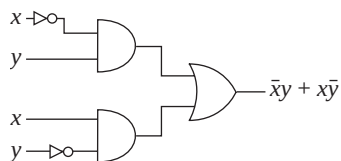
- We insert the 1 outputs in the truth table and By default all the other values of f must be 0.

x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1 $\bar{x}yz$
1	0	0	0
1	0	1	1 $x\bar{y}z$
1	1	0	1 $xy\bar{z}$
1	1	1	1 xyz

Step-by-Step Logic Circuit Design

- We usually go truth table-circuit however, consider truth table circuit

x	y	f	SOP term
0	0	0	
0	1	1	$\bar{x}y$
1	0	1	$x\bar{y}$
1	1	0	

**Looking for the Simplest Expression**

- SOP and POS Boolean expression may NOT be in the simplest form :
- However, it is possible to reduce a Boolean expression to its simplest form using the simple identities (we will study simplification of a circuit using a graphical technique called the karnaugh map).

Boolean Theorems

- The Boolean functions AND and OR are commutatives.

$$x \cdot y = y \cdot x$$

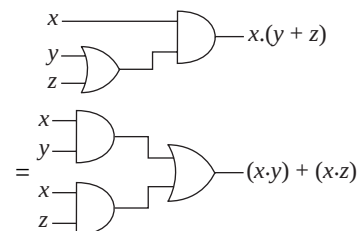
$$\text{And } x + y = y + x$$

- The **Distributivity**, the AND and OR functions are both **distributive** :

$$x \cdot (y + z) = (x \cdot y) + (x \cdot z)$$

$$\text{and } x + (y \cdot z) = (x + y) \cdot (x + z)$$

We can illustrate the AND version of this relation as—



Similarly OR version can be illustrated.

Other Fundamental Boolean Theorems**OR Form AND Form**

$$x + 0 = x \quad x \cdot 1 = x$$

$$x + \bar{x} = 1 \quad x \cdot \bar{x} = 0$$

$$x + x = x \quad x \cdot x = x$$

$$x + 1 = 1 \quad x \cdot 0 = 0$$

And, Finally, we note that :

$$\bar{\bar{x}} = x$$

De Morgan's Laws—OR and AND versions of the law.

- The OR version of De Morgan's law states that the complement of $(x \text{ OR } y)$ is equal to x -complement and y -complement, or—

$$\overline{x + y} = \bar{x} \cdot \bar{y} \quad \left| \begin{array}{l} \text{This may also be} \\ \text{stated as—} \\ x + y = \overline{\bar{x} \cdot \bar{y}} \end{array} \right.$$

- Likewise, the AND version states that the complement of $(x \text{ AND } y)$ is equal to x -complement OR y -complement, or—

$$\overline{x \cdot y} = \bar{x} + \bar{y}$$

(also stated as $x \cdot y = \overline{\bar{x} + \bar{y}}$)

Simplifying a Boolean Expression—

Consider $f = abc + ab\bar{c}$

SOP form, [distributive]

$$f = ab(c + \bar{c}) \\ = ab(1) = ab$$

(Equivalent expression much simpler, therefore easier and cheaper to build)

More complicated example Assume

$$f = (((abc + ab + bc) \cdot a + b) \cdot c) + abc$$

NOT in SOP or POS form.

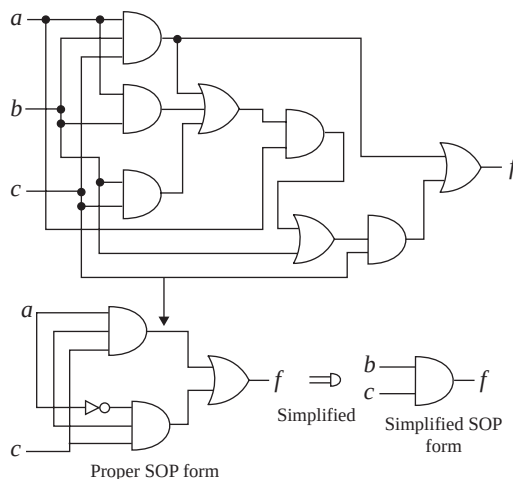
Using the truth table, the equivalent SOP expression in—

$$\begin{aligned} f &= \bar{a}bc + abc \\ &= (\bar{a} + a)bc \\ &= bc \end{aligned}$$

b' because $\bar{a} + a = 1$

a	b	c	f	SOP
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{a}bc$
1	0	0	0	
1	0	1	0	
1	1	0	0	
1	1	1	1	abc

Implemented—



Summary

- Digital circuits are electronic implementations of Boolean expressions.
- Computer circuits are all composed of logic gates (AND, OR, NOT, NAND, NOR etc.)
- Logic circuits whose outputs only depend on their inputs are called **combinational logic**.
- Combinational logic with more than two levels of logic may always be simplified into SOP or POS form using a truth table.
- An SOP and POS form may not always be in the simplest possible form. We can use De-

Morgan's law or other Boolean identities to simplify an expression, if simplification is possible.

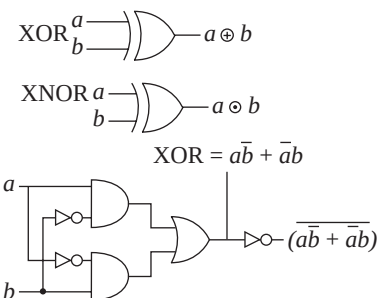
Exclusive OR/Exclusive NOR (XOR/XNOR)

- There are useful logic functions.
- Both have two or more inputs.
- (Two input truth table shown).

a	b	$a \text{ XOR } b$	$a \text{ XNOR } b$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

- For > 2 inputs, the **XOR output is 1 for an odd number of 1 inputs; XNOR has a 1 output for an even number of 1 inputs.**

XOR and XNOR are **not** basic Boolean functions, but can be made from AND, OR and NOT



Decoders

- An n - to 2^n decoder is a **combinational logic** circuit that has n inputs and **upto** 2^n output. **That is, it can have 2^n outputs**, but it may have less.
- Each output of a decoder will normally be true (i.e., go to logic 1) **for only one combination of the n inputs**.
- Consider the case of an $n = 2$ decoder. The decoder will have 2 inputs and upto $2^n = 2^2 = 4$ outputs.
- Assume that the decoder has the maximum possible number of outputs (4). Then the truth table for the 2-input decoder will show that for each combination of y and x (00, 01, 10, 11), one of the outputs will go high (logic 1).
- Let us call the inputs y and x and the outputs a, b, c and d (here, y is the more significant bit). Then let us define $a = 1$ for $y = 0, x = 0$; $b = 1$ for $y = 0$ and $x = 1$; $c = 1$ for $y = 1$ and $x = 0$ and $d = 1$ for $y = 1$, and $x = 1$.

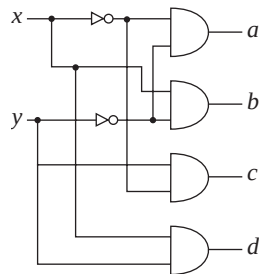
Truth tables for a - d in our 2 to 4 decoders are—

Input		Output			
y	x	a	b	c	d
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

⇒ Using the truth table above, we can define a – d in terms of x and y

$$\left. \begin{array}{l} a = \bar{y}\bar{x} \\ b = \bar{y}x \\ c = y\bar{x} \\ d = yx \end{array} \right\} = \begin{array}{l} \text{Boolean exp. for} \\ a, b, c \text{ \& } d \text{ in} \\ \text{terms of } x \text{ \& } y \end{array}$$

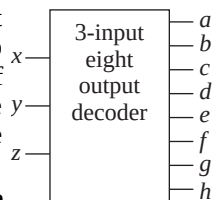
Now circuit



In the same way, a decoder with 3 inputs will have up to $2^3 = 8$ outputs and each outputs will have a unique address that represents one of the eight possible combinations of the three inputs.

Like wise, a 4-input decoder would have upto $2^4 = 16$ outputs, each of which has a unique address in terms of the four inputs.

Note again that any n -p input decoder can have upto 2^n outputs, but it may have less.



Multiplexer

Definition of a Multiplexer—A multiplexer is a combinational logic circuit that has upto 2^n inputs; an n -bit address and **one output**.

The multiplexer connects one of the inputs to the outputs, depending on the value of the n -bit address.

The n -bit address is **decoded**, just as before.

Thus the multiplexer uses a decoder and selector circuit to tie one of its inputs to its output.

The multiplexer is usually symbolized by the abbreviation **MUX** as the symbol for its function.

Multiplexer : An “Input to Output” selector.

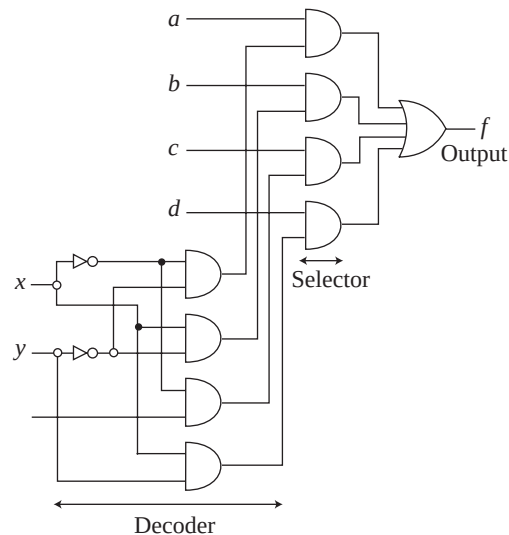
The truth table for a multiplexer is shown below.

Assume a 4-input MUX, with inputs labelled a, b, c, d .

Then there must be two address lines, y (MSB) and x (LSB).

The output is denoted as f

y	x	f
0	0	a
0	1	b
1	0	c
1	1	d



Note—That the multiplexer has 1-bit output.

Differences in Decoder and Multiplexer

Decoder—A decoder has n inputs, which are called the address.

A decoder has up to 2^n outputs (it can have that many, max, but it might have less). Each output line is true (or 1) for a specific combination of the input lines, **called the address**.

Multiplexer—A multiplexer has two sets of inputs— n address lines (fuse like decoder) as many as 2^n inputs, one of which is selected by each address for outputs.

A multiplexer has **only one** output. **The output is the value of the input selected by the address.**

Thus we see that a **decoder makes up a part of a multiplexer**.

Binary Arithmetic Circuits—A binary adder is a large part of a computer CPU. The CPU “figuring” unit is sometimes called the ALU, or arithmetic/logic unit.

The ALU is a **combinational logic unit** that can add subtract or do logical operation such as AND, OR, NOT etc.

Consider the “rules of addition” (adding two no’s only).

No’s are added on a **Columnar basis**, starting or the right.

If the sum of one column is a 2-digit number, the right no goes in current column and the left no. (always a 1) become a “carry” **to the next column to the left**.

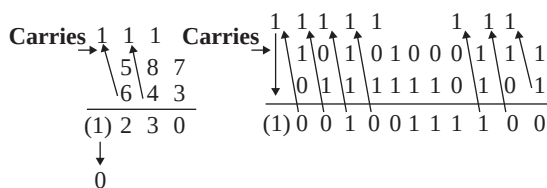
Column addition always includes the carry **from the column to the right**.

The resulting sum in any column will be **only a single digit**.

Binary no. added in same way—

Basic principles for an n -column **addition**.

Add column i plus carry from column $i - 1$ (c_{i-1})



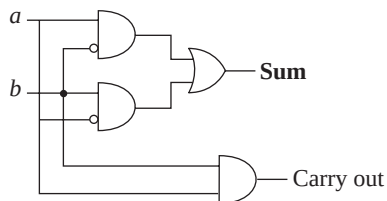
If a one digit result, that number is the sum of column i , s_i .

If a 2-two digit result, the right digit is s_i : the left digit is i or c_i .

The column i carry, c_i , will be added to column $(i + 1)$.

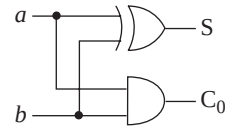
A Half Adder

a	b	s	c_0	$e_{\text{exp.}}$	$c_0 \text{ exp.}$
0	0	0	0		
0	1	1	0	$\bar{a}b$	
1	0	1	0	$a\bar{b}$	
1	1	0	1		ab



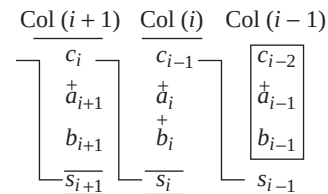
Thus, to add two 8-bit numbers, we simply use eight column **adder**.

Alternate adder—



The “Full Adder”—The half adder circuit is an important step in building a digital adder, but it is not the total solution to the binary addition problem.

When adding a binary no’s on a columnar basis, in addition to the column bits a_i and b_i , there is also the possibility of a **carry bit** c_{i-1} from the column to the right.



Thus the 2 bit column adder must be a 3-bit adder.

This 3-bit adder is a **full adder**.

In adding two n bit binary number, the **right-most** bits of the two no’s of two number could be added with a half adder (no carry in).

However most adder circuits use all full-adder, since carry in for the least significant bit can be useful we will see that use later.

Full Adder

a	b	c_i	s	c_0	s
0	0	0	0	0	$\bar{a}\bar{b}\bar{c}$
1	0	0	1	0	$\bar{a}\bar{b}c$
0	1	0	1	0	$\bar{a}b\bar{c}$
1	1	0	0	1	$\bar{a}bc$
0	0	1	1	0	$a\bar{b}\bar{c}$
1	0	1	0	1	$a\bar{b}c$
0	1	1	0	1	$ab\bar{c}$
1	1	1	1	1	abc

$$S = \bar{a}\bar{b}\bar{c} + \bar{a}\bar{b}c + \bar{a}b\bar{c} + \bar{a}bc + abc$$

$$C_0 = \bar{a}b\bar{c} + \bar{a}bc + \bar{a}bc + abc$$

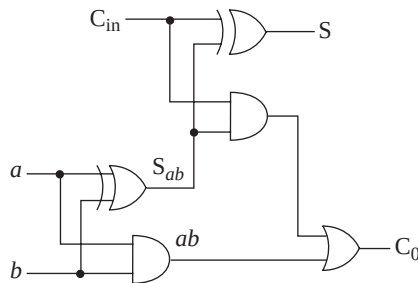
Alternate constructions—

For the full adder

$$S = (a \oplus b) \oplus c$$

$$C = a \cdot b + (a \oplus b) \cdot c$$

Where $C = C_{in}$



Simplifying Logic Circuits with Karnaugh Maps—

$$f = abc + \bar{a}bc + ab\bar{c}$$

We use graphical technique to solve this which is called **Karnaugh map. k-map.**

A k-map is a truth table graph, which aids in visually simplifying logic

It is useful for up to 5 or 6 variables, and a good tool to help understand the process of logic simplification.

A 2-variable k-map.

This is very simple k-map demonstrates that an n -variable k maps contains all combinations of the n variables in the k-map space.

	$\bar{y}$	y
$\bar{x}$	00 0	01 1
x	10 2	11 3

← This minterm express by $f = \bar{x}y$

3 variables—Each square represents 3-variable minterms or maxterm.

All the 8-possible 3 variables terms are represented on the k-map.

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{x}$	000 0	001 1	011 3	010 2
x	100 4	101 5	111 7	110 6

As an example, this minterm cell (011) represents the minterm $f = \bar{x}yz$

When moving horizontally or vertically, only 1 variable changes between adjacent square **never 2**. This property of k-map, is unique and accounts for **unusual numbering** system.

The k-map shown is one labeled for a **SOP**

As an example, this minterm cell (011) represents the minterm $f = \bar{x}yz$.

Similarly 4-variable—

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$	0000 0 0	0001 1 1	0011 3 3	0010 2 2
$\bar{w}x$	0100 4 4	0101 5 5	0111 7 7	0110 6 6
$w\bar{x}$	1100 C 12	1101 D 13	1111 F 15	1110 E 14
wx	1000 8 8	1001 9 9	1011 B 11	1010 A 10

SOP K map.

Karnaugh Map Comments—

k-maps can be labeled many ways, by

We can label it in **binary, decimal, or hexadecimal**. We can also designate the Boolean function by the **k-map squares it occupies**.

The minterms on the k-map can be labelled as

$$f = \Sigma m(5, 7, 13, 15) \text{ in decimal, or}$$

$$f = \Sigma m(5, 7, D, F) \text{ in hexa decimal}$$

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$	0000 0 0	0001 1 1	0011 3 3	0010 2 2
$\bar{w}x$	0100 4 4	0101 5 5	0111 7 7	0110 6 6
$w\bar{x}$	1100 C 12	1101 D 13	1111 F 15	1110 E 14
wx	1000 8 8	1001 9 9	1011 B 11	1010 A 10

If we were given the sigma (Σ) co-ordinates above, we could immediately deduce that SOP function was

$$f = \bar{w} \times \bar{y}z + \bar{w}xyz + wx\bar{y}z + wxyz$$

Any square or rectangular group of cells that is a power of 2 (1, 2, 4, 8, 16) is called **implicant**.

All of the groups of squares in the k-map to the left (including the single square) represent **implicants of different sizes**.

0000	0001	0011	0010
0 0	1 1	3 3	2 2
0100	0101	0111	0110
4 4	5 5	7 7	6 6
1100	1101	1111	1110
C 12	D 13	F 15	E 14
1000	1001	1011	1010
8 8	9 9	B 11	A 10

Prime Implicants—

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$				
$\bar{w}x$		1	1	
$w\bar{x}$		1	1	
wx				

So, the prime implicants at right may be represented by the Boolean expression

$$f = xz$$

k-map as follows—

For an **SOP expression**, find the *k*-map cell for each minterm of function *f*, and place a one (1) in it. Ignore 0's.

Circle groups of cells that contains 1's

The no. of cells enclosed in a circle must be a power of 2 and square or rectangular.

It is okay for groups of cells of overlap.

Each circled group of cells corresponds to a prime implicant of *f*.

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$				
$\bar{w}x$		1	1	
$w\bar{x}$		1	1	
wx				

Note that the more cells a given circle enclose, the fewer variables needed to specify the implicant,

$$f = \bar{w}x + xz$$

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$				
$\bar{w}x$	1	1	1	1
$w\bar{x}$	1			1
wx				

Another example—

The implicants of *f* are 4, 6, 12, 14.

This corresponds to

$$f = \bar{w}x\bar{y}\bar{z} + \bar{w}xy\bar{z} + w\bar{x}y\bar{z} + wx\bar{y}z = 0$$

$$\Rightarrow f = x\bar{z}$$

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$				
$\bar{w}x$	1			1
$w\bar{x}$	1			1
wx				

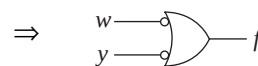
A POS *k*-Map—

The procedure is the same, except that we map 0's

$$\text{Let } f = (\bar{w} + \bar{x} + \bar{y} + \bar{z}) \cdot (\bar{w} + \bar{x} + \bar{y} + z) \cdot (\bar{w} + x + \bar{y} + \bar{z}) \cdot (\bar{w} + x + \bar{y} + z)$$

Simplified expression is

$$f = \bar{w} + \bar{y}$$



	$y + z$	$y + \bar{z}$	$\bar{y} + \bar{z}$	$\bar{y} + z$
$w + x$	0000 0 0	0001 1 1	0011 3 3	0010 2 2
$w + \bar{x}$	0100 4 4	0101 5 5	0111 7 7	0110 6 6
$\bar{w} + \bar{x}$	1100 C 12	1101 D 13	1111 F 15	1110 E 14
$\bar{w} + x$	1000 8 8	1001 9 9	1011 B 11	1010 A 10

Summary of Karnaugh Map Procedure—

(1) Start with the truth table of Boolean expression.

(2) If starting from the truth table, write the Boolean expression for each truth table term that is 1 (if SOP) or 0 (if POS).

(3) Develop the full Boolean expression, if necessary, by ORing the AND terms together if SOP or ANDing OR terms is POS).

(4) On the *k*-map, plot 1's (for SOP) or 0's (for POS).

(5) Group implicants together to get the largest set of prime implicants possible. Prime implicants may overlap each other. They will always be square or rectangular groups of cells that are power of 2 (1, 2, 4, 8).

(6) The variables that make up the term (s) of the new expression will be those which do not vary in value over the extent of each prime implicant.

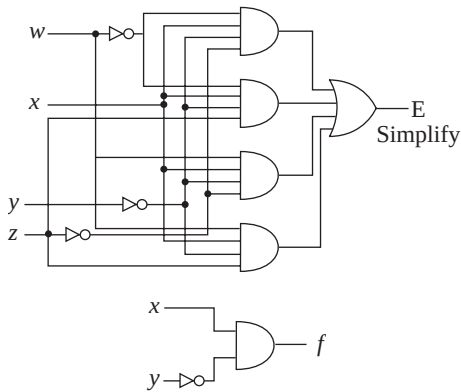
(7) Write the Boolean expression for each prime implicant and then OR (for SOP) or AND (for POS) terms together to get the new expression.

Example—Reduce the Boolean expression

$$f = \bar{w}x\bar{y}z + \bar{w}xyz + \bar{w}xy\bar{z} + wx\bar{y}z + wxyz + wxy\bar{z} + w\bar{x}y\bar{z} + w\bar{x}yz$$

Ans. $f = xy + xz + wz$

Original Circuit—



(3) Biggest Prime Implicant

w	x	y	z	f
0	0	0	0	
0	0	0	1	1
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	1
0	1	1	0	
0	1	1	1	1
1	0	0	0	
1	0	0	1	1
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	1
1	1	1	0	
1	1	1	1	1

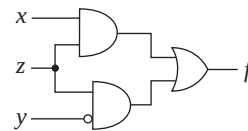
	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{w}\bar{x}$		1		
$\bar{w}x$		1	1	
wx		1	1	
$w\bar{x}$		1		

$$f = xyz + \bar{y}z$$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{w}\bar{x}$		1		
$\bar{w}x$		1	1	
wx		1	1	
$w\bar{x}$		1		

$$f = xz + \bar{y}z$$

Prime Implicant should overlap, if this mean that they can be made larger resulting gate



The simplified expression (and not very simplified at that) is—

$$f = \bar{w}z + \bar{w}x\bar{y} + x\bar{y}z + \bar{x}yz$$

Sometimes Major Simplification is Not Possible—Using the Σ notation

$$f = \Sigma m(1, 2, 3, 5, 7, 11, 13)$$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{w}\bar{x}$		1	1	1
$\bar{w}x$		1	1	
wx		1		
$w\bar{x}$			1	

The concept of “Don’t Cares”—The six implicants on the k -map shown be represented by the simplified exp. $f = xy + xz$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{w}\bar{x}$	×	1	1	1
$\bar{w}x$	×	1	1	1
wx	×	1	1	1
$w\bar{x}$	×			

	$y\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{w}\bar{x}$	x			
$\bar{w}x$	x	1	1	1
$w\bar{x}$	x	1	1	1
wx	x			

Suppose for our particular logic system, we know that y and z will **never be 0 together**.

Then the $\bar{y}\bar{z}$ implicants do not matter, **since they cannot happen**.

To show the condition cannot occur, we put X's in the column—they are “don't cares”—conditions that cannot happen.

Since the function will never get into the left column, we “don't care” how those minterms are represented.

Why not make them 1's—Doing that, we can make a much larger prime implicant and a simpler Boolean expression. For this implicant, $f = x$.

The expression is valid, since the forbidden condition with never allow the two left squares to be occupied.

“Don't Cares” let us **further simplify an expression**.

“Don't Cares”—Summary—“Don't Cares” occurs when there are variable combinations, represented by squares on a Karnaugh map, **that cannot occur in a digital circuit or Boolean expression**.

Such a square is called a “Don't Care”. Since it can never happen, we can assign a value of 1 to the square and use that 1 to (possibly) construct larger prime implicants, further simplifying the circuits.

We will see later in sequential logic that the concept of “Don't Cares” will help in simplifying counter circuits.

Sequential Logic and Clocked Circuits

A combinational logic circuit operates such that the **output** of the circuit **depends only on the input (s)**.

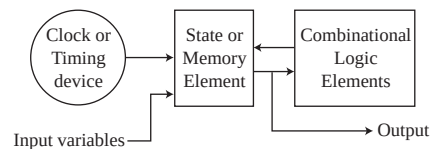
Sequential logic differs from combinational logic in several ways—

⇒ Its outputs depend not only on logic inputs but also the **internal state of the logic**.

⇒ The output of sequential logic does not necessarily change when an input changes, but is synchronized to some “triggering event”.

⇒ Sequential logic is often synchronized or triggered by a train of regular pulses on a serial input line, which is referred to as a “clock”.

Representation of sequential logic

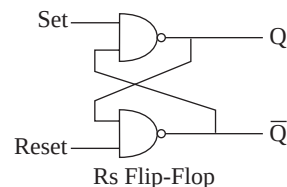


Sequential logic has inputs and outputs like combinational logic. It also has a timing (synchronizing) mech. and state or memory elements.

Sequential logic may (usually will) have combinational parts.

Sequential logic outputs depend on the timing and state elements as well as the input variables. That is, the outputs normally change **as a function of the timing element**.

The Simple Latch or R-S Flip-Flop—The simplest example of a sequential logic device is the R-S Latch or R-S flip-flop (R-S FF).



This is a non-clocked device that simply consists of two cross-connected 2-input NAND gates (**it may also be represented as combinations of other gates**).

Note that per our definition, the R-S FF has an output that depends on its **current state** as well as the inputs.

This most basic form of the R-S FF uses “negative-true” input logic. This is, it changes states based on inputs that **transition to 0 not to 1**.

When the Q output is 1 and $\bar{Q} = 0$, the RS-FF is said to be “set”.

Likewise, when $Q = 0$ and $\bar{Q} = 1$, the flip-flop is said to be “reset”.

Q and $\bar{Q}$ will always have **opposite** states.

The R-S FF is **bistable**. That is, it is “at rest” in one of its two states (“set” or “reset”) until an input forces it to change.

If the RS FF is in the “set” state, it will not go “reset” until the **Reset** line goes “true” (in this case, to 0).

Likewise, when “reset”, it will not go “set” unless the **set** line goes to 0.

Note also that once “set”, if **set** goes to 0 more than once, the FF simply stays “set”.

Likewise, when “reset”, more Reset’s do not affect the circuit; it remains “reset”.

Thus the RS FF has an output that depends not only on the inputs but the **current state**.

Note—The triggering signal of an input (set or reset) is always assumed to be **momentary**. That is, a “Set” or “Reset” signal is considered to as a very short time (in case of real circuits ~ nanosecond or less).

Assume the FF is Reset ($Q = 0$). Also, since the Set and Reset inputs are not active, both input are at 1.

Thus set = 1, reset = 1, $Q = 0$, $\bar{Q} = 1$.

Then the cycle is—

1. Set goes active (Set $\rightarrow 0$).
2. Then Q must $\rightarrow 1$ (output of a NAND = 1 if any input = 0).
3. Then both input to bottom NAND are 1 and $\bar{Q} \rightarrow 0$.
4. The other input to the upper NAND is now 0. Thus, when the set signal goes back high, Q remains at 1, since the other input is still 0.
5. Likewise, since both inputs to the lower NAND are now 1, then the value of $\bar{Q}$ remains 0.
6. The reverse cycle (set-to-reset) occurs in the identical way, except that the **change is initiated by reset going low**.

The truth table for RS FF

Inputs		Current Outputs		New Outputs	
S	R	Q	$\bar{Q}$	Q	$\bar{Q}$
1	1	1 or 0	0 or 1	Same	Same
0	1	0	1	1	0
0	1	1	0	Same	Same
1	0	1	0	0	1
1	0	0	1	Same	Same
0*	0*	1 or 0	0 or 1	1*	1*

This is race condition and not stable. The S-R input 0 -0 is forbidden.

As noted on the t - t , 0-input to both R and S is forbidden.

Note the race condition that is triggered by $R = S = 0$ —

Then $Q = \bar{Q} = 1$, so both are 2-NAND inputs are 1.

If S and R go to/simultaneously, then all 4 inputs of the two 2-input NAND gates are 1 and both outputs go to 0.

The result is a “race” to see which output gets to 0 first, getting one 2-NAND input to 0, and therefore forcing the NAND output to 1. The result of the “race” cannot be predicted.

Thus R and S = 0 together is forbidden, since the output is not stable.

RSFF—Many exotic flip-flops are based on the simple R-S FF.

Used in

The data storage register.

The binary counter.

The shift register.

Most “status indicators”

Many types of computer memory.

Note— Q can be set to 1 or 0. Thus the output Q can be said to “remember” are bit.

When an RS FF has its Q output changed to 1 or 0, the output stays in that state **until the opposite input is triggered**.

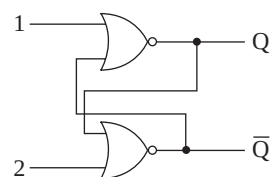
Thus the flip-flop or later has the property of “remembering” a one-bit binary no. It can be set to 1 or 0.

The flip-flop therefore becomes one storage memory choice *i.e.* we can use flip-flops to store binary no’s in a computer.

Simplified Version

S	R	Q
0	0	Indeterminate
0	1	Set (1)
1	0	Reset (0)
1	1	Hold

RS FF using NOR gate



Simplified Version

S	R	Q
0	0	Hold
0	1	Reset (0)
1	0	Set (1)
1	1	Intermediate

Inputs		Current Outputs		New Outputs	
$\textcircled{S}_1$	$\textcircled{R}_2$			Q	$\bar{Q}$
0	0	1 or 0	0 or 1	Same	Same
0	1	0	1	0	1
0	1	1	0	Same	Same
1	0	1	0	1	0
1	0	0	1	1	0
1*	1*	1 or 0	1 or 0	0	0

Intermediate state

Clocked

The majority of all sequential logic circuits are **clocked logic circuits**.

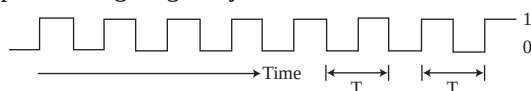
Clocked circuits are circuits that are regulated by a clocking element, which determines when state changes are made in the sequential logic circuit.

In a **clocked sequential** circuit, in general, the circuit can only change states on a “tick of the clock element. Thus we say that all operation in this type of circuits are “**clocked**”.

Clock does not mean any kind of watch.

We refer to a circuit as a “clocked circuit” when all the sequential elements in the circuit change states in synchronization to a train of pulses.

Such a pulse train is shown below. The clock pulse change regularly from 0 to 1 and back.



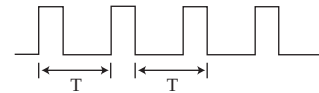
Facts—The period T of a sequential logic clock is the distance between two identical points on the pulse train, *e.g.*, two rising edges or two falling edges.

The clock has only two states : 0 and 1, and clock alternate between these states.

The amount of time that the clock spends in each of the two states is called the **duty cycle**.

The clock above has a 50/50 duty cycle; it spends equal time each period in the 1 and 0 states.

Equation the clock on next page does not have a 50/50 duty cycle. It stays in the “1” state about 35% of time, and in the “0” state about 65% of the time. Thus



Thus we say that the clock has a “35/65” duty cycle. In the same way, a clock can have a 70/30 cycle time (*i.e.*, it stays in the “1” state 70% of the time) and so on.

Clocked Flip-Flop

All ff’s have the same basic configuration—

Both true and false outputs (“Q” and “Q-not”)

“Set” is when $Q = 1$.

Triggered by “Set” and “reset” input.

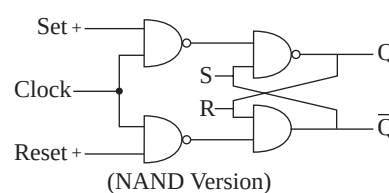
The most useful ff’s are not simple asynchronous (non-clocked) ff’s, however, but synchronous (“clocked”) ff’s.

Clocked ff’s are very similar to non-clocked ff’s the main difference is that in addition to a “set” or “reset” input to cause the outputs to change, there must also be the presence of a clock signal in its true state (normally 1).

Thus clocked ff’s do not change states, regardless of the set or reset inputs, until the “clock ticks”.

Clocked RS Flip-Flop

In addition to the set and reset inputs, the clock input is present.



Since when clock is low (0), neither set or reset affect the circuit, we say that the clock “gates” the set or reset signal to the RSFF.

In this case, the set or reset input must be high (1) to set or reset the ff when the clock goes true ($0 \rightarrow 1$).

Having set and reset 1 **at the same time** is forbidden as for the RSFF, simultaneous set and reset true causes a **race condition** when clock is high.

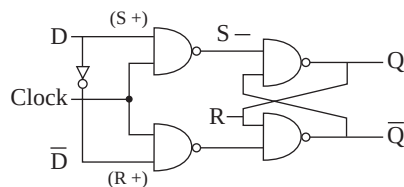
Clock	S	R	Q	$\bar{Q}$	New Q	Output $\bar{Q}$
0	X	X	1 or 0	0 or 1	Same	Same
1	0	0	1 or 0	0 or 1	Same	Same
1	1	0	0	1	1	0
1	1	0	1	0	Same	Same
1	0	1	1	0	0	1
1	0	1	0	1	Same	Same
1	1*	1*	1 or 0	0 or 1	1*	1*

Race condition

More sophisticated Flip-flops.

The D Flip-Flop

The **DFF** is bistable circuit with one input plus the clock. The term “D” refers to “data”.



Since the DFF is limited to one input, there is no chance that a race condition will occur.

The clocked DFF can be created simply by replacing the “reset” input with an inverted “set” input in the clocked RS Flip-Flop.

In this configuration, we can see that if the D input = 1, when the clock is high, the FF will go into the “set” state (“set” input = 1, “reset” = 0).

When the D input = 0, then the FF goes into the “reset” state when clock goes high (“set” input = 0, “reset” = 1).

When clock = 0, the FF is Idle.

There is no “forbidden” state here—the DFF output is defined for All inputs.

Inputs				New Output	
Clock	D	Q	$\bar{Q}$	Q	$\bar{Q}$
0	X	1 or 0	1 or 0	Same	Same
1	1	0	1	1	0
1	1	1	0	Same	Same
1	0	1	0	0	1
1	0	0	1	Same	Same

So a flip-flop is useful as “**memory**”. It can be set to 1 or 0 and retain that value until triggered to the opposite state.

There are many ff’s, (Summary), the RS FF is the simplest example of sequential logic.

(1) Non clocked; made of crossed-connected 2-input NANDs. (or NOR’s etc.)

(2) Output depends on **current state** as well as the inputs.

(3) Uses “negative-true” input logic. Output Q and $\bar{Q}$ are always opposite.

(4) When Q = 1, $\bar{Q}$ = 0, the RS FF is “set”, when Q = 0, $\bar{Q}$ = 1, ff is “reset”.

(5) Will not change states unless **opposite input** triggered.

(6) Only **opposite state trigger** causes state change.

(7) Race condition triggered by R = S = 0 (output cannot be predicted).

Timing Diagrams

It is important when designing sequential circuits to understand the timing relationships between circuit elements.

Normally done by plotting the “transitions”—the change between logic “1” and logic “0”—of the element.

All the timing normally related to the clock.

In the fundamental timing diagram, unless specifically instructed to do otherwise, always start with the clock.

Always assume a “50/50” clock, if duty cycle not specified.

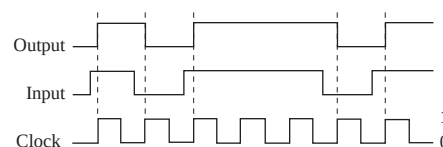
To illustrate DFF timing by plotting the DFF “Q” output.

Remember that a DFF output is normally triggered when clock = 1.

Assume DFF is originally “reset”, then on the rising edge of the clock, the DFF output goes to “set” (1), when the D input is 1.

On successive clock pulses, the DFF “Q” changes as D changes.

Principle—Output “Q” of a simple DFF tracks D **when the clock ticks**.



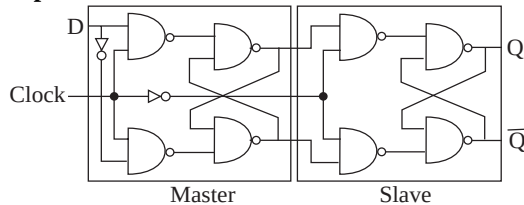
“Master-Slave”, or Delay Flip-Flops

It is often desirable to have a flip-flop whose output does not change immediately when its internal state is altered from “set” ($Q = 1$) to “reset” ($Q = 0$) or *vice-versa*.

This sort of ff is called a “master-slave” or “delay” ff.

The idea behind the master slaves ff is to have a “master” (*i.e.*, controlling) ff change states one edge of a clock pulse (normally the leading edge) and have a second ff connected to the first change to the same state as the “master” on the **trailing edge**, or “back side” of a clock pulse.

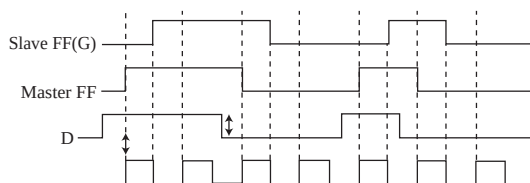
In this way, the internal state of the ff changes **one half clock cycle prior to the time in which the changed state appears on the circuit outputs.**



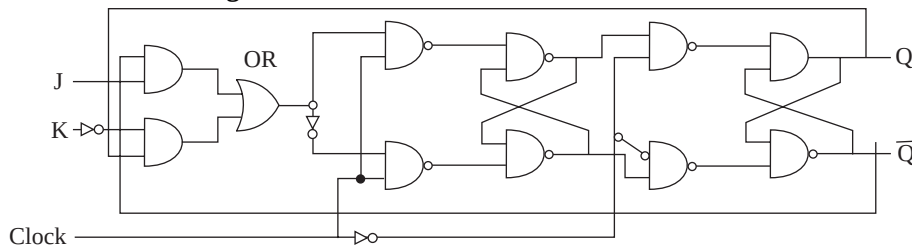
DFF converted to master-slave type.

The slave (basically a clocked RSFF) always mirrors the state of the master.

The slave circuit **changes state $\frac{1}{2}$ cycle after the master. The device still operates as a D-FF; no indeterminate state.**



JK master slave FF designed as—



The key states are $J = K = 1$, for either output state (set or reset).

If $Q = 1$ (“set”) and $J = K = 1$, output of the OR = 0, so the ff will be set.

(For $J = 1$ and $K = 0$ or $J = 0$ and $K = 1$) normal set of reset occurs.

Then for $J = K = 1$, when the clock ticks **Q-the output state.**

D Flip-Flop Symbols

Flip-Flop details is not usually shown in diagrams.

One symbol for a DFF is shown.

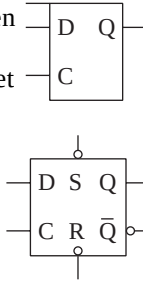
⇒ **There is no small circle on the either input.**

∴ “1” is the active state (when clock and $D = 1$, output will → 1).

DFF’s with asynchronous Set and Reset are also available.

Circles on S and R inputs mean that Set and Reset are “negative-true” signals (active at level 0).

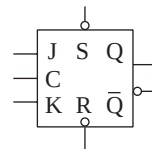
⇒ “Q-not” out is also available.



Set and Reset have the same problem, before if $S = R = 0$, output may be indeterminate.

The JK Master-Slave FF

Not useful to have a FF that will not have indeterminate outputs when S and R inputs are both 1 simultaneously.



The JKFF, sits those. The J input corresponds to “set” while K corresponds to “Reset”.

The JK FF is designed so that the of $J = K = 1$ does **not** result in an indeterminate output when clocked.

There may still be asynchronous RS inputs with the usual cautions. However, **the JK inputs are not restricted.**

Input		Outputs		AND output		OR output		New output	
J	K	Q	$\bar{Q}$	1	2	3	4	Q	$\bar{Q}$
0	0	0	1	0	0	0	1	Same	Same
0	0	1	0	0	1	1	0	Same	Same
0	1	0	1	0	0	0	1	Same	Same
0	1	1	0	0	0	0	1	0	1
1	0	0	1	1	0	1	0	1	0
1	0	1	0	1	0	1	0	Same	Same
1	1	0	1	1	0	1	0	1	0
1	1	1	0	0	0	0	1	0	1



JK +ve edge triggered

J	K	C	Outputs		
			Q	$\bar{Q}_0$	
0	0	↑	Q ₀	$\bar{Q}_0$	No change
0	1	↑	0	1	Reset
1	0	↑	1	0	Set
1	1	↑	$\bar{Q}_0$	Q ₀	toggle

Solved Questions

Q. 1. The load resistance of a full wave rectifier is $10^3 \Omega$. And the input AC voltage (rms) is 200-0-200. Calculate average d.c. voltage, average d.c. current and ripple factor. Diode do not show any significant resistance itself.

Solution : The peak value of d.c. current can be calculated by

$$\begin{aligned}
 I_m &= \frac{V_m}{R_L} \\
 &= \frac{V_{rms} \sqrt{2}}{R_L} \\
 &= \frac{200 \sqrt{2}}{10^3} \\
 &= 0.2828 \text{ amp.}
 \end{aligned}$$

The average d.c. current is given by

$$\begin{aligned}
 I_{dc} &= \frac{2I_m}{\pi} \\
 &= \frac{2 \times 0.2828}{3.14} \\
 &= 0.18 \text{ amp}
 \end{aligned}$$

Average d.c. voltage

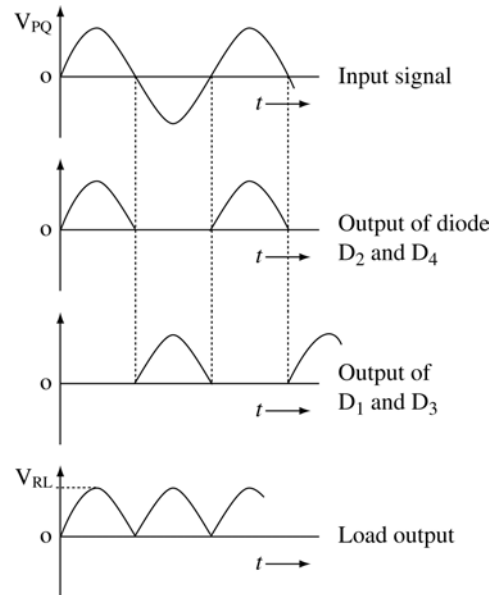
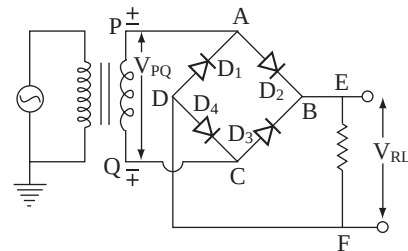
$$\begin{aligned}
 V_{dc} &= I_{dc} \times R_L \\
 &= 0.18 \times 10^3 \\
 &= 180 \text{ volt}
 \end{aligned}$$

and Ripple factor, $\gamma = \sqrt{\frac{V_{rms}^2}{V_{dc}^2} - 1}$

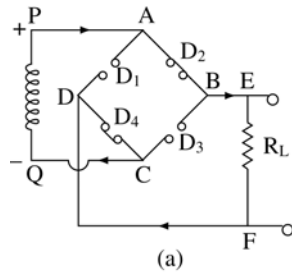
$$\begin{aligned}
 &= \sqrt{\left(\frac{200}{180}\right)^2 - 1} \\
 &= 0.4843
 \end{aligned}$$

Q. 2. Explain the function of bridge rectifier. Also write its advantage over single face full wave rectifier.

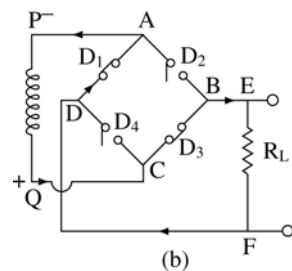
Solution : A bridge rectifier consists of four identical diodes arranged in a cyclic manner as shown in the figure—



Working—During the positive half cycle diode D_2 and D_4 conducts as 'P' is +ve, and 'Q' is -ve. D_1 and D_3 shows open circuit due to reverse biased the current cycle will be PABEFDCQ fig. (a).



During next half cycle 'P' will be negative whereas 'Q' positive so D_1 and D_3 conducts and D_2 and D_4 open circuited. The current cycle be QCBFEDAP, fig (b). As we have seen 'E' is

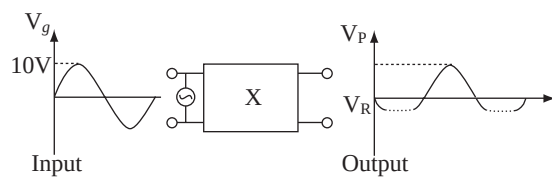


always positive with respect to 'F'. So the voltage across 'RL' always shows positive voltage (Fully rectified).

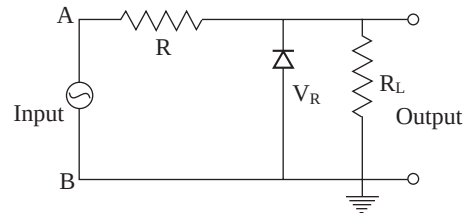
Advantage—In Bridge rectifier the transformer is not centre tapped. Also it can be used for high voltage operation. The transformer used is also small.

Q. 3. A sinusoidal waveform signal is applied to the input of box 'X'. What electronic circuit will you construct in box X to get the output signal as shown in the figure ?

Also calculate the value of V_p . If you utilise a diode of barrier voltage $0.7V$, a resistance of 100Ω and another resistance of $1k$ as load. The input signal is $10V$.



Solution : The circuit shown in the box is a negative clipper i.e., a circuit which clips-out a fraction of negative half cycle of input signal applied. The circuit can be constructed as follows :



During the positive half cycle diode behaves as open circuited. So the applied voltage drops across R_L . The load resistance provides an output voltage equal to $V_{in} \left(\frac{R_L}{R + R_L} \right)$.

During negative half cycle of input signal 'B' will be positive so the diode conducts. Since diode and load resistance are parallel the voltage delivered at load R_L will not exceed diode barrier voltage. In other words only the barrier voltage will appear at output and rest of the negative signal clips out.

$$V_p = V_{in} \left[\frac{R_L}{R + R_L} \right]$$

$$V_{in} = 10V$$

$$R_L = 1k$$

$$R = 100\Omega$$

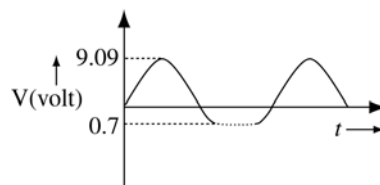
$$V_p = 10 \left(\frac{1}{1 + 0.1} \right)$$

$$= 9.09V$$

and

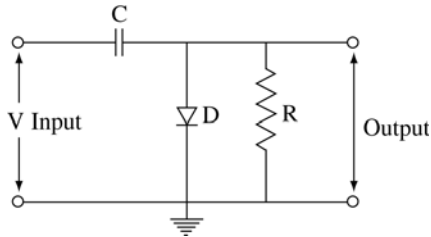
$$V_R = 0.7V$$

The output waveform will follow the given figure—



Q. 4. Design a diode clamper circuit to clamp the positive peaks of the input signal at zero level. The input frequency is 1000 Hz and $R_f = 1\text{ k}\Omega$ $R_r = 200\text{ k}\Omega$.

Solution : A negative clamper circuit is shown in the figure.



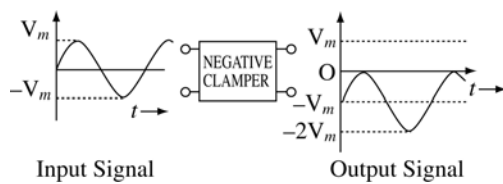
Diode forward resistance R_f and diode reverse resistance are given the R will be $\sqrt{R_f R_r}$

$$\begin{aligned} R &= \sqrt{1 \times 10^3 \times 200 \times 10^3} \Omega \\ &= 14140 \Omega \\ &= 14.14 \text{ k}\Omega \\ &\approx 15 \text{ k}\Omega \end{aligned}$$

Then the value of capacitor needed

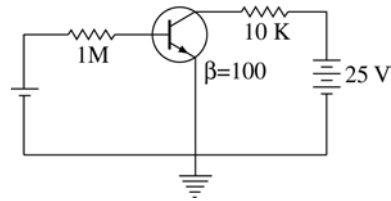
$$\begin{aligned} C &= \frac{20}{f R} \\ &= \frac{20}{1000 \times 15} \mu\text{F} \\ &= 1.33 \mu\text{F} \end{aligned}$$

The input and output wave forms can be shown as follows :

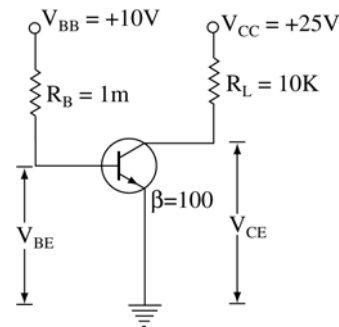


Q. 5. For the common emitter configuration of transistor, as shown in the figure find—

- (i) base current
- (ii) collector and emitter current
- (iii) voltage across collector and emitter



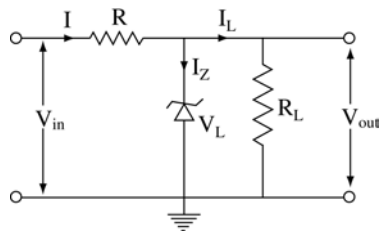
Solution : The above figure can be simplified as—



- (i) Base current $= I_B \cong \frac{V_{BB}}{R_B}$
 $= \frac{10\text{V}}{1\text{M}} = 10\mu\text{A}$
- (ii) Collector current $= \beta I_B$
 $= 100 \times 10\mu\text{A} = 1\text{mA}$
 Emitter current $= I_E = I_C + I_B$
 $= 10\text{mA} + 10\mu\text{A}$
 $= 1.01 \text{ mA}$
- (iii) $V_{CE} = V_{CC} - I_C R_L$
 $= 25 - (1 \times 10) \text{ V}$
 $= 15 \text{ volt}$

Q. 6. Explain the functioning of zener diode as shunt regulator. What is the effect of zener impedance on this regulated power supply ?

Solution : When a zener operates in breakdown region, the voltage across its terminals remains fairly constant for a large variation of current through it. This condition is maintained as long as the input voltage V_{in} is greater than the zener voltage V_Z . During this period zener operates in breakdown region and delivers constant voltage across load resistance R_L . A schematic diagram of the zener regulated power supply is shown below. In this circuit the series resistance R limits the input current.



$$V_{out} = V_{in} - IR$$

The output voltage V_{out} remains constant for (i) change in load current and/or (ii) change in input voltage.

The zener impedance Z_Z plays a significant role in the supply of regulated output voltage V_{out} may slightly change due to zener impedance.

$$V_{out} = V_Z + I_Z Z_Z$$

The zener shunt regulated supply is limited only for such applications where variation in load current and input voltage are small.

Q. 7. A Hartley oscillator is to span a frequency range between 50 kHz to 150 kHz. The variable capacitance has the values in the range 10pF to 500pF. The transistor to be used has $h_{fe} = 50$ and $\Delta h_{re} = 0.5$. Determine the values of the inductances. Neglect inductances between the coils and use CE circuit configurations.

Solution : The frequency of a Hartley oscillator is given by

$$f = \frac{1}{2\pi \sqrt{(L_1 + L_2)C}}$$

using only one pair of coils

$$50 \times 10^3 = \frac{1}{2 \times 3.14 \times \sqrt{(L_1 + L_2) \times 500 \times 10^{-12}}}$$

or $(L_1 + L_2)$

$$= \frac{1}{2 \times 3.14^2 \times (50)^2 \times (103)^2 \times 500 \times 10^{-12}}$$

$$= \frac{10^{12}}{4 \times (3.14)^2 \times (50)^2 \times 10^6 \times 500 \times 500}$$

$$= \frac{1}{49.3} = 0.0203H$$

$$(L_1 + L_2) = 20.3mH \quad \dots(1)$$

$$\text{Again} \quad \frac{L_2}{L_1} = \frac{h_{fe}}{\Delta h_{re}} = \frac{50}{0.5} = 100 \quad \dots(2)$$

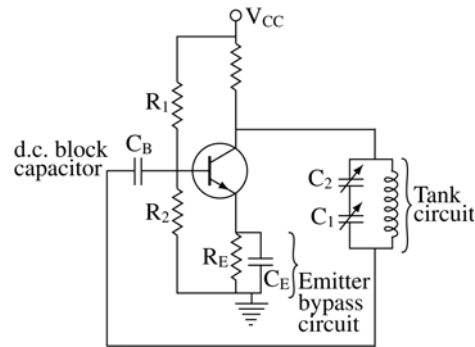
From equation (1) and (2)

$$L_2 = 100 L_1$$

$$\text{and} \quad L_1 + 100 = L_1 = 20.3 \quad \begin{cases} L_1 \approx 0.2mH \\ L_2 = 20mH \end{cases}$$

Q. 8. Draw the Colpitts' oscillator and explain its function. What is the advantage of Colpitts' oscillator over Hartley oscillator ?

Solution : Following is a figure of a colpitts' oscillator in which a NPN transistor is used in common emitter configuration.



The resistors shown in the figure provides proper biasing. Capacitor C_E provides negligible reactance at the frequency of operation. The capacitor C_B blocks the d.c. components arises from collector. Two capacitors C_1 and C_2 in series along with conductor in parallel are used as tank circuit or frequency selector device.

The feedback is π out of phase with the output which itself π out of phase with input. Hence total phase change is 0° or 2π . (CE mode provides a phase shift of π) The condition of sustained oscillation can be achieved by the selection of C_1 and C_2 such that

$$\frac{C_1}{C_2} \approx \frac{h_{fe}}{h_{re}}$$

and the frequency of oscillation is

$$f = \frac{1}{2\pi \sqrt{LC}}$$

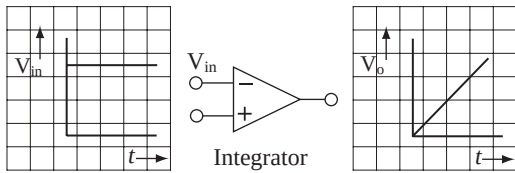
$$\text{where} \quad C = \frac{C_1 C_2}{C_1 + C_2}$$

The advantage of Colpitts' oscillator over Hartley's is that in the former the coil is not tapped and two capacitors are used which is practically feasible to construct.

Q. 9. (a) Draw a diagram for integrator operational amplifier and explain its function.

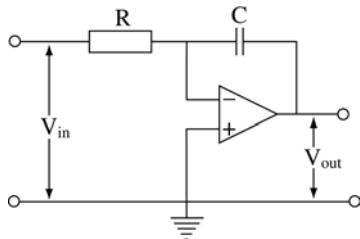
(b) A 5mV 1 kHz sinusoidal signal is applied to the input of an integrator. Calculate the waveform if $R = 100K$ and $C = 1\mu F$ for the integrator concerned.

Solution : (a) As its name suggests an integrator is a signal processing device which integrates the input signal, i.e., the integrator provides the output voltage which is directly proportional to the input voltage applied.



To construct the integrator, the feedback resistor of an operational amplifier is replaced by a capacitor. The constant value of $-1/RC$ is known as scale factor.

A schematic diagram of an integrator is given below—



The current through R is same as current through C so that

$$i = V_{in}/R \quad \dots(i)$$

$$\text{Voltage through } C = V_o$$

and the current can be given by

$$i = -C \frac{dV_o}{dt} \quad \dots(ii)$$

From equation (i) and (ii)

$$\frac{V_{in}}{R} = -C \frac{dV_o}{dt}$$

or

$$V_o = -\frac{1}{RC} \int V_{in} dt$$

The output varies linearly with time. This ability to integrate a given signal helps the analog computer to solve a differential equation. If the input of an integrator is step voltage the output is a linear ramp.

$$(b) \text{ Scale factor } \left(-\frac{1}{RC}\right)$$

$$= \frac{1}{100 \times 10^3 \times 1 \times 10^{-6}} \text{ per sec}$$

$$\therefore = -10 \text{ per sec}$$

$$\text{Now } V_{in} = V \sin \omega t$$

$$= 5 \sin 2\pi \times 10^3 t$$

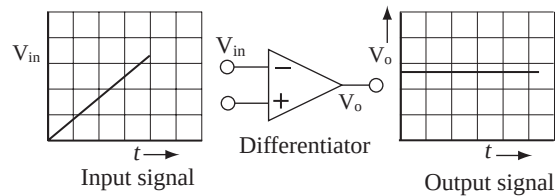
$$\text{and } V_o = -\frac{1}{RC} \int V_{in} dt \quad (\because \text{ at } t = 0, V_{in} = 0)$$

$$= -10 \int_0^t 5 \sin 2 \times 10^3 t_1 dt_1$$

$$V_o = -\frac{1}{40\pi} (\cos 2 \times 10^3 t_1 - 1)$$

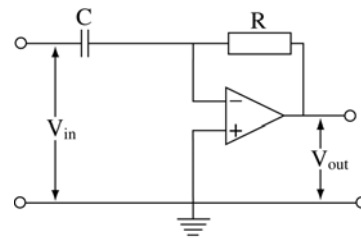
Q. 10. How will you construct the differentiator operational amplifier ? Derive an equation for the output of a differentiator.

Solution : The function of a differentiation is to provide an output voltage which is proportional to the rate of change of input voltage.



A differentiating amplifier can be obtained by interchanging the resistor of feedback and capacitor. Here the scale factor is given by RC .

The schematic diagram of a differentiator is given below—



The input voltage V_{in} may be taken across the capacitor C . The change on the capacitor

$$g = C V_{in}$$

and the changing current

$$i = \frac{d}{dt}(q) = C \frac{dV_{in}}{dt} \quad \dots(i)$$

Applying Ohm's law input terminal

$$V_o = -iR \quad \dots(ii)$$

where negative sign appears because of polarity several of amplifier

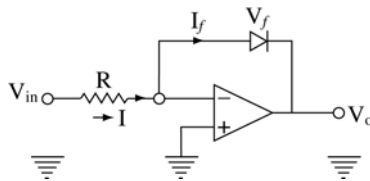
$$V_o = -RC \frac{dV_{in}}{dt}$$

From equation (i) and (ii).

The output voltage is negative time gradient of input voltage with a scale factor.

Q. 11. An operational amplifier produce the output waveform which is proportional to the logarithmic of input signal. Draw its construction and find an expression for output voltage.

Solution : The amplifier which generates the output waveform proportional to the log of input waveform is known as log operational amplifier. A schematic diagram is shown below—



The non-linear current voltage relationship of ($p-n$) junction diode can be used to get the logarithmic waveform for a diode it is given by—

$$I_f = I_0 [e^{V_f/kT} - 1] \text{ for Ge diode}$$

where I_f and I_0 denote the forward current and saturation current whereas V_f is forward voltage drop.

We can select the operating position of V_f such that $e^{V_f/kT} \gg 1$

$$\text{or} \quad I_f = I_0 e^{V_f/kT}$$

$$\text{or} \quad V_f = \frac{kT}{e} [\log I_f - \log I_0]$$

$$\text{Also} \quad I_f = I = \frac{V_{in}}{R}$$

$$\text{and} \quad V_o = V_f$$

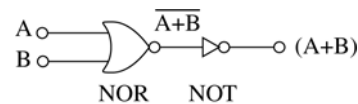
$$\text{then} \quad V_o = -\frac{kT}{e} \left[\log \frac{V_{in}}{R} - \log I_0 \right]$$

For the constant temperature T and saturation current I_0 , the output voltage is directly proportional to logarithmic of input voltage.

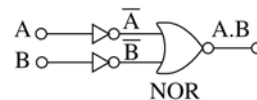
Q. 12. Which gates are known as universal building blocks and why ? Construct three basic logic functions with the help of these building blocks only.

Solution : NOR and NAND gates are known as universal gates or universal building blocks. These two gates has an ability to get utilised as three basic logic functions, i.e., OR, AND and NOT gates.

(a) **NOR gate as OR function**—For two inputs A and B , a NOR output will be $(\overline{A+B})$ which can be further inverted to give $(A+B)$ or OR output



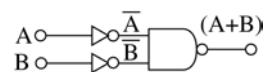
(b) **NOR gate as AND function**—Here the output taken from two inverters is NORed to give output $(\overline{\overline{A} + \overline{B}})$, which can be De-morganize in $(A \cdot B)$



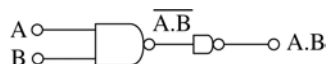
(c) **NOR gate as NOT function**—The two inputs of a NOR gate are tied together to give an output as an inverter (NOT function).



(d) **NAND gate as OR function**—For two inputs A and B we can get the ORed output with the help of NAND gate in which two inverted outputs are taken as input by a NAND gate to result OR function. This OR function is De-morganized form of $(\overline{\overline{A} \cdot \overline{B}})$



(e) **NAND gate as AND function**—This can be constructed by a two input NAND gates followed by a single input NAND gate.



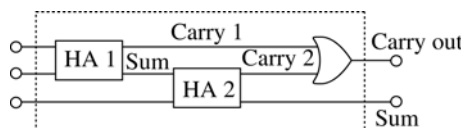
(f) **NAND gate as an inverter (NOT function)**—Two similar inputs are tied together before applying to input of a NAND gate. The output will be an inverter.



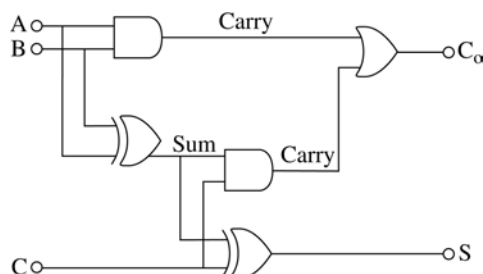
Q. 13. How will you construct a full adder with the help of half adders ? Describe its function with truth table.

Solution : As we know that the usefulness of a half adder is limited to the lowest order bit of a multiple stage adder circuit. It cannot be used for higher order bits since it does not have a carry input.

To overcome this problem a full adder is used. A full adder is constructed when two half adders (HA) are combined in such a manner that the sum output of first HA and another input are given to second HA. And the carries from both HA are ORed. The two output represents the sum and carry respectively. So full adder is a device which adds three bits (inputs) at a time producing their sum and carry.



Block diagram of a full adder



A full adder needs two XOR two AND and a single OR gate

The sum output S is the binary added output of inputs A, B and C respectively and carry out C_o is the final carry generated in this procedure.

The truth table given below illustrate the sum and carry for all possible combinations.

Inputs			Output	
A	B	C	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Q. 14. Design a parallel adder and explain its working.

Solution : As we know that a full adder can be used to add 1 bit binary number, by cascading a large number of full adders we can obtain an n -bit binary adder circuit commonly called a parallel adder. A major disadvantage of parallel adder is its relatively long delay time.

For example we can consider the addition of binary number 1111 and 1011.

$$\begin{array}{r}
 \text{Carry} \rightarrow 111 \\
 \text{Input A} \rightarrow 1111 \\
 \text{Input B} \rightarrow 1011 \\
 \hline
 11010 \\
 \uparrow \\
 \text{Overflow carry}
 \end{array}$$

Overflow carry if sum exceeds four bits

or in general

$C_3 \ C_2 \ C_1$

$m_4 \ m_3 \ m_2 \ m_1$

$n_4 \ n_3 \ n_2 \ n_1$

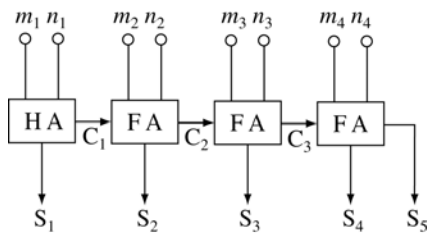
$S_5 \ S_4 \ S_3 \ S_2 \ S_1$

$\uparrow$

Overflow carry

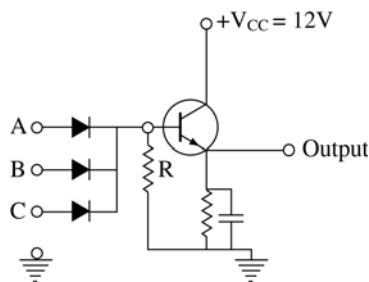
In the construction of parallel adder one half adder is followed by a set of full adders. (first adder does not takes a carry previously)

In our example first half adder adds m_1 and n_1 and forward the carry to next full adder next adder forward the carry to another and at last a overflow carry OFC, S_5 appears with sum.

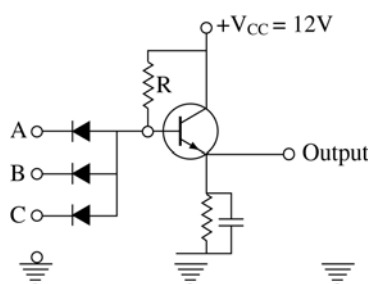


Q. 15. Construct the OR and AND gates of DTL family and describe their function.

Solution : The circuit diagram of diode transistor logic DTL-OR and DTL-AND gates are shown below.



(a) DTL - OR gate



(b) DTL-AND gate

Function : (a) **DTL - OR gate**—If any of the inputs are at ON position (logic 1), the diode starts conducting and points OX is at 12 volt the transistor runs in saturation mode and + 12 volt appear at output (logic 1). Conversely if all the inputs are at ground (logic 0) the point X is at zero value.

The transistor is off providing zero (logic 0) at output.

(b) **DTL-AND gate**—If any of the input is at ground (logic 0) the diode start conducting through R. This forward bias of diode lower the voltage to zero at X. The transistor runs in cut off and output is at zero.

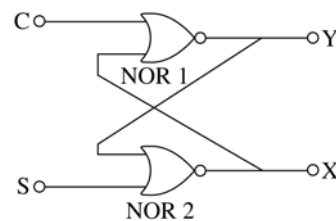
Whereas for all inputs at 12 V. (logic 1) diode do not conduct and point X is at 12V. Transistor runs in saturation which gives output to 12V (logic 1).

Q. 16. Explain the function of a flip-flop as a 1-bit memory. Plot the graph to show the trigger action.

Solution : Flip-flops are the binary or bistable multivibrators which have to stable states. These are used in memory elements in digital systems. The output of a flip-flop remains in either of the two binary states (0 or 1) until changed by applying an input or triggered by an external source.

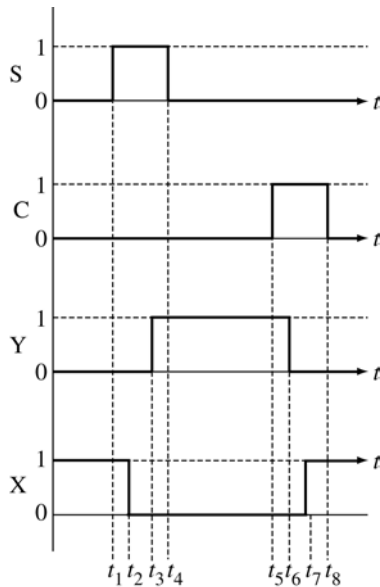
Since a flip-flop can retain or remember a certain condition indefinitely after the removal of trigger. It is said to have 1-bit memory.

A simple flip-flop circuit constructed from two NOR gates is shown below.



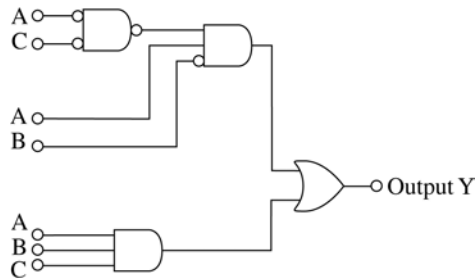
The operation of this flip-flop may be understood as follows. We start the operation with both inputs at zero level (low level).

So $Y = 0$ and $Y = 1$ NOR - 1 gate has input $C = 0$, $X = 1$ giving $Y = 0$ as expected. At $t = t_1$ $S = 1$, thus the input of S are changing from 0 to 1. After a small delay X changes from 1 to 0 at $t = t_2$. This changes the input of gate 1 from 1 to 0 change in Y from 0 to 1 at $t = t_3$. This again changes the input S from 1 to 0 but leaves no effect on the outputs. The change of S from 1 to 0 at $t = t_4$ has also no effect on the output. When C goes to 1 Y goes to 0 after a small delay, during X to 1 after a delay as shown in the figure. This locks the Y and C has no further effect.

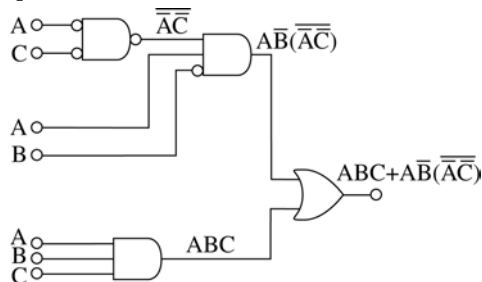


Here the condition of output determines which of its input was lost at 1 level. Two inputs cannot be high (1) simultaneously. That is why it is called 1-bit memory.

Q. 17. Simplify the logic circuit shown below—



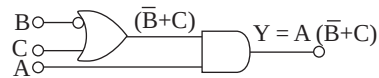
Solution : The Boolean expression for Y [output] can be written as—



$$\begin{aligned}
 Y &= ABC + \bar{A}\bar{B}\left(\overline{\overline{A}C}\right) \\
 &= ABC + \bar{A}\bar{B}(A + C) \quad (\text{De-Morgan's theorem})
 \end{aligned}$$

$$\begin{aligned}
 &= ABC + A\bar{A}\bar{B} + \bar{A}\bar{B}C \\
 &= ABC + \bar{A}\bar{B} + \bar{A}\bar{B}C \quad (A\bar{A} = 0) \\
 &= AC(\bar{B} + B) + \bar{A}\bar{B} \\
 &= AC + \bar{A}\bar{B} \quad (B + \bar{B} = 1) \\
 &= A(\bar{B} + C)
 \end{aligned}$$

The simplified logic circuit is shown below—

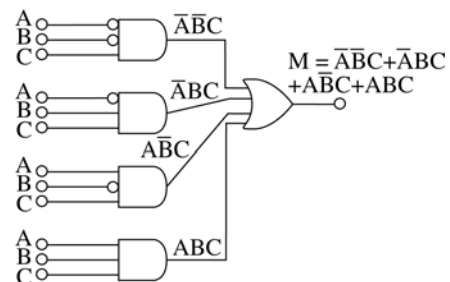


Q. 18. Simplified the Boolean expression given below and draw the logic circuits equivalent to both given expression and simplified expression.

$$\text{Output } M = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC$$

Solution : The gate equivalent of the expression

$$M = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC \text{ is}$$



We shall write and solve the Boolean expression first.

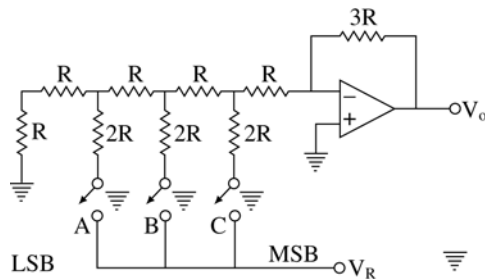
$$\begin{aligned}
 M &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC \\
 &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + AC(\bar{B} + B) \\
 &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + AC \quad (\because B + \bar{B} = 1) \\
 &= \bar{A}C(\bar{B} + B) + AC \quad (\because B + \bar{B} = 1) \\
 &= \bar{A}C + AC \\
 &= C(A + \bar{A}) \quad (A + \bar{A} = 1) \\
 &= C
 \end{aligned}$$

This single input does not need any logic function.

Q. 19. Taking the example of a R-2R ladder D/A converter, explain the functioning of D/A conversion.

Solution : A digital to analog converter (D/A converter) is an electronic device which takes the input in binary form and produce and output voltage proportional to the decimal equivalent of its binary input.

A R/2R ladder D/A converter is shown in the figure in which the non-inverting terminal of operational amplifier is grounded.



In such type of digital to analog converters an arrangements (array) of several resistance is used to form a ladder and terminal A, B and C supplies voltage V_R corresponding to the binary values (high or low). Terminal A corresponds to least significant bit (LSB) and C corresponds to most significant bit (MSB) for three bit input. The feedback of amplifier is given by $3R$ resistor and output voltage V_o gives the required output.

$$V_o = \left(\frac{a_{n-1}}{2} + \frac{a_{n-2}}{4} + \frac{a_{n-3}}{8} + \dots + \frac{a_0}{2} \right) V_R$$

[for n bit input]

where a_i is the binary state of i th bit. So all a 's should be either high (1) or low (0) corresponding to input signal in binary form.

Q. 20. Define A/D conversion. Taking the reference level equals to 0 volt digitalize a small analog pulse of maximum height 7.86V.

(First sensor is at 5V)

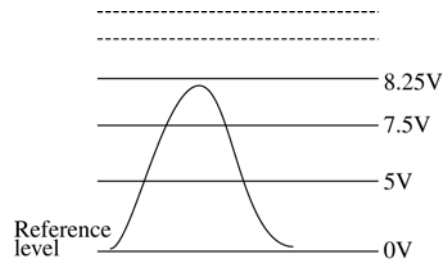
Solution : The electronic device which converts the input analog voltage into its equivalent binary form are known as analog to digital converters (A/D converters). This process is sometime referred as quantization of analog signal. For complete operation first the input voltage is decimalized then converted into binary input. Two types of A/D converters are popular—

(a) Peak sensing analog to digital converters in which the maximum voltage is digitalized.

(b) Change sensing analog to digital converters in which the total integrated current is digitalized.

The time required for digitalization is called conversion time which is of the order of few μs .

The small analog pulse of maximum height 7.86V can be digitalize as follows—



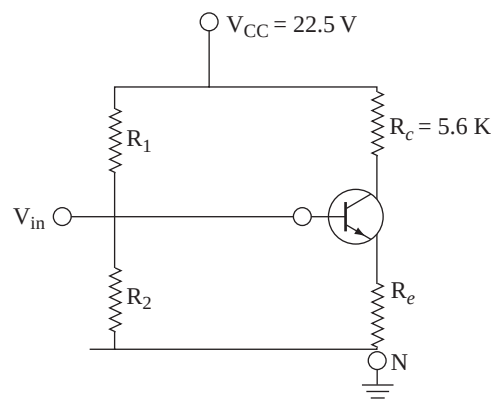
We can draw the sensors at (5V), $\left(5 + \frac{5}{2}\right)$ V, $\left(5 + \frac{5}{2^1} + \frac{5}{2^2}\right)$ V etc. as shown in the above figure.

Now the input pulse height is matched with these voltage level

For 5V sensor the pulse is on	1
For 7.5V sensor the pulse is on	1
For 8.25V sensor the pulse is off	0
$\vdots$	$\vdots$

The output will be 011, which is in the binary form.

Q. 21. Assume that a silicon transistor with $\beta = 50$, $V_{BE} = 0.6V$, $V_{CC} = 22.5 V$ and $R_C = 5.6 K$ as shown in the figure. It is desired to establish a Q-point at $V_{CE} = 12 V$, $I_C = 1.5 mA$ and a stability factor $S \leq 3$, find R_e , R_1 and R_2 .



Solution : The current in R_e is $I_C + I_B \simeq I_C$

$$\begin{aligned} \text{Hence } R_e + R_e &= \frac{V_{CC} - V_{CE}}{I_C} \\ &= \frac{22.5 - 12}{1.5} = 7.0 \text{ K} \end{aligned}$$

or $R_e = 7.0 - 5.6 = 1.4 \text{ K}$

$$\begin{aligned} \text{Also } S &= (1 + \beta) \frac{1 + \frac{R_b}{R_c}}{1 + \beta + \frac{R_b}{R_e}} \\ 3 &= 51 \frac{1 + \frac{R_b}{R_e}}{51 + \frac{R_b}{R_e}} \end{aligned}$$

$$\begin{aligned} \text{We have } \frac{R_b}{R_e} &= 2.12 \\ R_b &= 2.12 \times 1.4 = 2.96 \text{ K} \end{aligned}$$

If $R_b < 2.96 \text{ K}$ then $S < 3$.

The base current I_B will be

$$I_B \approx \frac{I_C}{\beta} = \frac{1.5}{50} \text{ mA} = 30 \mu\text{A}$$

we can solve R_1 and R_2 as

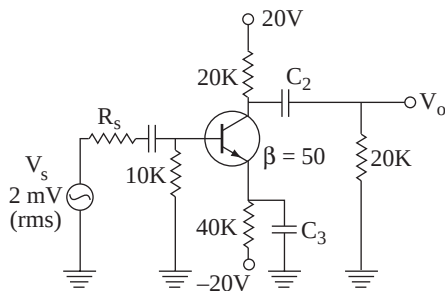
$$R_1 = R_b \frac{V_{CC}}{V} \text{ and } R_2 = \frac{R_1 V}{V_{CC} - V}$$

$$\begin{aligned} \text{Also } V &= I_B R_b + V_{BE} + (I_B + I_C) R_e \\ V &= (0.030)(2.96) + 0.6 + (0.030 + 1.5)(1.4) \\ &= 2.83 \text{ V} \end{aligned}$$

$$R_1 = \frac{2.96 \times 22.5}{2.83} = 23.6 \text{ K}$$

$$R_2 = \frac{23.6 \times 2.83}{22.5 - 2.83} = 3.38 \text{ K}$$

Q. 22. For the CE amplifier circuit shown in the figure find out R_{in} , R_c , A_v , A_p and V_{out} take $\beta = 50$ and $R_S = 0$



$$\begin{aligned} \text{Solution : } I_E &\cong \frac{V_{EE}}{R_E} \\ &= \frac{20}{40} \\ &= 0.5 \text{ mA} \\ R_e &= \frac{25}{0.5} \\ &= 50 \Omega \\ \beta R_e &= 50 \times 50 \Omega \\ &= 2500 \Omega \end{aligned}$$

(i) $R_{in} = 10 \text{ K} \parallel 2500 \Omega$

$$= 2000 \Omega$$

(ii) $R_L = 20 \text{ K} \parallel 20 \text{ K}$

$$= 10 \text{ K}$$

(iii) $A_v = \frac{R_L}{R_e} = \frac{10 \text{ K}}{50 \Omega}$

$$= 200$$

(iv) $A_p = A_v \cdot A_i$

$$= 200 \times 50$$

$$= 10,000$$

$$G_p = 10 \log_{10} 10,000$$

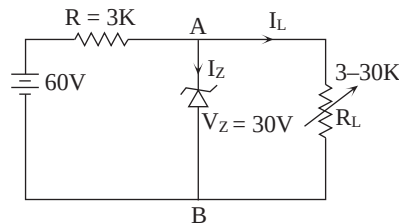
$$= 40 \text{ dB}$$

(v) $V_{out} = A_p V_{in}$

$$= 20 \times 2$$

$$= 40 \text{ mV (rms)}$$

Q. 23. Using the ideal zener approximations, find current through the diode shown below when load resistance R_L is (i) 30 K, (ii) 5 K, (iii) 3 K.



Solution : (i) $R_L = 30 \text{ K}$

$$V_{AB} = V_Z = 30 \text{ V}$$

drop across $R = 60 - 30 = 30 \text{ V}$

$$\therefore I = \frac{30}{3 \text{ K}} = 10 \text{ mA}$$

$$I_L = \frac{V_{AB}}{R_L} = \frac{30}{30 \text{ K}} = 1 \text{ mA}$$

$$\begin{aligned} I_Z &= I - I_L \\ &= 10 - 1 = 9 \text{ mA} \end{aligned}$$

- (ii) $R_L = 5 \text{ K}$
 $I = 10 \text{ mA}$ (as in previous case)
 $I_L = \frac{30}{5} \text{ K} = 6 \text{ mA}$
 $I_Z = 10 - 6 = 4 \text{ mA}$
- (iii) When
 $R_L = 3 \text{ K}$
 $I = 10 \text{ mA}$ (as in previous case)
 $I_L = \frac{30}{3} \text{ K} = 10 \text{ mA}$
 $I_Z = I - I_L = 10 - 10 = 0$

In third case diode is on the verge of breakdown region. If R_L is reduced further, the diode will come out of breakdown region and no longer be a regulator.

Q. 24. Simplify the following Boolean expressions—

(i) $\overline{A}BC + AB\overline{C} + ABC$

(ii) $A[B + C(\overline{AB + AC})]$

Solution : (i) $\overline{A}BC + AB\overline{C} + ABC$

$$= AC(B + \overline{B}) + AB\overline{C}$$

(First and third term together)

$$= AC + AB\overline{C}$$

$$= A(C + B\overline{C})$$

$$= A(B + C)$$

(ii) $A[B + C(\overline{AB + AC})]$

$$= A[B + C(\overline{AB} \cdot \overline{AC})]$$

(De-Morganization)

$$= AB + AC \cdot \overline{AB} \cdot \overline{AC}$$

$$= AB + 0$$

$$= AB$$

Q. 25. Prove the following Boolean identity and draw its gates equivalent—

$$y = \overline{B}(A + C) + C(\overline{A} + B) + AC$$

Solution : Given :

$$y = \overline{B}(A + C) + C(\overline{A} + B) + AC$$

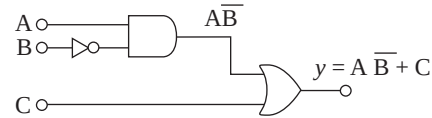
$$= \overline{A}\overline{B} + \overline{B}C + \overline{A}C + BC + AC$$

$$= \overline{A}\overline{B} + C(\overline{B} + \overline{A} + B + A)$$

$$= \overline{A}\overline{B} + C \cdot 1$$

$$= \overline{A}\overline{B} + C$$

The equivalent gates combination is shown below—



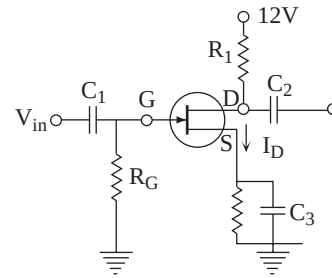
Q. 26. Determine the quiescent value of V_{GS} , I_d and V_{DS} for the JFET shown in the figure below. Given that—

$$I_{DSS} = 10 \text{ mA}$$

$$R_S = 5 \text{ K}$$

and

$$V_P = -5 \text{ V}$$



Solution : Since $I_S \cong I_D$

$$V_{GS} = -I_D R_S$$

$$= -5000 I_D \quad \dots(1)$$

$$\text{Now, } I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\Rightarrow I_D = 10 \times 10^{-3} \left(1 - \frac{V_{GS}}{-5}\right)^2$$

$$= 10 \times 10^{-3} (1 + 0.2 V_{GS})^2 \quad \dots(2)$$

Substituting the value of I_D from equation (2) into equation (1)

$$V_{GS} = -5000 [10 \times 10^{-3} (1 + 0.2 V_{GS})^2]$$

Rearranging the equation for V_{GS}

$$2V_{GS}^2 + 21 V_{GS} + 50 = 0$$

On solving gives

$$V_{GS} = -3.65 \text{ V or } -6.85 \text{ V}$$

Rejecting the higher value because it is more than V_p , we have $V_{GS} = -3.65$ V

$$\therefore -3.65 = -5 \times 10^3 I_D$$

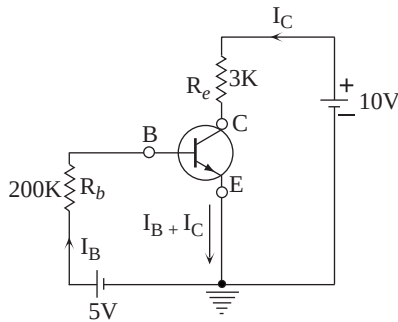
$$\therefore I_D = 0.73 \text{ mA}$$

$$\begin{aligned} V_D &= V_{DD} - I_D R_L \\ &= 12 - 0.73 \times 2 = 10.54 \text{ V} \end{aligned}$$

$$\begin{aligned} V_S &= I_D R_S \\ &= 0.73 \times 5 = 3.65 \text{ V} \end{aligned}$$

$$\begin{aligned} \therefore V_{DS} &= V_D - V_S = 10.54 - 3.65 \\ &= 6.89 \text{ V} \end{aligned}$$

Q. 27. A silicon transistor with $\beta = 100$ and $I_{CO} = 20 \text{ nA} = 2 \times 10^{-5} \text{ mA}$ is shown in the figure. Find the transistor currents and determine whether the transistor is in active region.



Solution : Since the base is forward biased, the transistor is not cut off. Hence, it must be either in active or in saturation region. Assuming it in active region and applying KVL to base-loop,

$$-5 + 200 I_B + V_{BE} = 0$$

As for silicon transistor reasonable value of $V_{BE} = 0.7$ V in the active region,

$$\begin{aligned} \text{Hence } I_B &= \frac{5.0 - 0.7}{200} \\ &= 0.0215 \text{ mA} \end{aligned}$$

Since $I_{CO} \ll I_B$ then $I_C \approx \beta I_B = 2.15 \text{ mA}$

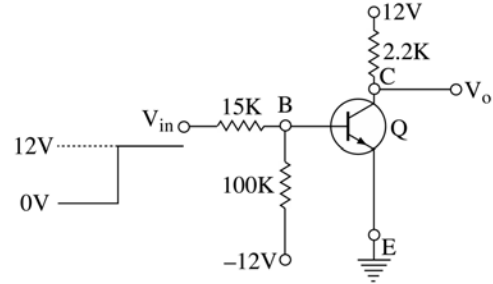
we now justify our assumption that the transistor is in active region, by verifying that the collector junction is reversed-biased. Applying KVL to the collector loop, we obtain

$$-10 + 3 I_C + V_{CB} + V_{BE} = 0$$

$$\text{or } V_{CB} = (-3)(2.15) + 10 - 0.7 = +2.85 \text{ V}$$

For an n - p - n device a positive value of V_{CB} represents reversed biased collector junction. Hence, the transistor is indeed in the active region.

Q. 28. If the silicon transistor shown in the figure has a minimum value of h_{FE} of 30, find the output levels for input levels of 0 V and 12 V.



Solution : For $V_{in} = V(0) = 0$ the open circuited base voltage V_B is

$$V_B = -12 \frac{15}{100 + 15} = -1.56 \text{ V}$$

Since a bias of about zero volt is adequate to cut-off a silicon emitter junction then Q is indeed cut-off.

Hence $V_o = 12 \text{ V}$ for $V_{in} = 0$

For $V_{in} = V(1) = 12 \text{ V}$ let us verify the assumption that Q is in saturation. The minimum base current required for saturation is

$$(I_B)_{\min} = \frac{I_C}{h_{FE}}$$

It is usually sufficiently accurate to use the approximate values for the saturation junction voltages *e.g.*, for silicon $(V_{BE})_{\text{sat}} = 0.8 \text{ V}$ and $(V_{CE})_{\text{sat}} = 0.2 \text{ volt}$. with these values

$$I_C = \frac{12 - 0.2}{2.2}$$

$$= 5.36 \text{ mA}$$

$$(I_B)_{\min} = \frac{5.36}{30} = 0.18 \text{ mA}$$

$$I_1 = \frac{12 - 0.8}{15} = 0.75 \text{ mA}$$

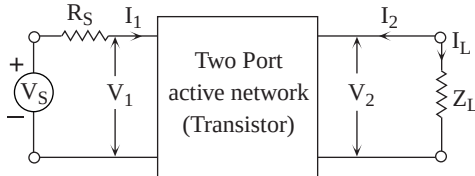
$$\begin{aligned} I_2 &= \frac{0.8 - (-12)}{100} \\ &= 0.15 \text{ mA} \end{aligned}$$

$$\text{and } I_B = I_1 - I_2 = 0.75 - 0.15 = 0.62 \text{ mA}$$

Since the value exceeds $(I_B)_{\min}$, Q is indeed in saturation and the drop across the transistor is $(V_{CE})_{\text{sat}}$. Hence, $V_o = 0.2 \text{ V}$ for $V_{in} = 12 \text{ V}$ and the circuit performs NOT operation.

Q. 29. A two port active network is shown in the figure. In the network a transistor is connected as a CE amplifier. If $R_L = 10\text{ K}$ and $R_S = 1\text{ K}$, find the various gains and the input and output impedances.

Given : $h_{fe} = 50$, $h_{oe} = 25 \times 10^{-6}\text{ }\Omega^{-1}$
 $h_{ie} = 1,100\text{ }\Omega$, $h_{re} = 2.5 \times 10^{-4}$

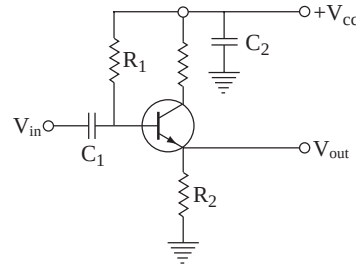


Solution :

$$\begin{aligned}
 A_I &= -\frac{h_{fe}}{1 + h_{oe} R_L} \\
 &= -\frac{50}{1 + 25 \times 10^{-6} \times 10^4} \\
 &= -40.0 \\
 R_i &= h_{ie} + h_{re} A_I R_L \\
 &= 1,100 - 2.5 \times 10^{-4} \times 40.0 \times 10^4 \\
 &= 1000\text{ }\Omega = 1\text{ K} \\
 A_V &= \frac{A_I R_L}{R_i} \\
 &= -\frac{40 \times 10^3}{1000} \\
 &= -400 \\
 A_{V_S} &= \frac{A_V R_i}{R_i + R_S} \\
 &= -\frac{400 \times 1}{1 + 1} \\
 &= -200 \\
 A_{I_S} &= \frac{A_I R_S}{R_i + R_S} \\
 &= -\frac{40.0 \times 1}{1 + 1} \\
 &= -20.0 \\
 Y_0 &= h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_S} \\
 &= 25 \times 10^{-6} - \frac{50 \times 2.5 \times 10^{-4}}{1,100} \\
 &= 19.0 \times 10^{-6}\text{ }\Omega \\
 &= 19.0\text{ }\mu\text{A/V} \\
 \text{So, } Z_0 &= \frac{1}{Y_0} = \frac{1}{19.0 \times 10^{-6}}\text{ }\Omega \\
 &= 52.6\text{ K}
 \end{aligned}$$

Q. 30. Draw an emitter follower and explain its various characteristics.

Solution : An emitter follower is shown in the figure



An emitter follower has the following interesting characteristics. A voltage gain close to unity, high input resistance and low output resistance. It is also observed that an emitter follower provides no phase shift between output and input in either voltage or current (since both A_v and A_i are positive and real).

Voltage gain $A_v \simeq 1$

$$\text{Current gain } A_I = \frac{1}{1 - \alpha}$$

$$\therefore \text{Power gain } A_p = \frac{1}{1 - \alpha}$$

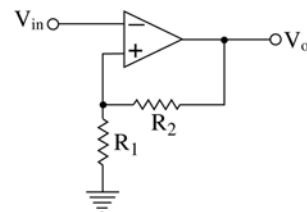
$$\text{Input resistance } R_{in} = \frac{R_L}{1 - \alpha}$$

and output resistance $R_{out} = r_e + (1 - \alpha)(R_G + r_b)$

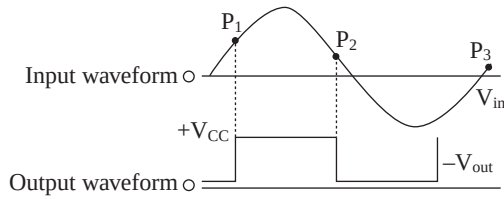
where r_e , r_b and R_G are emitter resistance, base resistance and internal resistance of the signal source respectively.

Q. 31. Draw an OP-AMP Schmidt trigger. What waveform will be produced at output if a sine wave input is applied to it?

Solution : A Schmidt trigger has two stable states : Q_1 may be on and Q_2 off or Q_2 may be on and Q_1 off the switching action may be started by raising or lowering the bias of Q_1 . An OP-AMP Schmidt trigger is shown below—



If a sinusoidal input is applied to Schmidt trigger it will produce a square wave output.



The positive feedback voltage applied to the non inverting terminal of Schmidt trigger is called the upper trip voltage UTP. The negative voltage feedback is the lower trip voltage.

$$UTP = \frac{R_1}{R_1 + R_2} V_{SAT}$$

$$LTP = -\frac{R_1}{R_1 + R_2} V_{SAT}$$

Q. 32. What are relaxation oscillators ? Draw an OP-AMP relaxation oscillator which produce a square wave output from an exponential input ?

Solution : A relaxation oscillator is a circuit whose frequency depends on the charging and discharging of capacitor (or Inductor).

An OP-AMP relaxation oscillator is shown below in which an exponential input signal is being converted into a square wave output. Mathematical analysis shows that the frequency of the output square wave is

$$f = \frac{0.455}{RC}$$

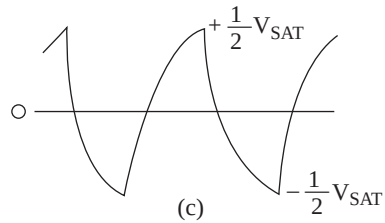
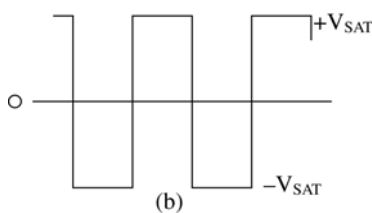
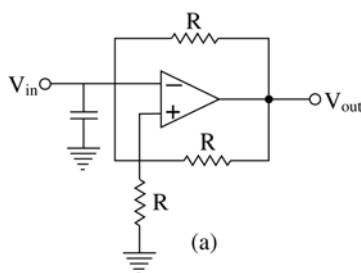


Figure (a), (b) and (c) represents OP-AMP relaxation oscillator circuit, output waveform and capacitor voltage respectively.

Q. 33. What logic function is represented by the expression

$$Y = A\bar{B} + \bar{A}B$$

Draw the circuit equivalent for this function and write truth table.

Solution : The given logic expression $Y = A\bar{B} + \bar{A}B$ represents the exclusive—OR (XOR) function. Its truth table is shown below—

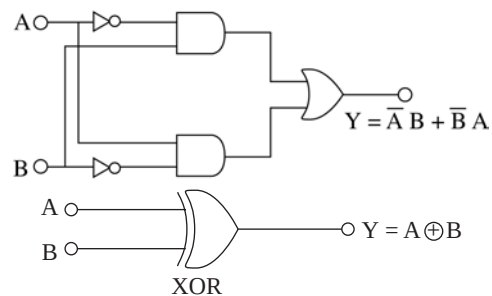
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

It is clear from the truth table that output Y will be high if A is low and B is high or the reverse is true. Output Y is low whenever the both inputs are either zero or one (low or high). In other words the output exists only for different inputs. The outputs are low when the inputs agrees.

The Boolean operator for exclusive OR is $\oplus$. As such another for of above expression will be

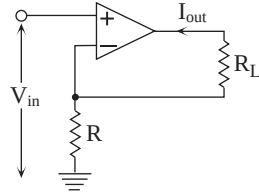
$$Y = A \oplus B$$

The logic schematic diagram for exclusive OR is shown



Q. 34. Construct a voltage to current converter OP-AMP and explain its functioning.

Solution : A voltage to current OP-AMP is a device in which output current depends only on input voltage and feedback resistor only.



A voltage to current converter. Here the feedback is negative because the returning voltage opposes the input voltage. Because $V^- = V^+$, the value of V_R , the voltage across resistor R is equal to V_{in} . With the input impedance (Z_{in}) infinite, the current in R_L must be equal to current in R (as a series circuit). A mathematical analysis of this circuit shows that it acts approximately like a perfect V-to-C converter.

$$I_{out} = \frac{V_{in}}{R}$$

$$Z_{in} = \infty$$

In a perfect voltage to current converter the output current is a function of input voltage and feedback resistance only. *e.g.*

$$V_{in} = 2 \text{ V}$$

$$\text{and } R = 1 \text{ k}\Omega$$

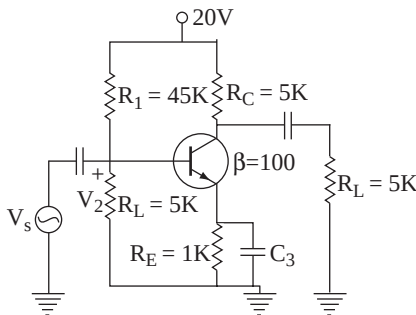
$$\text{then } I_{out} = \frac{2 \text{ V}}{1 \text{ k}\Omega} = 2 \text{ mA}$$

The circuit acts like a current source producing exactly 2 mA through any load resistance.

Q. 35. For the single stage CE amplifier shown below find approximate value of

- | | |
|--------------|------------|
| (i) r_{in} | (ii) r_L |
| (iii) A_v | (iv) A_p |
| (v) G_p | |

Take transistor $\beta = 100$ use $r_e' = 50 \frac{\text{mV}}{I_E}$.



$$\text{Solution : } V_2 = 20 \times \frac{50}{5 + 45} = 2 \text{ V};$$

$$I_E \approx \frac{V_2}{R_E} = \frac{2}{1} = 2 \text{ mA}$$

$$r_e' = \frac{50}{2} = 25 \Omega$$

(AC junction resistance of the emitter)

$$\beta r_e' = 100 \times 25 = 2.5 \text{ K}$$

$$(i) \quad r_{in} = R_1 \parallel R_2 \parallel \beta r_e' = 45 \parallel 5 \parallel 2.5 = 1.6 \text{ K}$$

$$(ii) \quad r_L = 5 \text{ K} \parallel 5 \text{ K} = 2.5 \text{ K}$$

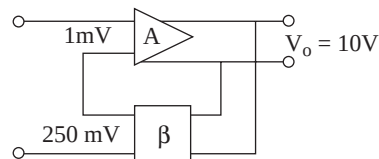
$$(iii) \quad A_v = \frac{r_L}{r_e'} = \frac{2.5 \text{ K}}{25 \Omega} = 100$$

$$(iv) \quad A_p = A_v \cdot A_i = 100 \times 100 = 10,000$$

$$(v) \quad G_p = 10 \log_{10} 10,000 = 40 \text{ dB}$$

Q. 36. In the series-parallel (SP) feedback amplifier shown below calculate—

- Open loop gain of amplifier
- Gain of the feedback network
- Close loop gain of amplifier
- Sacrifice factor, S



Solution : (a) Since 1 mV goes into the amplifier and 10 V comes out

$$\therefore A = \frac{10 \text{ V}}{1 \text{ mV}} = 10,000$$

(b) The feedback network is driven by the output voltage of 10 V.

$\therefore$ Gain of feedback network

$$= \frac{\text{Output}}{\text{Input}} = \frac{250 \text{ mV}}{10 \text{ V}} = 0.025$$

(c) So far as the feedback amplifier is concerned, input is $(250 + 1) = 251 \text{ mV}$ and final output is 10 V. Hence, gain with feedback is

$$A' = \frac{10 \text{ V}}{251 \text{ mV}} = 40$$

(d) The sacrifice factor

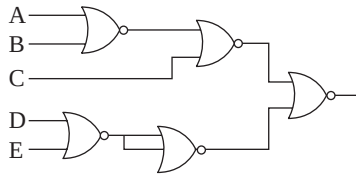
$$S = \frac{A}{A'} = \frac{10,000}{40} = 250$$

Objective Questions

1. Convert $(937)_{BCD}$ to Gray Code—

(A) 110100100100 (B) 100110110010
(C) 101110010110 (D) 100110111001

2. The circuit shown in figure realizes the function—



(A) $(A + B + C)(D \overline{E})$
(B) $(A + \overline{B + C})(\overline{D} E)$
(C) $(A + B + C)(\overline{DE})$
(D) $(\overline{A + B} + C)(\overline{D} \overline{E})$

3. What is the dual of $X + \overline{X}Y = X + Y$ —

(A) $X + Y = XY$
(B) $\overline{X} + XY = XY$
(C) $X(\overline{X} + Y) = XY$
(D) $X(\overline{X} + Y) = X + Y$

4. What is boolean expression for a getting network that will have output 0 only, when :

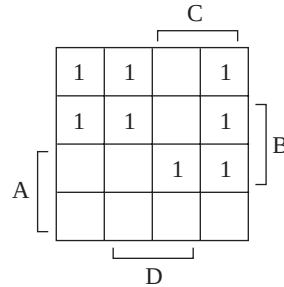
$X = 1, Y = 1, Z = 1;$
 $X = 0, Y = 0, Z = 0;$
 $X = 1, Y = 0, Z = 0;$

(A) $XYZ + \overline{X}YZ + \overline{X}\overline{Y}Z$
(B) $(\overline{X} + \overline{Y} + Z)(X + Y + Z)(\overline{X} + Y + Z)$
(C) $(XY\overline{Z})(\overline{X} + \overline{Y} + Z)(X + \overline{Y} + Z)$
(D) $(\overline{X}\overline{Y}Z) + XYZ + \overline{X}YZ$

5. Which one of the following is equivalent to AND–OR realization ?

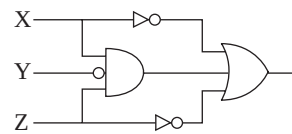
(A) NAND–NOR realization
(B) NOR–NOR realization
(C) NOR–NAND realization
(D) NAND–NAND realization

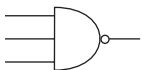
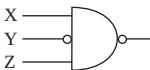
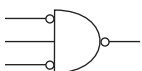
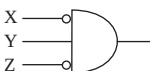
6. The simplified Boolean expression from the Karnaugh map given in the fig. is—



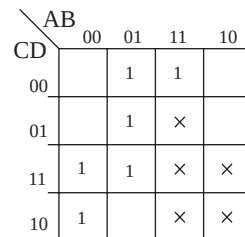
(A) $\overline{A} \overline{C} + \overline{A} \overline{D} = ABC$
(B) $\overline{A} \overline{C} + \overline{A} \overline{D} + ABC$
(C) $AC + ACD + ABC + \overline{BCD}$
(D) $\overline{A} \overline{B} + \overline{C} \overline{D} + \overline{A} \overline{D}$

7. The minimised logic circuit for the circuit shown in fig. is—



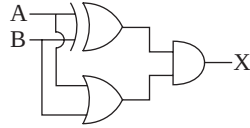
(A)  (B) 
(C)  (D) 

8. The minimised expression for the given K-map is (X-don't care)—



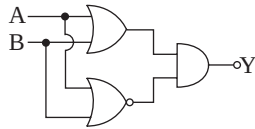
(A) $\overline{C} \overline{B} + BD + CD$
(B) $AB + \overline{C} \overline{B} + \overline{BC}$
(C) $\overline{C} \overline{B} + AC + \overline{BC}$
(D) $\overline{C} \overline{B} + CD + \overline{C} \overline{B}$

9. The output X of the circuit shown in the figure will be—



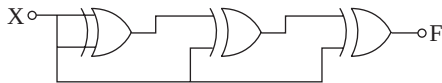
- (A) AB (B) $\overline{A}B$
(C) $A\overline{B}$ (D) $\overline{A}\overline{B}$

10. The circuit given below is functionally equivalent to—



- (A) EX-OR gate (B) NOR gate
(C) AND gate (D) NAND gate

11. For the circuit shown below, the output F is given by—



- (A) $F = 1$ (B) $F = 0$
(C) $F = X$ (D) $F = \overline{X}$

12. Which of the following flip-flop cannot be converted to D-type (delay) flip-flop?

- (A) S-R flip-flop
(B) J-K flip-flop
(C) Master-slave flip-flop
(D) None of these

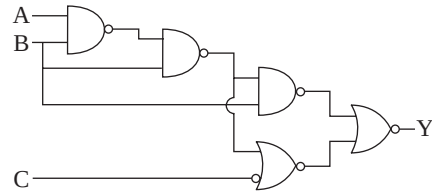
13. Race condition always arises in a—

- (A) Combination circuit
(B) Asynchronous circuit
(C) Synchronous circuit
(D) Digital circuit

14. Number of flip-flop needed to divide the input frequency by 32 is—

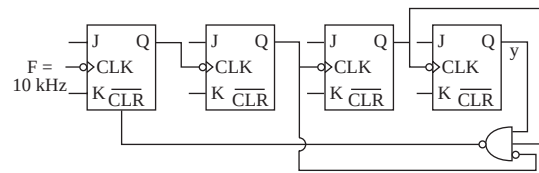
- (A) 2 (B) 4
(C) 5 (D) 8

15. Find the logic circuit shown in figure the simplified Boolean expression for the output Y is—



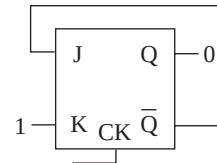
- (A) $A + B + C$ (B) A
(C) B (D) C

16. In fig., J & K inputs of all the four flip-flops are made high. The freq. of the signal at output Y is—



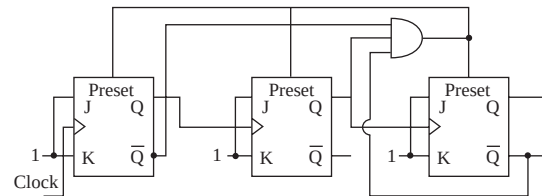
- (A) 0.833 KHz
(B) 1.0 KHz
(C) 0.91 KHz
(D) 0.77 KHz

17. In a JK flip-flop we have $J = Q$ and $K = 1$ (shown in fig. Assuming the flip-flop was initially cleared and then clocked for 6 pulses, the sequence at the Q-output will be—



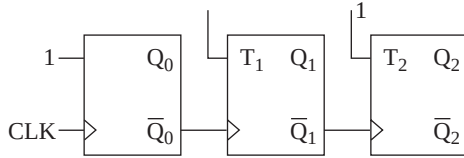
- (A) 010000 (B) 011001
(C) 010010 (D) 010101

18. The ripple counter shown in fig. works as a—

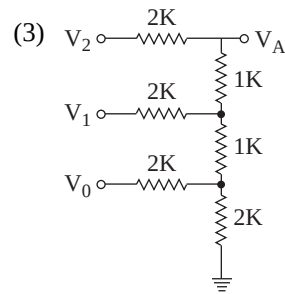
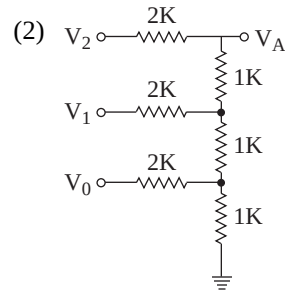
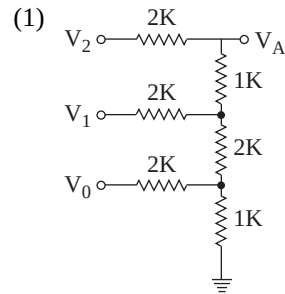


- (A) Mod - 3 up counter
(B) Mod - 5 up counter
(C) Mod - 3 down counter
(D) Mod - 5 down counter

19. Fig. below shows a ripple counter using positive edge triggered flip-flops. If the present state of the counter is $Q_2Q_1Q_0 = 011$, then its next state ($Q_2Q_1Q_0$) will be—



- (A) 010 (B) 100
(C) 111 (D) 101
20. The resolution of a D/A converter is approximately 0.4% of its full-scale range. It is—
(A) an 4-bit converter
(B) an 8-bit converter
(C) a 12-bit converter
(D) a 16-bit converter
21. A 10-bit ADC with a full scale V voltage of $10.24V$ is designed to have a $\pm \text{LSB}/2$ accuracy. If the ADC is calibrated at 25°C and the operating temp. range from 0°C to 50°C then the maximum net temp. coeff. of the ADC should not exceed—
(A) $\pm 200 \mu\text{V}/^\circ\text{C}$
(B) $\pm 400 \mu\text{V}/^\circ\text{C}$
(C) $\pm 600 \mu\text{V}/^\circ\text{C}$
(D) $\pm 800 \mu\text{V}/^\circ\text{C}$
22. An 8-bit D/A converter has a full scale output voltage of $20V$. The output voltage when the input is 11011011 is—
(A) 160 mV
(B) 78 mV
(C) $20V$
(D) $17V$
23. An 8-bit DAC produces an output voltage of $2V$ for input code of 01100100 . The value of output V_0 for an input code of 10110011 would be—
(A) $3.59 V$ (B) $4.49 V$
(C) $3.28 V$ (D) $4.92 V$
24. Which of the resistance networks of fig. below, can be used as 3-bit R-2R ladder DAC. Assume V_0 corresponds to LSB—



- (A) Both (1) and (2) (B) Both (2) and (3)
(C) Only 3 (D) Only 2
25. An 8-bit successive approximation analog to digital converter has full scale reading of $2.55 V$ and its conversion time for an analog input of $1V$ is $20 \mu\text{s}$. The conversion time for a $2V$ input will be—
(A) $10 \mu\text{s}$ (B) $20 \mu\text{s}$
(C) $40 \mu\text{s}$ (D) $50 \mu\text{s}$
26. The number of comparators required in a 3-bit comparator type ADC is—
(A) 2 (B) 3
(C) 7 (D) 8
27. Peak inverse voltage for a full wave rectifier is, peak value of input voltage—
(A) E_m (B) $2 E_m$
(C) $3 E_m$ (D) $4 E_m$

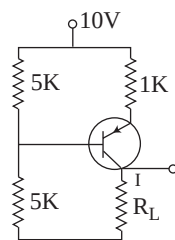
28. The transistors are—

- (A) low voltage and low current devices
- (B) high voltage and high current devices
- (C) low voltage and high current devices
- (D) only low current devices

29. FET is a device which has—

- (A) high input impedance and is voltage controlled
- (B) high input impedance and is current controlled
- (C) low input impedance and is voltage controlled
- (D) low input impedance and is current controlled

30. The circuit shown below is a—



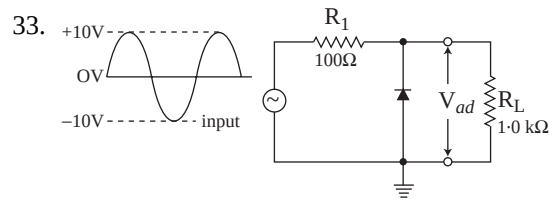
- (A) Constant voltage source
- (B) Constant current source
- (C) Common emitter
- (D) Oscillator

31. A full wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer *r.m.s.* secondary voltage from centre tap to each end of secondary is 50 V and load resistance is $980\ \Omega$, the mean load current is—

- (A) 70.7 m amp
- (B) 45 m amp
- (C) 50 m amp
- (D) 60.6 m amp

32. For a half wave rectifier—

- (A) $I_{diode} = I_{dc}$
- (B) $I_{diode} = \frac{1}{2} I_{dc}$
- (C) $I_{diode} = 2 I_{dc}$
- (D) $I_{diode} = 4 I_{dc}$



than the output is—

- (A)
- (B)
- (C)
- (D)

34. If you are checking a 60 Hz full wave bridge rectifier and observe that the output has a 60 Hz ripple—

- (A) the circuit is working properly
- (B) there is an open diode
- (C) the transformer secondary is shorted
- (D) the filter capacitor is leaky

35. A no load condition means that—

- (A) the load has infinite resistance
- (B) the load has zero resistance
- (C) the output terminals are open
- (D) Only (A) and (C)

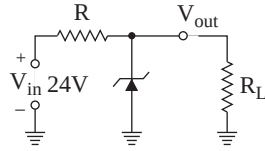
36. Compared to a visible red LED, an infrared LED—

- (A) Produces light with shorter wavelengths
- (B) Produces light of all wavelength
- (C) Produces only on color of light
- (D) Produces light with longer wavelength

37. The internal resistance of a photo diode—

- (A) increases with light intensity when reverse-biased
- (B) decreases with light intensity when reverse-biased
- (C) increases with light intensity when forward-biased
- (D) decreases with light intensity when forward-biased

38. For the circuit Determine V_{OUT} at I_{ZK} and at I_{ZM} respectively—

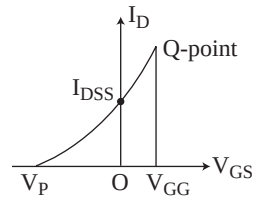


- (A) 14.76 V, 15.7 V
 (B) 15.7 V, 14.76 V
 (C) 6.07 V, 4.7 V
 (D) 7.7 V, 9.3 V
39. The positive part of the output signal in a transistor circuit is clipped, if Q-point moves—
 (A) towards the saturation point
 (B) towards the cut off point
 (C) towards the centre of the load line
 (D) None of these
40. The biasing circuit which gives best stability to the Q-point is—
 (A) feedback resistor biasing
 (B) base resistor biasing
 (C) emitter resistor biasing
 (D) potential divider biasing
41. Calculate the values of **drain to source** voltage and **gate-to-source** voltage for a JFET, if $I_b = 2 \text{ mA}$, $V_{DD} = 10\text{V}$, $R_S = 1 \text{ k}\Omega$ and $R_D = 2 \text{ k}\Omega$ —
 (A) 4V, -4V (B) 4V, -2V
 (C) 2V, -4V (D) 2V, -2V
42. For a JFET, $V_{GS(on)} = 8\text{V}$, $I_{b(on)} = 10 \text{ mA}$ and $V_T = 2\text{V}$, I_b for $V_{GS} = 4\text{V}$ is—
 (A) 0.8 mA (B) 1.11 mA
 (C) 12.1 mA (D) 0.3 mA
43. The, JFET is often called square-law device, because its—
 (A) transconductance curve is parabolic
 (B) AC resistance from drain to source varies inversely as square of the drain current
 (C) I_b varies as square of drain voltage for a fixed gate to source voltage
 (D) reverse gate leakage current varies as square of the reverse gate voltage

44. The positive gate operation of an n -channel MOSFET is known as—

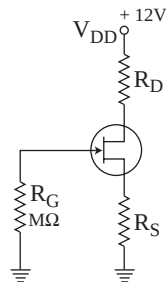
- (A) Depletion mode
 (B) Enhancement mode
 (C) Normal
 (D) Neither depletion nor enhancement mode

45. The plot is the graphical solution of—

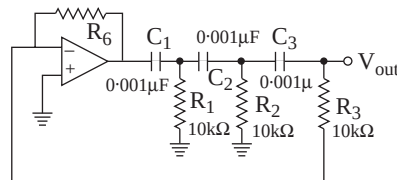


- (A) JFET ($V_{GSO} = 0$)
 (B) JFET ($R_D = 0$)
 (C) D-MOSFET fixed bias
 (D) D-MOSFET voltage divider bias
46. For what value of V_{GS} , $I_D = 0$
 (A) $V_{GS} = \frac{V_P}{2}$
 (B) $V_{GS} = V_P$
 (C) $V_{GS} = 2V_P$
 (D) $V_{GS} = 0$
47. Each stage of a four stage amplifier has a voltage gain of 15. The overall voltage gain is—
 (A) 60 (B) 15
 (C) 50, 625 (D) 3078
48. A CE amplifier is driving a $10 \text{ k}\Omega$ load. If $R_C = 2.2 \text{ k}\Omega$ and $r_e' = 10\Omega$, the voltage gain is approx—
 (A) 220 (B) 1000
 (C) 10 (D) 180
49. In Q – 47, the overall gain can be expressed in decibels as—
 (A) 94.1 dB (B) 47.0 dB
 (C) 35.6 dB (D) 69.8 dB
50. The value of R_S required to self bias a p -channel JFET with $I_{DSS} = 25 \text{ mA}$ and $V_{GS(off)} = 15\text{V}$ in to be 5V—
 (A) 800Ω (B) 450Ω
 (C) 600Ω (D) 700Ω

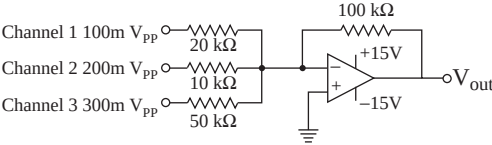
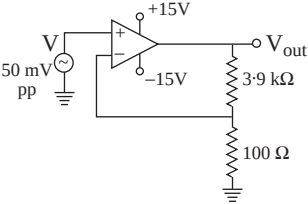
51. Select resistor values for R_D and R_S in fig. to set up an approx. mid point bias. For this particular JFET, the parameters are $I_{DSS} = 12 \text{ mA}$ and $V_{GS(off)} = -3 \text{ V}$. V_D should be approx. 6 V (one half of V_{DD})—

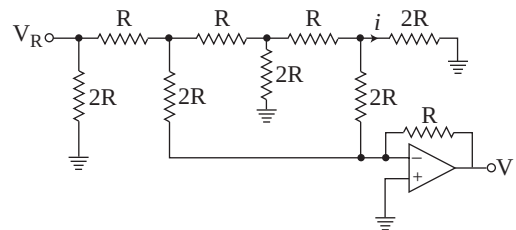


- (A) $R_D = 1 \text{ k}\Omega$, $R_S = 147 \text{ }\Omega$ (B) $R_D = 147 \text{ }\Omega$, $R_S = 1 \text{ k}\Omega$
 (C) $R_D = 200 \text{ }\Omega$, $R_S = 10 \text{ k}\Omega$ (D) $R_D = 10 \text{ k}\Omega$, $R_S = 200 \text{ }\Omega$
52. The constant-current area of a FET lies between—
 (A) Cut off and saturation
 (B) Cut off and pinch off
 (C) 0 and I_{DSS}
 (D) Pinch off and breakdown
53. A certain p -channel E-MOSFET has a $V_{GS(th)} = -2 \text{ V}$. If $V_{GS} = 0 \text{ V}$, the drain current is—
 (A) 0 A (B) $I_{D(on)}$
 (C) Max (D) I_{DSS}
54. A certain common source amplifier has a voltage gain of 10. If the source bypass capacitor is removed—
 (A) the voltage gain will increase
 (B) the transconductance will increase
 (C) the voltage gain will decrease
 (D) the Q-point will shift
55. If there is an internal open between the drain and source in a CS amplifier, the drain voltage is equal to—
 (A) 0 V
 (B) V_{DD}
 (C) A value less than normal
 (D) V_{GS}
56. For both good voltage gain and high input resistance, we must use a—
 (A) CS amplifier (B) CD amplifier
 (C) CG amplifier (D) None of these
57. The efficiency for class A and Class B power amplifier respectively—
 (A) 25%, 50% (B) 25%, 79%
 (C) 98%, 25% (D) 50%, 79%
58. The transistor in a class B amplifier are biased—
 (A) into cut off
 (B) in saturation
 (C) at midpoint of the load line
 (D) right is cut off
59. The efficiency of class C amplifier is—
 (A) less than Class A
 (B) less than Class B
 (C) less than Class AB
 (D) greater than Class A, B or AB
60. The gain of a particular amplifier at a given freq. decreases by 6 dB when the freq. is doubled. The roll-off is—
 (A) -12 dB/decade (B) -20 dB/decade
 (C) -6 dB/Octave (D) Only (B) and (C)
61. When the voltage gain of an amplifier is increased, the bandwidth—
 (A) is not affected (B) increases
 (C) decreases (D) becomes distorted
62. In a wien-bridge oscillator, if the resistances, in the positive feedback circuit are decreased, the frequency—
 (A) decreases
 (B) increases
 (C) remains the same
 (D) None of these
63. Colpitts Clapp and Hartley are names that refer to—
 (A) types of RC oscillators
 (B) inventors of the transistor
 (C) types of LC oscillator
 (D) types of filters
64. Determine the value of R_f necessary for the circuit to operate as an oscillator—



- (A) $29 \text{ k}\Omega$ (B) $290 \text{ k}\Omega$
 (C) $29 \text{ }\Omega$ (D) $290 \text{ }\Omega$

65. In (Q-64) the freq. of oscillation—
 (A) 6.5 Hz (B) 6.5 kHz
 (C) 65 Hz (D) 65 kHz
66. The closed loop voltage gain of an inverting amplifier equals—
 (A) The ratio of the input resistance to the feedback resistance
 (B) The open-loop voltage gain
 (C) The feedback resistance divided by the input resistance
 (D) The input resistance
67. The voltage follower has—
 (A) closed loop voltage gain of unity
 (B) small open-loop voltage gain
 (C) closed-loop bandwidth of zero
 (D) large closed-loop output impedance
68. A summing amplifier can have—
 (A) No more than two input signals
 (B) Two or more input signals
 (C) A closed loop input impedance of infinity
 (D) A small open-loop voltage gain
69. Channel 1 100m V_{pp} Channel 2 200m V_{pp} Channel 3 300m V_{pp}
- 
- What is the ac output voltage ?
 (A) -3.1 V_{pp} (B) 3.1 V_{pp}
 (C) -3.1 mV_{pp} (D) 3.1 mV_{pp}
70. 
- The closed loop voltage gain and band width is [unity gain]—
 (A) 39 (B) 40
 (C) 41 (D) 50
71. The output voltage at 250 kHz (In Q-70)—
 (A) 200 V_{pp} (B) 200 mV_{pp}
 (C) 2 mV_{pp} (D) 2 V_{pp}
72. The center freq. of a band pass filter is always equal to the—
 (A) Bandwidth
 (B) Geometric average of the cut-off freq.
 (C) Bandwidth divided by Q
 (D) 3-dB freq.
73. The Q of a narrow band filter is always—
 (A) small
 (B) equal to Bw divided by f_0
 (C) less than 1
 (D) greater than 1
74. A first order-active filter stage has—
 (A) one capacitor (B) two op-Amps
 (C) three-resistors (D) a high Q
75. To build a tenth-order filter, we should cascade—
 (A) 10 first order stages
 (B) 5 second order stages
 (C) 3 third order stages
 (D) 2 fourth order stages
76. A second order all-pass filter can vary the output phase from—
 (A) 90° to -90° (B) 0 to -360°
 (C) 0 to -180° (D) 0 to -720°
77. The region between the pass band and the stop band is called the—
 (A) Attenuation (B) Centre
 (C) Transition (D) Ripple
78. In the Digital to Analog converter circuit shown in the figure below, $V_R = 10 \text{ V}$ and $R = 10 \text{ k}\Omega$ —



The current i is—

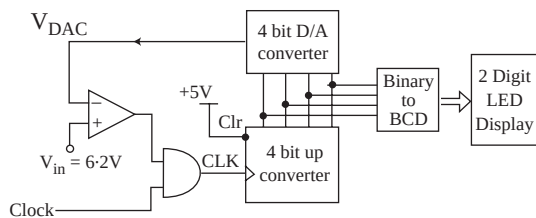
- (A) $31.25 \mu\text{A}$ (B) $62.5 \mu\text{A}$
 (C) $125 \mu\text{A}$ (D) $250 \mu\text{A}$

79. In (Q. 78) the voltage V_0 is—

- (A) -0.781 V
- (B) -1.562 V
- (C) -3.125 V
- (D) -6.250 V

80. In the following circuit, the comparator output is logic '1' if $V_1 > V_2$ and is logic '0' otherwise. The D/A conversion is done as per the relation

$$V_{\text{DAC}} = \sum_{r=0}^3 2^{n-1} b_n \text{ volts, where } b_3 \text{ (MSB), } b_2, b_1 \text{ and } b_0 \text{ (LSB) are counter outputs. The counter starts from the clear state :}$$



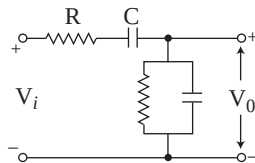
The stable reading of the LED display is—

- (A) 06
- (B) 07
- (C) 12
- (D) 13

81. In (Q-80) the magnitude of the error between V_{DAC} and V_{in} at steady state in volts is—

- (A) 0.2
- (B) 0.3
- (C) 0.5
- (D) 1.0

82. The RC circuit shown in the figure is—

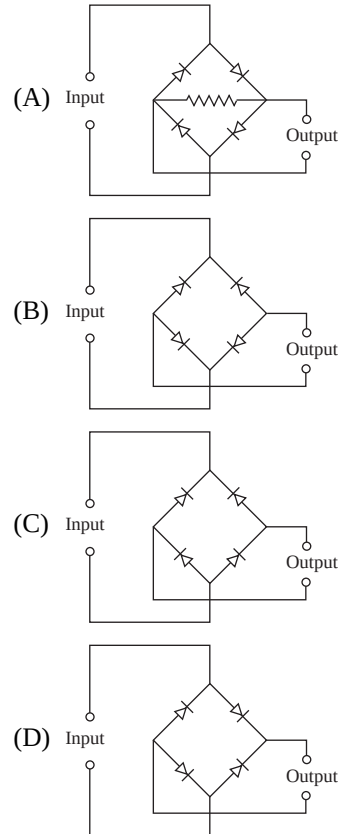


- (A) a low pass filter
- (B) a high pass filter
- (C) a band pass filter
- (D) a band reject filter

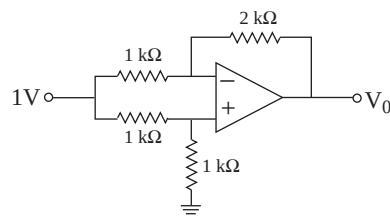
83. The electron and hole concentration in an intrinsic semiconductor are n_i per cm^3 at 300 K. Now, acceptor impurities are introduced with a concentration of N_A per cm^3 impurities are introduced with a concentration of N_A per cm^3 (where $N_A \gg n_i$), the electron concentration per cm^3 at 300 K will be—

- (A) n_i
- (B) $n_i + N_A$
- (C) $N_A - n_i$
- (D) $\frac{n_i^2}{N_A}$

84. The correct full wave rectifier circuit is—

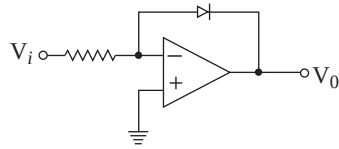


85. For the Op-amp circuit shown below, output V_0 is :



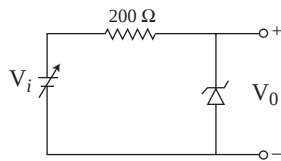
- (A) -2 V
- (B) -1 V
- (C) -0.5 V
- (D) 0.5 V

86. In the op-amp circuit shown, assume that the diode current follows the equation $I = I_s \exp(V/V_T)$. For $V_i = 2V$, $V_0 = V_{01}$ and for $V_i = 4V$, $V_0 = V_{02}$. The relationship between V_{01} and V_{02} is—



- (A) $V_{02} = \sqrt{2} V_{01}$
 (B) $V_{02} = e^2 V_{01}$
 (C) $V_{02} = V_{01} \ln 2$
 (D) $V_{01} = V_{02} = V_T \ln 2$

87. For the zener diode shown in the figure, the zener voltage at knee is 7V, the knee current is negligible and the zener dynamic resistance is 10Ω . If the input voltage V_i range is from 10 to 16V, the output voltage (V_0) range from—

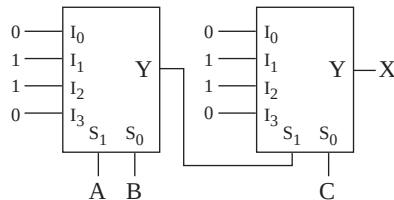


- (A) 7.00 to 7.29 V (B) 7.14 to 7.29 V
 (C) 7.14 to 7.43 V (D) 7.29 to 7.43 V

88. The Boolean expression $Y = \overline{A}BCD + \overline{A}BC\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}B\overline{C}\overline{D}$ can be minimized to—

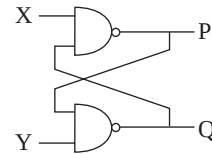
- (A) $Y = \overline{A}BCD + \overline{A}BC\overline{D} + \overline{A}CD$
 (B) $Y = \overline{A}BCD + BCD + \overline{A}BC\overline{D}$
 (C) $Y = \overline{A}BCD + \overline{B}CD + \overline{A}BC\overline{D}$
 (D) $Y = \overline{A}BCD + \overline{B}CD + \overline{A}BC\overline{D}$

89. In the following circuit, X is given by—



- (A) $X = \overline{A}BC + \overline{A}BC + \overline{A}BC + \overline{A}BC$
 (B) $X = \overline{A}BC + \overline{A}BC + \overline{A}BC + \overline{A}BC$
 (C) $X = AB + BC + AC$
 (D) $X = \overline{A}B + \overline{B}C + \overline{A}C$

90. The following binary values were applied to the X and Y inputs of the NAND latch shown in the figure in the sequence indicated below :
 $X = 0, Y = 1; X = 0, Y = 0; X = 1, Y = 1$
 The corresponding stable P, Q outputs will be—



- (A) $P = 1, Q = 0; P = 1, Q = 0; P = 1, Q = 0$
 or $P = 0, Q = 1$
 (B) $P = 1, Q = 0; P = 0, Q = 1$ or $P = 0, Q = 1; P = 0, Q = 1$
 (C) $P = 1, Q = 0; P = 1, Q = 1; P = 1, Q = 0$
 or $P = 0, Q = 1$
 (D) $P = 1, Q = 0; P = 1, Q = 1; P = 1, Q = 1$

91. The concentration of minority carriers in an extrinsic semi-conductor under equilibrium is—

- (A) directly proportional to the doping concentration
 (B) inversely proportional to the doping concentration
 (C) directly proportional to the intrinsic concentration
 (D) inversely proportional to the intrinsic concentration

92. The phenomenon known as “Early effect” in a bipolar transistor refer to a reduction of the effective base-width caused by—

- (A) electron-hole recombination at the base
 (B) the reverse biasing of the base-collector junction
 (C) the forward biasing of the emitter-base junction
 (D) the early removal of stored base charge during saturation to-cut-off switching

93. The input impedance (Z_i) and the output impedance (Z_0) of an ideal transconductance (Voltage controlled current source) amplifier are—

- (A) $Z_i = 0, Z_0 = 0$ (B) $Z_i = 0, Z_0 = \infty$
 (C) $Z_i = \infty, Z_0 = 0$ (D) $Z_i = \infty, Z_0 = \infty$

94. The majority carriers in an n -type semiconductor have an average drift velocity V in the direction perpendicular to a uniform magnetic field B . The electric field E induced due to Hall effect acts in the direction—

(A) $V \times B$
 (B) $B \times V$
 (C) Along V
 (D) Opposite to V

95. A heavily doped n -type semiconductor has the following data :

Hole–electron mobility ratio = 0.4

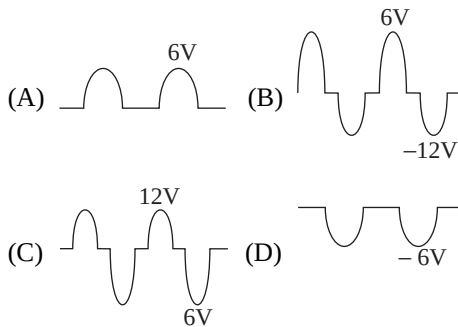
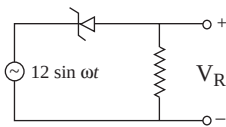
Doping concentration = 4.2×10^8 atoms/m³

Intrinsic concentration = 1.5×10^4 atoms/m³

The ratio of conductance of the n -type semiconductor to that of the intrinsic semiconductor of same material and at the same temperature is given by—

(A) 0.00005 (B) 2,000
 (C) 10,000 (D) 20,000

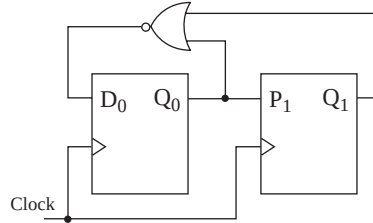
96. For the circuit shown, assume that the zener diode is ideal with a breakdown voltage of 6 volts. The wave observed across R is—



97. A silicon PN junction diode under reverse bias has depletion region of width $10 \mu\text{m}$. The relative permittivity of silicon, $\epsilon_r = 11.7$ and the permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12}$ F/m. The depletion capacitance of the diode per square meter is—

(A) $100 \mu\text{F}$ (B) $10 \mu\text{F}$
 (C) $1 \mu\text{F}$ (D) $20 \mu\text{F}$

98. For the circuit shown, the counter state (Q_1, Q_0) following sequence—

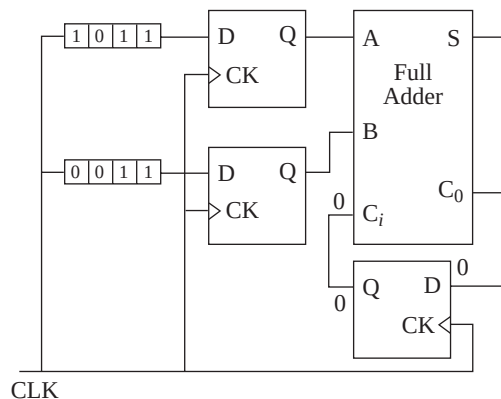


(A) 00,01,10,11,00.....
 (B) 00,01,10,00,01.....
 (C) 00,01,11,00,01.....
 (D) 00,10,11,00,10.....

99. In a non-linear op-amp circuit, the—

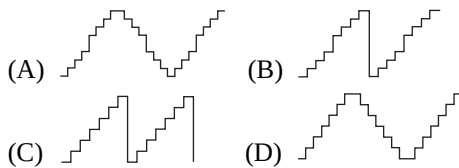
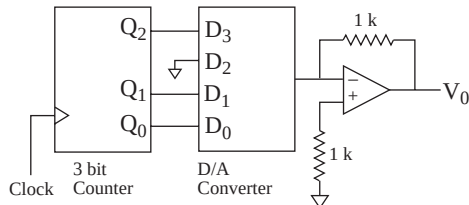
(A) Op-amp never saturates
 (B) Feedback loop is never opened
 (C) Output shape is the same as the input shape
 (D) Op-amp may saturate

100. For the circuit shown in figure below, two 4-bit parallel-in serial-out shift registers loaded with, the data shown are used to feed the data to a full adder. Initially, all the flip-flops are in clear state. After applying two clock pulses, the outputs of the full adder should be—

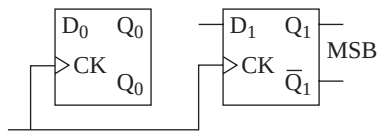


(A) $S = 0, C_0 = 0$
 (B) $S = 0, C_0 = 1$
 (C) $S = 1, C_0 = 0$
 (D) $S = 1, C_0 = 1$

101. A 4-bit D/A converter is connected to a free-running 3-bit up-counter, as shown in the following figure which of the following waveform will be observed at V_0 . In the fig. above, the ground has been shown by the symbol ∇ —



102. Two D-flip-flops as shown below, are to be connected as a synchronous counters that goes through the following $Q_1 Q_0$ sequence $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00 \rightarrow \dots$. The inputs D_0 and D_1 respectively should be connected :



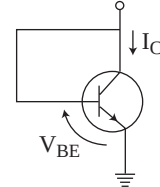
- (A) $\overline{Q_1}$ and Q_0
 (B) $\overline{Q_0}$ and Q_1
 (C) $\overline{Q_1} Q_0$ and $\overline{Q_1} Q_0$
 (D) $\overline{Q_1} Q_0$ and $Q_1 Q_0$

103. A Silicon Sample A is doped with 10^{18} atoms/cm³ of Boron. Another Sample B of identical dimensions is doped with 10^{18} atoms/cm³ of phosphorus. The ratio of electrons to hole mobility is 3. The ratio of conductivity to the Sample A to B is—

- (A) 3 (B) $\frac{1}{3}$
 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

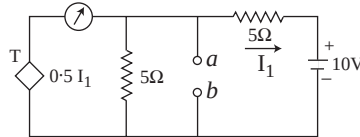
104. For an npn transistor connected as shown in the figure, $V_{BE} = 0.7$ volts. Given that reverse saturation currents of the junction at

room temperature 300 K is 10^{-13} A, the emitter current is—



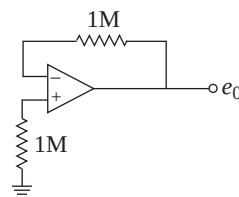
- (A) 30 mA (B) 39 mA
 (C) 49 mA (D) 20 mA

105. For the circuit shown in figure. Thevenin's voltage and Thevenin's equivalent resistance at terminals $a - b$ is—



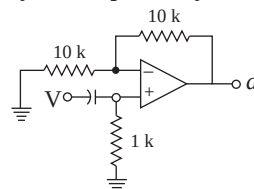
- (A) 5V & 2Ω (B) 7.5 V & 2.5Ω
 (C) 4V & 2Ω (D) 3V & 2.5Ω

106. The voltage e_0 indicated in the figure has been measured by an ideal voltmeter, which of the following can be calculated—



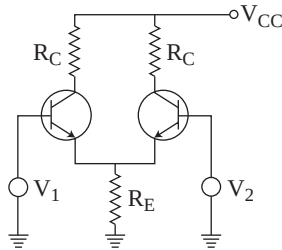
- (A) Bias current of the inverting input only
 (B) Bias current of the inverting and non-inverting inputs only
 (C) Input offset current only
 (D) Both the bias currents and the input offset currents

107. The op-Amp circuit shown in the figure is a filter. The type of filter and its cut off frequency are respectively—

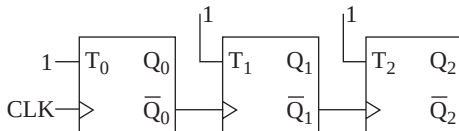


- (A) High pass, 1000 rad/sec
 (B) Low pass, 1000 rad/sec
 (C) High pass, 10000 rad/sec
 (D) Low pass, 10000 rad/sec

108. In an ideal differential amplifier shown in figure, a large value of (R_E)—



- (A) increases both the differential and common mode gains
 (B) increases the common-mode gain only
 (C) decreases the differential mode gain only
 (D) decreases the common mode gain only
109. The present output Q_n of an edge triggered JK flip-flop is logic 0. If $J = 1$, then Q_{n+1} —
 (A) cannot be determined
 (B) will be logic 0
 (C) will be logic 1
 (D) will race around
110. The given figure shows a ripple counter using positive edge triggered flip-flops. If the present state of the counter is $Q_2Q_1Q_0 = 011$, then its next stage ($Q_2Q_1Q_0$) will be—



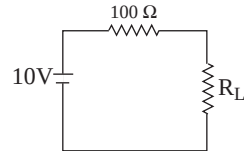
- (A) 010 (B) 100
 (C) 111 (D) 101
111. The Boolean expression for truth table shown is—

A	B	C	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

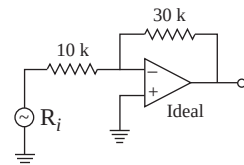
- (A) $B(A + C)(\overline{A} + \overline{C})$

- (B) $B(A + \overline{C})(\overline{A} + C)$
 (C) $\overline{B}(A + \overline{C})(\overline{A} + C)$
 (D) $\overline{B}(A + C)(\overline{A} + \overline{C})$

112. The maximum power that can be transferred to the load resistor R_L from the voltage source in the figure is—

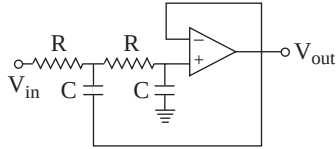


- (A) 1 W (B) 10 W
 (C) 0.25 W (D) 0.5 W
113. The effect of current shunt feedback in an amplifier is to—
 (A) increase the input resistance and decreases the output resistances
 (B) increase both input and output resistances
 (C) decreases both input and output resistances
 (D) decreases the input resistance and increases the output resistances
114. The input resistance R_i of the amplifier shown in the figure is—



- (A) $\frac{30}{4} k\Omega$ (B) 10 k Ω
 (C) 40 k Ω (D) Infinite
115. If for a silicon *npn* transistor, the base to emitter voltage (V_{BE}) is 0.7 V and the collector to base voltage (V_{CE}) is 0.2V, then the transistor is operating in the—
 (A) normal active mode
 (B) saturation mode
 (C) inverse active mode
 (D) cut off mode
116. An ideal op-amp is an ideal—
 (A) Voltage controlled current source
 (B) Voltage controlled voltage source
 (C) Current controlled voltage source
 (D) Current controlled current source

117. The circuit in the given figure is a—

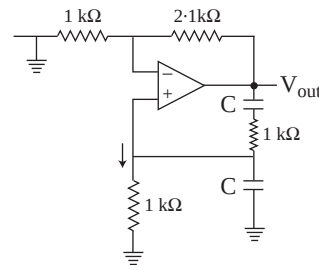


- (A) low pass filter
(B) high pass filter
(C) band pass filter
(D) band reject filter
118. A digital system is required to amplify a binary-encoded audio signal. The user should be able to control the gain of the amplifier from a minimum to a maximum in 100 increment. The number of bits required to encode, in straight binary, is—
(A) 8 (B) 6
(C) 5 (D) 7
119. The drain of an n -channel MOSFET is shorted to the gate so that $V_{GS} = V_{DS}$. The threshold voltage (V_T) of MOSFET is 1V. If the drain current (I_D) is 1 mA for $V_{GS} = 2V$, then for $V_{GS} = 3V$, I_D is—
(A) 2 mA (B) 3 mA
(C) 9 mA (D) 4 mA
120. The longest wavelength that can be absorbed by silicon, which has the band gap of 1.12 eV, is $1.1 \mu\text{m}$. If the longest wavelength that can be absorbed by another material is $0.87 \mu\text{m}$, then the band gap of this material is—
(A) 1.416 eV (B) 0.886 eV
(C) 0.854 eV (D) 0.706 eV
121. The Boolean expression $AC + B\overline{C}$ is equivalent to—
(A) $\overline{A}C + B\overline{C} + AC$
(B) $\overline{B}C + AC + B\overline{C} + \overline{A}C\overline{B}$
(C) $AC + B\overline{C} + \overline{B}C + ABC$
(D) $ABC + \overline{A}B\overline{C} + ABC\overline{C} + ABC\overline{C}$
122. The minimum number of 2 to 1 multiplexers required to realize 4 to 1 multiplexer is—
(A) 1 (B) 2
(C) 3 (D) 4

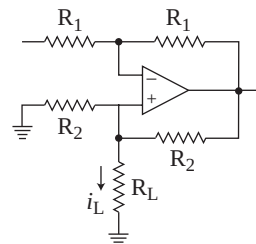
123. In a full wave rectifier using two ideal diodes, V_{dc} and V_m are the dc and peak values of the voltage respectively across a resistive load. If PIV is the peak inverse voltage of the diode, then the appropriate relationship for this rectifier are—

- (A) $V_{dc} = \frac{V_m}{\pi}$, $\text{PIV} = 2V_m$
(B) $V_{dc} = \frac{2V_m}{\pi}$, $\text{PIV} = 2V_m$
(C) $V_{dc} = \frac{2V_m}{\pi}$, $\text{PIV} = V_m$
(D) $V_{dc} = \frac{V_m}{\pi}$, $\text{PIV} = V_m$

124. The value of C required for sinusoidal oscillations of frequency 1 kHz in the circuit of given figure—



- (A) $\frac{1}{2\pi} \mu\text{F}$
(B) $2\pi \mu\text{F}$
(C) $\frac{1}{2\pi\sqrt{6}} \mu\text{F}$
(D) $2\pi\sqrt{6} \mu\text{F}$
125. In the op-amplifier circuit given in the figure, the load current i_L is—



- (A) $\frac{-V_S}{R_2}$ (B) $\frac{V_S}{R_2}$
(C) $\frac{-V_S}{R_L}$ (D) $\frac{V_S}{R_1}$

126. The type of a register, in which data is entered into it only one bit at a time, but has all data bits available as output is—
 (A) serial-in parallel out register
 (B) serial in serial out register
 (C) parallel in serial out register
 (D) parallel in parallel out register
127. A mod-16 counter can function as a divide by 16 frequency divider by taking the output from the—
 (A) 2^2 output (B) 2^3 output
 (C) 2^4 output (D) 2 output
128. To convert a 4-bit mod-16 counter to a mod-12 counter, the flip-flop must be Reset when the counter reaches the number—
 (A) 11 (B) 12
 (C) 16 (D) None of these
129. A ring counter consisting of five flip-flops has—
 (A) 10 states (B) 5 states
 (C) 2^5 states (D) None of these
130. A twisted ring counter consisting of four flip-flops will have—
 (A) 4 states (B) 8 states
 (C) 2^4 states (D) None of these
131. The number of states in a decade counter is—
 (A) 5 (B) 10
 (C) 15 (D) 20
132. A 3-bit synchronous counter uses flip-flops with propagation delay time of 20 ns each. The maximum possible time required for change of state will be—
 (A) 60 ns (B) 40 ns
 (C) 20 ns (D) None of these
133. The maximum possible number of states in a ripple counter consisting of five flip-flops is—
 (A) 32 (B) 15
 (C) 10 (D) 5
134. The maximum module number that can be obtained by a ripple counter using five flip-flops is—
 (A) 32 (B) 16
 (C) 15 (D) 5
135. The number of flip-flops required for a mod-16 ring counter are—
 (A) 4 (B) 8
 (C) 12 (D) 16
136. A BCD counter has—
 (A) 12 distinct states
 (B) 10 distinct state
 (C) 6 distinct states
 (D) 3 distinct state
137. The output frequency of a decade counter, clocked from a 50 kHz signal is—
 (A) 500 kHz (B) 50 kHz
 (C) 5 kHz (D) None of these
138. The counter which requires maximum number of flip-flops for a given mode number is—
 (A) ripple Counter
 (B) BCD Counter
 (C) ring Counter
 (D) programmable Counter
139. A mod-2 counter followed by a mod-5 is the same as—
 (A) mod 7 counter
 (B) a decade counter
 (C) a mod 3 counter
 (D) None of these
140. When a flip-flop is reset, the output will be—
 (A) $Q = 0, \overline{Q} = 1$
 (B) $Q = 1, \overline{Q} = 0$
 (C) $Q = 0, \overline{Q} = 0$
 (D) $Q = 1, \overline{Q} = 1$
141. In a T flip-flop that output frequency is—
 (A) same as the input frequency
 (B) one-half its inputs frequency
 (C) double the input frequency
 (D) None of these
142. The equivalent weight of the LSB in a 4-bit variable resistive divider DAC is—
 (A) $\frac{1}{4}$ (B) $\frac{1}{16}$
 (C) $\frac{1}{15}$ (D) $\frac{1}{32}$

143. For a 4 bit input resistive divider ($0 = 0V$, $1 = +10V$) the analog output voltage for a digital input of 1101 is—

(A) $7\frac{1}{3}V$ (B) $13V$
(C) $8\frac{2}{3}V$ (D) $11.01V$

144. A 4-bit resistor divider DAC uses $80k\Omega$ resistor for MSB. The resistor value for LSB is—

(A) $160k\Omega$ (B) $40k\Omega$
(C) $20k\Omega$ (D) $10k\Omega$

145. For a 5-bit resistive divider network, the weight assigned to MSB is—

(A) $\frac{1}{31}$ (B) $\frac{1}{32}$
(C) $\frac{8}{31}$ (D) $\frac{16}{32}$

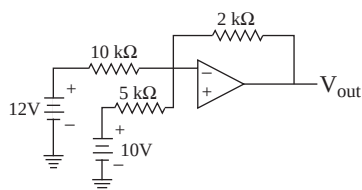
146. In a binary $\frac{R}{2R}$ ladder DAC, the input resistance for each input is—

(A) R (B) $2R$
(C) $3R$ (D) $4R$

147. The percentage resolution of a 10-bit ADC is nearly—

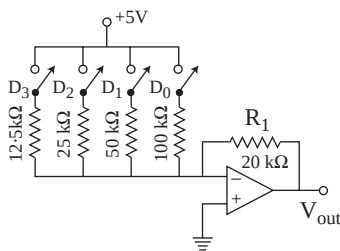
(A) 1% (B) 0.01%
(C) 0.1% (D) 10%

148. The V_{out} is—



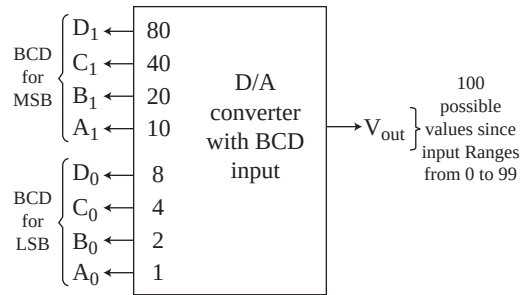
(A) $6.4V$ (B) $-6.4V$
(C) $3.2V$ (D) $-3.2V$

149. Determine the voltage at V_{out} in figure if the binary equivalent 10_{10} is input to switches D_3 to D_0 —



(A) $10V$ (B) $-10V$
(C) $5V$ (D) $-5V$

150. If the weight of A_0 is $0.1V$ in figure find the V_{out} for $D_1C_1B_1A_1 = 0101$ and $D_0C_0B_0A_0 = 1000$ —



(A) $5V$ (B) $5.8V$
(C) $4.2V$ (D) $2.8V$

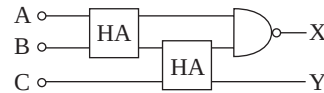
151. The minimum conversion interval of an ADC using an eight stage counter with a clock frequency of $2MHz$ —

(A) $0.5\mu s$ (B) $64\mu s$
(C) $128\mu s$ (D) $100\mu s$

152. Which of the following is known as a half adder?

(A) XOR gate (B) XNOR gate
(C) NAND gate (D) NOR gate

153. For the logic network, the outputs X & Y are given by—



(A) $X = (AB + AB)C$, $Y = (B \oplus C)A + BC$
(B) $X = (A \oplus B)C + AB$, $Y = A \oplus B \oplus C$
(C) $X = AB + BC + AC$, $Y = (A \oplus B)C$
(D) None of these

154. An integrator is a—

(A) low pass RC circuit with a large time constant
(B) high pass RC circuit with a large time constant
(C) low pass RC circuit with a short time constant
(D) high pass RC circuit with a short time constant

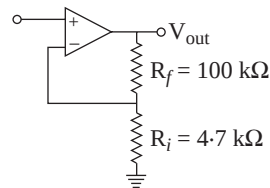
155. A differentiator is a—

- (A) high pass RC circuit with a large time constant
- (B) low pass RC circuit with short time constant
- (C) low pass RC circuit with large time constant
- (D) high pass RC circuit with short time constant

156. A clamping circuit is also called a—

- (A) limiter
- (B) dc restorer
- (C) stabiliser
- (D) amplitude Selector

157. The gain of the amplifier shown in figure. The open loop voltage gain of the op-Amp is 1,00,000—



- (A) 20.3
- (B) 24.3
- (C) 22.3
- (D) 28.3

158. A certain op-amp has an open-loop voltage gain of 1,00,000 and a common mode gain of 0.2. The CMRR is—

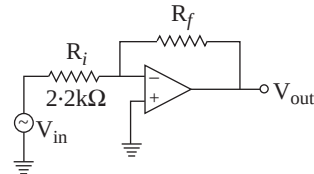
- (A) 5,00,000
- (B) 14 dB
- (C) Both
- (D) None of these

159. Common-mode rejection ratio is given by—

1. $\text{CMRR} = \frac{A_{ol}}{A_{cm}}$
2. $\text{CMRR} = 20 \log \left(\frac{A_{ol}}{A_{cm}} \right)$

- (A) Only 1
- (B) Only 2
- (C) Both 1 and 2
- (D) Neither 1 and 2

160. Given the op-amp configuration in given figure below, the value of the R_f required to produce a closed-loop voltage gain of -100—

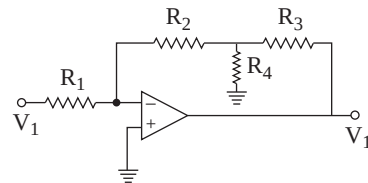


- (A) 480 kΩ
- (B) 220 kΩ
- (C) 440 kΩ
- (D) 110 kΩ

161. For an op-amp with negative feedback, the output is—

- (A) equal to the input
- (B) increased
- (C) feedback to the inverting input
- (D) feedback to the non-inverting input

162. The transistor gain of the circuit shown in the figure is given by—



- (A) $\left[\frac{R_2 + R_3 + \frac{R_2 R_3}{R_4}}{R_1} \right]$
- (B) $\left[\frac{\frac{R_3 R_4}{R_3 + R_4} + R_2}{R_1} \right]$
- (C) $\left[\frac{\frac{R_2 R_4}{R_2 + R_4} + R_3}{R_1} \right]$
- (D) $\left[\frac{R_2 + R_3}{R_1} \right]$

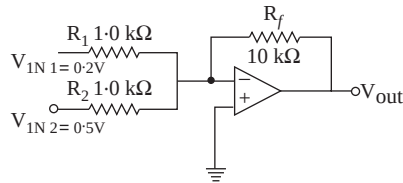
163. When negative feedback is used, the gain bandwidth product of an op-amp—

- (A) increases
- (B) decreases
- (C) stays the same
- (D) fluctuates

164. The certain inverting amplifier has a closed-loop gain of 25. The op-amp has an open loop gain of 1,00,000. If another op-amp with an open-loop gain of 2,00,000 is substituted in the configuration, the closed loop gain—

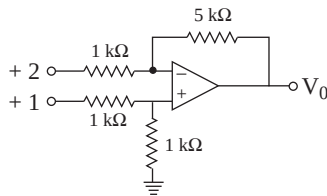
- (A) Doubles
- (B) Drops to 12.5
- (C) Remains at 25
- (D) Increases slightly

165. The output voltage for the summing amplifier shown in figure is—



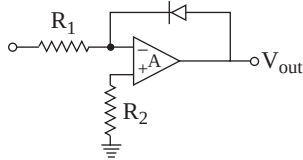
- (A) 7 V (B) -7 V
(C) 5 V (D) -5 V

166. The output V_0 of the ideal op-amp circuit shown in the figure—

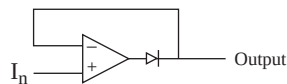


- (A) -7 V (B) -5 V
(C) 5 V (D) 7 V

167. The circuit connection of op-amp given in the fig. (I) and (II) represent—

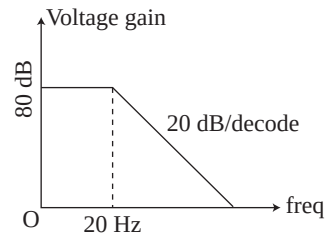


(I)

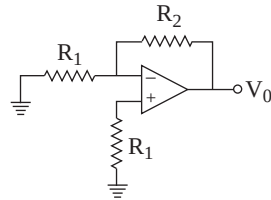


(II)

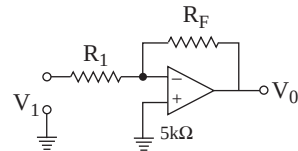
- (A) Logarithmic amplifier for both figures I and II
(B) Detectors for both figures I and II
(C) Detector for figure I and logarithmic amplifier for figure II
(D) Logarithmic amplifier for figure II and detector for figure II
168. The voltage gain versus frequency curve of an op-amp is shown in the given figure. The gain band width product of the op-amp is—



- (A) 200 Hz (B) 200 MHz
(C) 200 kHz (D) 2 MHz
169. Which one of the following conditions would give $V_e = 0$ in the circuit shown in figure ?



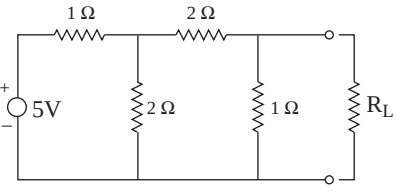
- (A) $R = R_1 + R_2$ (B) $R = \frac{R_2}{R_1}$
(C) $R = R_2 - R_1$ (D) $R = R_1 || R_2$
170. A circuit using an op-amp shown below, has—



- (A) Voltage series feedback
(B) Voltage shunt feedback
(C) Current series feedback
(D) Current shunt feedback
171. Active devices used in digital circuits generally operate as—
(A) amplifiers
(B) switches
(C) rectifiers
(D) wave form generators
172. The devices commonly used for making digital circuits are—
(A) mechanical Switches
(B) relays
(C) vacuum Tubes
(D) semiconductor devices

173. The time required to switch a p - n junction from ON to OFF is equal to—
 (A) zero (B) storage time
 (C) switching time (D) transition time
174. The storage time of a p - n junction—
 (A) decreases with increased reverse-bias voltage
 (B) decreases with increased forward-bias voltage
 (C) increases with increased reverse-bias voltage
 (D) increases with increased forward-bias voltage
175. Faster switching of a p - n junction requires—
 (A) a large current in the reverse direction
 (B) zero current in the reverse direction
 (C) reverse saturation current in the reverse direction
 (D) None of the above
176. The maximum operating frequency of a diode used as a switch—
 (A) depends on the diode characteristics and switching voltages
 (B) depends only on switching voltages
 (C) depends only on diode characteristics
 (D) none of the above
177. The voltage out of a schmitt trigger is—
 (A) A low voltage
 (B) A high voltage
 (C) Either a low or a high voltage
 (D) A sine wave
178. For fast switching operation, it is preferred to use—
 (A) Normal p - n junction diodes
 (B) Vacuum diodes
 (C) Zener diodes
 (D) Schottky diodes
179. The Schottky diode is a—
 (A) metal semiconductor junction
 (B) p - n junction
 (C) MOS devices
 (D) vacuum device
180. The switching speed of a Schottky diode—
 (A) is lower than that of a p - n junction diode
 (B) is the same as that of a p - n junction diode
 (C) is higher than that of a p - n junction diode
 (D) may be lower than or higher than that of a p - n junction diode
181. The most commonly used configuration of a transistor used as a switch is—
 (A) CB (B) CC
 (C) CE (D) CB or CC
182. The delay in the switching mode operation of a p - n junction diode is mainly due to—
 (A) excess minority charge stored in the junction when forward biased
 (B) metallic contact
 (C) different doping levels on the two sides of the junction
 (D) none of the above
183. A junction transistor used as a switch, switches between—
 (A) cut-off and saturation region
 (B) cut-off and active region
 (C) active and saturation region
 (D) none of the above
184. A BJT with $h_{FE} = 200$, $I_B = 10 \mu A$ and $I_C = 4 mA$ is operating in—
 (A) active region
 (B) cut off region
 (C) saturation region
 (D) (A) and (C)
185. npn transistors are preferred over p - n - p transistors for digital applications because—
 (A) they consume less power
 (B) of the requirements of positive logic system
 (C) the mobility of electrons is higher than that of holes
 (D) none of the above
186. In switching applications CE configuration is preferred because—
 (A) it requires low voltage or low current
 (B) it requires only one power supply
 (C) it has negligible I_{CEO}
 (D) it is easy to understand

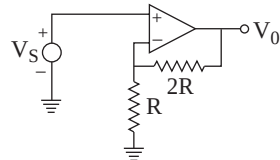
187. When a BJT is turned OFF, the transistor comes to the OFF state—
 - (A) as soon as the input signal is reversed
 - (B) as soon as the power is switched OFF
 - (C) after excess charge stored in the base region is removed
 - (D) as soon as the input signal is removed
188. When used as a switch, a Schottky transistor switches between—
 - (A) cut-off and active regions
 - (B) cut off and saturation region
 - (C) active and saturation region
 - (D) different operating points in the active region
189. Schottky transistors are preferred over normal transistors in digital applications because of their—
 - (A) higher propagation delay
 - (B) lower propagation delay
 - (C) higher power dissipation
 - (D) lower power dissipation
190. Higher switching speed is possible in Schottky transistors over normal transistors because—
 - (A) the transistors is not allow to go to cut off
 - (B) the transistors is not allow to go to saturation
 - (C) the transistors operates in cut off and saturation regions
 - (D) the transistors operates in active and saturation regions
191. For an n -channel enhancement mode MOSFET, the drain current—
 - (A) decreases with increase in drain voltage
 - (B) decreases with decrease in drain voltage
 - (C) increases with increase in gate voltage
 - (D) decreases with increase in gate voltage
192. For a MOSFET, the gate current—
 - (A) is dependent on drain current
 - (B) is negligible small
 - (C) is independent of gate voltage
 - (D) increases with increase in gate voltage
193. n -channel MOS devices are preferred over p -channel MOS devices for digital application because of—
 - (A) higher mobility of electrons than that of holes
 - (B) lower power dissipation
 - (C) higher power dissipation
 - (D) none of the above
194. A square wave can be generated from a sinusoidal wave by using a—
 - (A) double-diode clamper
 - (B) double-diode clipper
 - (C) transistor clipper
 - (D) none of the above
195. The amplifier used in the miller voltage sweep should have a gain of—
 - (A) $-\infty$
 - (B) $+1$
 - (C) $+\infty$
 - (D) -1
196. The amplifier used in the boot strap voltage sweep should have a gain of—
 - (A) $+1$
 - (B) $+\infty$
 - (C) -1
 - (D) $-\infty$
197. RTL consists of—
 - (A) resistors, transistors and inductors
 - (B) resistors and transistors
 - (C) resistors, diode and bipolar junction transistors
 - (D) resistors, capacitors and diodes
198. Which of the following logic families dissipates minimum power ?
 - (A) CMOS
 - (B) DTL
 - (C) ECL
 - (D) TTL
199. In general, the propagation delay time of MOS devices—
 - (A) is of the same order as bipolar transistors
 - (B) is less than that of bipolar transistors
 - (C) is more than that of bipolar transistors
 - (D) may be more or less than that of bipolar devices
200. Bipolar logic families employ—
 - (A) junction field-effect transistors
 - (B) p - n junctions
 - (C) n - p - n transistors
 - (D) MOSFETs

201. Which of the following logic families has maximum fan out—
 (A) CMOS (B) PMOS
 (C) ECL (D) TTL
202. MOSFET is also known as—
 (A) UJT
 (B) CMOS
 (C) IGFET
 (D) Bipolar junction transistor
203. The Thevenin equivalent of the circuit is—

 (A) $\frac{8}{3} \Omega, \frac{10}{3} \text{ V}$ (B) $\frac{8}{11} \Omega, \frac{10}{11} \text{ V}$
 (C) $\frac{11}{8} \Omega, \frac{10}{11} \text{ V}$ (D) $\frac{3}{8} \Omega, \frac{3}{10} \text{ V}$
204. A single diode operates as a—
 (A) full wave rectifier
 (B) half wave rectifier
 (C) bridge rectifier
 (D) non-rectifier
205. The efficiency of a full wave rectifier is that of a halfwave rectifier.
 (A) twice (B) half
 (C) equal to (D) one and half
206. The D.C. output of a half wave rectifier is that of an equivalent full wave rectifier.
 (A) half (B) 0.318
 (C) 0.636 (D) 0.707
207. In a full-wave rectifier if a.c. supply frequency is at 50 Hz, then the a.c. ripple in the output is—
 (A) 100 Hz (B) 25 Hz
 (C) 50 Hz (D) 75 Hz
208. A zener diode is operated in the
 (A) forward bias
 (B) reverse bias
 (C) breakdown region
 (D) None of these
209. Ripple factor of a full wave rectifier is—
 (A) 0.24 (B) 0.96
 (C) 0.48 (D) 0.72
210. If a PN junction is heavily doped, breakdown voltage will be—
 (A) more
 (B) neither more or less
 (C) more or less
 (D) less
211. A junction transistor with $\beta = 49$ and $I_{CO} = I_{CBO} = 1 \mu\text{A}$ has $I_B = 10 \mu\text{A}$. The value of I_C is given in μA by—
 (A) 441 (B) 490
 (C) 539 (D) 540
212. To obtain a fixed bias in CE configuration a battery of 2V is used in the base with a series resistor R_B across the base emitter junction given $\beta = 100$ and $V_{BE} = 0.6 \text{ V}$, to obtain $I_C = 2 \text{ mA}$, the value of R_B is given by—
 (A) 700 Ω
 (B) 7 k Ω
 (C) 70 k Ω
 (D) 130 k Ω
213. The effect of introducing R_E in the CE amplifier is to—
 (A) increase the voltage gain
 (B) decrease the voltage gain
 (C) increase the current gain
 (D) decrease the current gain
214. In an FET,
 (A) both junctions are reverse biased
 (B) one junction is reverse biased and the other forward biased
 (C) one junction has reverse bias on both sides of the junction
 (D) one junction has reverse bias on one side and forward bias on the other
215. A CE amplifier has a resistor R_F connected between collector and base $R_F = 40 \text{ k}\Omega$. R_C (collector load resistance) = 4 k Ω given $h_{fe} = 50$, $r_\pi = 1 \text{ k}\Omega$, the output resistance R_0 is given by—
 (A) 40 k (B) 20 k
 (C) 4 k (D) 0.66 k

216. A differential amplifier has inputs $V_1 = 1050 \mu\text{V}$ and $V_2 = 950 \mu\text{V}$, $\text{CMRR} = 1000$, the error in the differential output is—

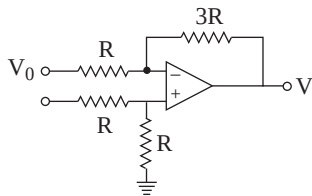
(A) 10 % (B) 1 %
(C) 0.1 % (D) 0.01 %

217. In the op-amp circuit V_0 is given by—



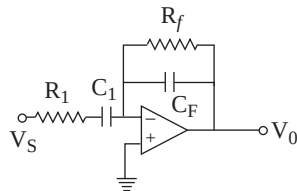
(A) $2 V_S$ (B) $-2 V_S$
(C) $3 V_S$ (D) $-3 V_S$

218. The value of V_0 is given by—



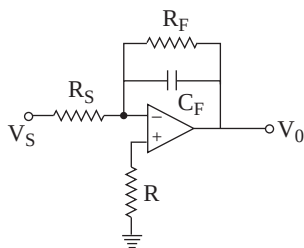
(A) $-3V_1 + \frac{1}{2}V_2$ (B) $-3V_2$
(C) $\frac{3}{2}V_2 - \frac{9}{4}V_1$ (D) $2V_2 - 3V_1$

219. In the practical differentiator circuit shown—



(A) C_F cuts off the high frequency noise
(B) R_1 cuts off low frequency noise
(C) $R_1 C_F$ forms the basic differentiation elements
(D) C_F ensures stable operation

220. In the practical integrator circuit shown—

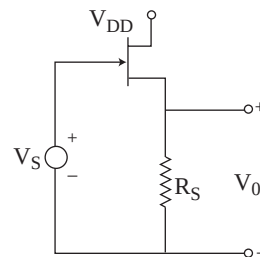


(A) R_F enables low frequency gain to be limited
(B) Larger the value of R_S higher the gain
(C) Effective integration occurs at ω less than $\frac{1}{R_F C_F}$
(D) The basic elements for integration are R and C_F

221. In a negative feedback amplifier, the input impedance is increased—

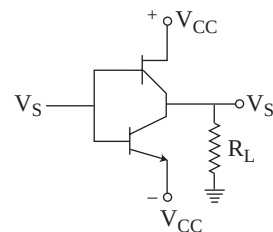
(A) if the signal sampled is a voltage
(B) if the signal sampled is a current
(C) if the feedback signal is a voltage
(D) if the feedback signal is a current

222. In the following figure, $R_S = 3\text{K}$. Given $\frac{V_0}{V_S} = 0.9$, the value of g_m is given by—



(A) 0.3 mA/V (B) 0.33 mA/V
(C) 3.0 mA/V (D) 3.0 A/V

223. Figure shows a push pull B amplifier using complementary symmetry transistors. Given $V_{CC} = 6\text{V}$ and peak output current amplitude is 1 amp. The DC power chain from each power supply is nearly—

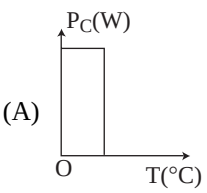


(A) 2 watts (B) 4 watts
(C) 6 watts (D) 12 watts

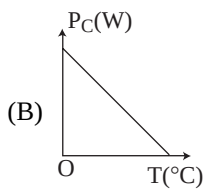
224. For a full wave rectifier—

(A) $V_{dc} = \frac{V_P}{\pi}$ (B) $V_{dc} = \frac{\pi}{V_P}$
(C) $V_{dc} = \frac{2V_P}{\pi}$ (D) $V_{dc} = \frac{2\pi}{V_P}$

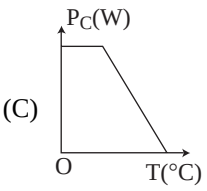
225. If the linear capacitance is increased, the ripple will—
 (A) decrease (B) stay the same
 (C) increase (D) None of these
226. If $R_L = \infty$ in an inductor filter, the value of ripple factor is approximately—
 (A) 0.48 (B) 1.21
 (C) 0.048 (D) 0.121
227. In an inductor filter filtering can be increased by—
 (A) decreasing circuit resistance
 (B) increasing circuit current
 (C) both the above
 (D) None of these
228. In an inductor filter for a given L it is approximately observed that—
 (A) $r \propto R_L$ (B) $r \propto R_L^2$
 (C) $r \propto \frac{1}{R_L}$ (D) $r \propto \frac{1}{R_L^2}$
229. Simplest voltage regulator is built with—
 (A) four-layer diode
 (B) zener diode
 (C) LED
 (D) Schottky diode
230. Shunt regulation characterised by—
 (A) large power losses
 (B) low efficiency
 (C) its simplicity
 (D) All the above
231. With usual notations, the percentage regulation is given by—
 (A) $\frac{V_{NL} - V_{FL}}{V_{NL} + V_{FL}} \times 100$
 (B) $\frac{V_{NL} - V_{FL}}{V_{NL}} \times 100$
 (C) $\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$
 (D) $\frac{V_{NL} + V_{FL}}{V_{NL}} \times 100$
232. A zener diode is used as a—
 (A) shunt regulator (B) series regulator
 (C) shunt (D) None of these
233. A bridge rectifier is preferable to an ordinary two diode full wave rectifier because—
 (A) it uses four diodes
 (B) its transformer has no centre tap
 (C) it needs much smaller transformer for the same output
 (D) it has higher safety factor
234. A power supply with low output impedance has low—
 (A) load regulation (B) current limiting
 (C) line regulation (D) efficiency
235. The relation between current gain α and β of a transistor is—
 (A) $\alpha = \beta$ (B) $\alpha = \frac{\beta}{1 + \beta}$
 (C) $\beta = \frac{\alpha}{1 + \alpha}$ (D) None of the above
236. In an npn transistor, the collector current is 10 mA. If 90% of the electrons emitted reach the collector, then—
 (A) $I_E = 9$ mA (B) $I_E = 11$ mA
 (C) $I_B = 9$ mA (D) $I_B = -1$ mA
237. A transistor is in active region, when—
 (A) $I_B = \beta I_C$ (B) $I_C = \beta I_B$
 (C) $I_C = I_E$ (D) $I_C = I_B$
238. In an npn transistor, both emitter and collector junctions are forward biased, the transistor operates in—
 (A) Cut-off region (B) Saturation region
 (C) Active region (D) Inverted region
239. A transistor will operate in cut-off region when—
 (A) collector junction is forward biased and emitter junction reverse biased
 (B) collector junction in reverse biased and emitter junction forward biased
 (C) both collector and emitter junctions are forward biased
 (D) both collector and emitter junctions are reverse biased
240. A transistor-terminal voltage is considered positive if—
 (A) the terminal is more positive than the common terminal
 (B) the common terminal is more positive than the terminal
 (C) the terminal is the output terminal
 (D) the terminal is connected to p -type terminal

241. A transistor-terminal current is considered positive if—
 (A) electrons flow out of the transistor at the terminal
 (B) electrons flow into the transistor at the terminal
 (C) the current is due to the flow of electron only
 (D) the current is due to the flow of holes only
242. When the collector junction in transistor is biased in the reverse direction and the emitter junction is forward biased, the transistor is said to be in the—
 (A) active region
 (B) cut-off region
 (C) saturation region
 (D) None of these
243. The DC load line of a transistor circuit—
 (A) is a graph between I_C and I_B
 (B) is a graph between I_C and V_{CE}
 (C) is a curved graph
 (D) does not contain the operating point
244. The positive part of the output signal in a transistor circuit is clipped, if Q point moves—
 (A) towards the saturation point
 (B) towards the cut-off point
 (C) towards the centre of the load line
 (D) None of the above
245. The negative part of the output signal in a transistor is clipped, if Q-point moves—
 (A) towards the saturation point
 (B) towards the cut-off point
 (C) towards the centre of load line
 (D) None of the above
246. A biasing circuit has a stability factor of 40. If due to temperature change, I_{CO} changes by $1 \mu A$, then I_C will be change by—
 (A) $20 \mu A$ (B) $40 \mu A$
 (C) $80 \mu A$ (D) $100 \mu A$
247. The correct statement is—
 (A) h_{fe} and h_{oe} can be determined from C-E output characteristics
 (B) h_{ie} and h_{re} can be determined from C-E input characteristics
 (C) Both the above
 (D) None of the above
248. The value of h -parameters depends upon—
 (A) the positive of Q-point
 (B) the temp
 (C) both the above
 (D) None of the above
249. The value of thermal resistance depends upon—
 (A) the size of the transistor
 (B) convection or radiation
 (C) thermal connection to the device to a heat sink
 (D) All the above
250. The typical power temperature derating curve for a power transistor is—
- 

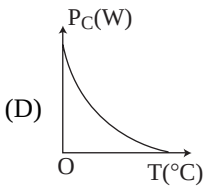
(A)

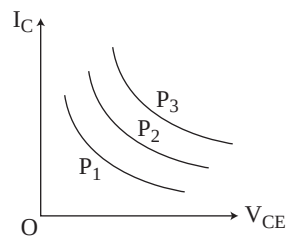


(B)



(C)



(D)
251. The effects of transistor self heating may be appreciated by referring to the figure, which shows three constant-power (P_1 , P_2 , P_3) hyperbolas—
- 
- Then—
 (A) $P_1 > P_2 > P_3$ (B) $P_2 > P_3 > P_1$
 (C) $P_3 > P_2 > P_1$ (D) $P_3 > P_1 > P_2$

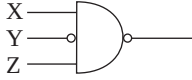
252. Compared to a bipolar transistor, the JFET has a much higher—
(A) voltage gain
(B) input resistance
(C) supply voltage
(D) current
253. The gate source diode of a JFET should be—
(A) forward-biased
(B) reverse-biased
(C) either forward or reverse biased
(D) none of the above
254. If the emitter resistance is reduced by one-half in a voltage divider bias circuit, the collector current will—
(A) double
(B) drop in half
(C) remain the same
(D) increase
255. The output voltage of a CE amplifier is—
(A) amplified
(B) inverted
(C) 180° out of phase with the input
(D) all of the above
256. A coupling capacitor is—
(A) a DC short
(B) an AC open
(C) a DC open and an AC short
(D) a DC short and an AC open
257. The capacitors of a CE amplifier appear to be—
(A) open to AC
(B) shorted to DC
(C) open to supply voltage
(D) shorted to AC
258. Reducing all DC sources to zero is one of the steps in getting the—
(A) DC equivalent circuit
(B) AC equivalent circuit
(C) Complete amplifier circuit
(D) Voltage-divider biased circuit
259. The ac equivalent circuit is derived from the original circuit by shorting all—
(A) resistors (B) capacitors
(C) inductors (D) transistors
260. When the AC base voltage is too large, the ac emitter current is—
(A) sinusoidal (B) constant
(C) distorted (D) alternating
261. In a CE amplifier with a large input signal, the positive half cycle of the ac emitter current is—
(A) equal to the negative half cycle
(B) smaller than the negative half cycle
(C) larger than the negative half cycle
(D) equal to the negative half cycle
262. AC emitter resistance equals 25 mV divided by the—
(A) Quiescent base current
(B) DC emitter current
(C) AC emitter current
(D) Change in collector current
263. To reduce the distortion in a CE amplifier, reduce the—
(A) DC emitter current
(B) Base-emitter voltage
(C) Collector current
(D) AC base voltage
264. If the AC voltage across the emitter diode is 1 mV and the AC emitter current is 100 μ A, the AC resistance of the emitter diode is—
(A) 1 Ω (B) 10 Ω
(C) 100 Ω (D) 1 k Ω
265. A graph of AC emitter current versus AC base-emitter voltage applies to the—
(A) resistor (B) emitter diode
(C) collector diode (D) power supply
266. The emitter of a CE amplifier has no AC voltage because of the—
(A) DC voltage on it
(B) by pass capacitor
(C) coupling capacitor
(D) load resistor

267. The voltage across the load resistor of a capacitor-coupled CE amplifier is—
 (A) DC and AC
 (B) DC only
 (C) AC only
 (D) Neither DC and AC
268. The AC collector current is approximately equal to the—
 (A) AC base current
 (B) AC emitter current
 (C) AC source current
 (D) AC bypass current
269. When the emitter resistance R_E doubles, the AC emitter resistance—
 (A) increases
 (B) decreases
 (C) remains the same
 (D) cannot be determined
270. Compared to the AC resistance of the emitter diode, the feedback resistance of a swamped amplifier should be—
 (A) small (B) equal
 (C) large (D) zero
271. If the emitter-bypass capacitor opens, the AC output voltage will—
 (A) decrease
 (B) increase
 (C) remain the same
 (D) equal zero
272. If the collector resistor is shorted, the AC output voltage will—
 (A) decrease
 (B) increase
 (C) remain the same
 (D) equal zero
273. If the input-coupling capacitor is open, the ac input voltage will—
 (A) decrease
 (B) remain the same
 (C) increase
 (D) equal zero
274. If the emitter-bypass capacitor is shorted, the ac input voltage at the base will—
 (A) decrease
 (B) increase
 (C) remain the same
 (D) equal zero
275. The DC current gain of an emitter follower is—
 (A) 0
 (B) ≈ 1
 (C) β_{dc}
 (D) Dependent on r'_e
276. The output voltage of an emitter follower is—
 (A) in phase with V_{in}
 (B) much greater than V_{in}
 (C) 180° out of phase
 (D) generally much less than V_{in}
277. An emitter follower buffer is generally used when—
 (A) $R_G < R_L$
 (B) $R_G = R_L$
 (C) $R_L < R_G$
 (D) R_L is very large
278. An emitter follower circuit always makes the source resistance—
 (A) β times smaller
 (B) β times larger
 (C) Equal to the load
 (D) Zero
279. If $\beta = 200$ and $r_e = 150 \Omega$, the input impedance of the base is—
 (A) $30 \text{ k}\Omega$ (B) 600Ω
 (C) $3 \text{ k}\Omega$ (D) $5 \text{ k}\Omega$
280. A common base amplifier can be used when—
 (A) matching low to high impedances
 (B) a voltage gain without a current gain is required
 (C) a high-frequency amplifier is needed
 (D) All of the above

281. An application of a common-base amplifier is when—
 (A) $R_{source} \gg R_L$
 (B) $R_{source} \ll R_L$
 (C) a high current gain is required
 (D) high frequencies need to be blocked
282. The zener current in a zener follower is—
 (A) equal to the output current
 (B) smaller than the output current
 (C) larger than the output current
 (D) prone to thermal run away
283. In the two-transistor voltage regulator, the output voltage—
 (A) is regulated
 (B) has much smaller ripple than the input voltage
 (C) is larger than the zener voltage
 (D) All of the above
284. For Class B operation, the collector current flows for—
 (A) The whole cycle
 (B) Half the cycle
 (C) Less than half a cycle
 (D) Less than a quarter of a cycle
285. Class C amplifiers are almost always—
 (A) Transformer-coupled between stages
 (B) Operated at audio frequencies
 (C) Tuned RF amplifiers
 (D) Wideband
286. The maximum efficiency of a Class B push-pull amplifier is—
 (A) 25% (B) 5%
 (C) 78.5% (D) 100%
287. If the load power is 300 mW and the dc power is 1.5 W, the efficiency is—
 (A) 0 (B) 3 per cent
 (C) 2 per cent (D) 20 per cent
288. A JFET—
 (A) is a voltage-controlled device
 (B) is a current-controlled device
 (C) has a low input resistance
 (D) has a very large voltage gain
289. R_{DS} equals pinch off voltage divided by—
 (A) Drain current
 (B) Gate current
 (C) Ideal drain current
 (D) Drain current for zero gate voltage
290. The transconductance curve is—
 (A) linear
 (B) similar to the graph of a resistor
 (C) non-linear
 (D) like a single drain curve
291. VHF covers frequencies from—
 (A) 300 kHz to 3 MHz
 (B) 3 to 30 MHz
 (C) 30 to 300 MHz
 (D) 300 MHz to 3 GHz
292. The transconductance increases when the drain current approaches—
 (A) 0 (B) $I_{D(sat)}$
 (C) I_{DSS} (D) I_S
293. When a JFET is cut off, the depletion layers are—
 (A) Far apart (B) Close together
 (C) Touching (D) Conducting
294. Self-bias produces—
 (A) Positive feedback
 (B) Negative feedback
 (C) Forward feedback
 (D) Reverse feedback
295. Transconductance is measured in—
 (A) Ohms (B) Amperes
 (C) Volts (D) Mhos or Siemens
296. Frequency response is a graph of voltage gain versus—
 (A) frequency (B) power gain
 (C) input Voltage (D) output voltage
297. At low frequencies, the coupling capacitors produces a decrease in—
 (A) input resistance
 (B) voltage gain
 (C) generator resistance
 (D) generator Voltage
298. At the lower or upper cut-off frequency, the voltage gain is—
 (A) $0.35 A_{V(mid)}$ (B) $0.5 A_{V(mid)}$
 (C) $0.707 A_{V(mid)}$ (D) $0.995 A_{V(mid)}$

299. If the power gain doubles the decibel power gain increases by—
 (A) A factor of 2
 (B) 3 dB
 (C) 6 dB
 (D) 10 dB
300. If the voltage gain doubles, the decibel voltage gain increases by—
 (A) A factor 2 (B) 3 dB
 (C) 6 dB (D) 10 dB
301. If the voltage gain is 10, the decibel voltage gain is—
 (A) 6 dB (B) 20 dB
 (C) 40 dB (D) 60 dB
302. If the voltage gain is 100, the decibel voltage gain is—
 (A) 6 dB (B) 20 dB
 (C) 40 dB (D) 60 dB
303. If the voltage gain is 2000, the decibel voltage gain is—
 (A) 40 dB (B) 46 dB
 (C) 66 dB (D) 86 dB
304. One frequency is 8 times another frequency. How many octaves apart are the two frequencies?
 (A) 1 (B) 2
 (C) 3 (D) 4
305. A common-mode signal is applied to—
 (A) the non-inverting input
 (B) the inverting input
 (C) both inputs
 (D) the top of the tail resistor
306. The common-mode rejection ratio is—
 (A) very low
 (B) often expressed in decibels
 (C) equal to the voltage gain
 (D) equal to the common-mode voltage gain
307. The input offset current is usually—
 (A) less than the input bias current
 (B) equal to zero
 (C) less than the input offset voltage
 (D) unimportant when a base resistor is used
308. The voltage gain of a loaded differential amplifier is—
 (A) larger than the unloaded voltage gain
 (B) equal to R_C/re'
 (C) smaller than the unloaded voltage gain
 (D) impossible to determine
309. The initial slope of a sine wave is directly proportional to—
 (A) Slew rate (B) Frequency
 (C) Voltage gain (D) Capacitance
310. If the two supply voltages are ± 12 V, the MPP value of an op-amp is closest to—
 (A) 0 (B) + 12 V
 (C) - 12 V (D) 24 V
311. Frequency is 10 MHz and mid-band open-loop voltage gain is 2,00,000, then the open-loop cut off frequency at the op-amp is—
 (A) 10 Hz (B) 20 Hz
 (C) 50 Hz (D) 100 Hz
312. The initial slope of a sine wave increases when—
 (A) Frequency decreases
 (B) Peak value increases
 (C) C_C increases
 (D) Slew rate decreases
313. If the frequency of the input signal is greater than the power bandwidth—
 (A) slew rate distortion occurs
 (B) a normal output signal occurs
 (C) output offset voltage increases
 (D) distortion may occur
314. An op-amp has an open base resistor. The output voltage will be—
 (A) zero
 (B) slightly different from zero
 (C) maximum positive or negative
 (D) an amplified sine wave
315. The voltage gain of an op-amp is unity at the—
 (A) Cut off frequency
 (B) Unity gain frequency
 (C) Generator frequency
 (D) Power bandwidth

Answers with Explanation

1. (A) _____
2. (D) $(\overline{A+B}) + C + (\overline{D+E})$
 $= (\overline{A+B} + C)(\overline{D} \cdot \overline{E})$
3. (C)
4. (C) Putting the values of X, Y, Z in all the four options.
5. (D)
6. (B)
- | | $\overline{C}\overline{D}$ | $\overline{C}D$ | CD | $C\overline{D}$ |
|----------------------------|----------------------------|-----------------|------|-----------------|
| $\overline{A}\overline{B}$ | 1 | 1 | 0 | 1 |
| $\overline{A}B$ | 1 | 1 | 0 | 1 |
| AB | 0 | 0 | 1 | 1 |
| $A\overline{B}$ | 0 | 0 | 0 | 0 |
- $\overline{A}\overline{C} + \overline{A}\overline{D} + ABC$
7. (B) $F = \overline{X} + \overline{XYZ} + \overline{Z}$
 $= \overline{X} + \overline{Z} + \overline{X} + Y + \overline{Z}$
 $= \overline{X} + \overline{Z} + Y$
 $= \overline{X \cdot Y \cdot Z}$
- 
8. (D) $\overline{C}B + CD + C\overline{B}$
9. (C) $(\overline{A}B + A\overline{B})(A + \overline{B})$
 $= 0 + \overline{A}B + \overline{A}B + 0$
 $= \overline{A}B$
10. (A) $Y = (A+B) \cdot (\overline{AB})$
 $= (A+B) \cdot (\overline{A} + \overline{B})$
 $= \overline{A}B + A\overline{B}$
- XOR gate.
11. (B) 12. (D)
13. (B) Race around condition always arises in asynchronous circuit.
14. (C) $2^5 = 32$, each flip-flop divides the input frequency to half.
15. (C)
16. (B) Mod 10 counter.

17. (D)

$J = \overline{Q_n}$	$K = 1$	Q_n	$\overline{Q_n}$	CLK
		0	1	
1	1	1	0	1
0	1	0	1	2
1	1	1	0	3
0	1	0	1	4
1	1	1	0	5
0	1	0	1	6

18. (C) C B A

1	1	1	← preset 3
1	1	0	MOD-3 down counter
1	0	1	
1	1	1	

19. (B)

$$20. (B) \quad \text{Resolution} = \frac{1}{2^N - 1} \times 100$$

$$\text{As } \frac{1}{2^N - 1} \leq 0.004$$

$$2N \geq 251$$

$$\text{or } \boxed{N = 8}$$

$$21. (B) \quad 1 \text{ LSB} = \frac{1}{2^{10}}$$

$$= 0.000977$$

$$\text{Accuracy} = \pm \frac{\text{LSB}}{2}$$

$$= \pm 0.000488$$

$$= \pm 400 \mu\text{V}/^\circ\text{C}$$

$$22. (D) \quad K = \frac{4}{51}$$

$$(11011011) \frac{4}{51} = 17 \text{ V}$$

23. (A) Convert binary into decimal representation.

$$24. (C) \quad V_A = \frac{V_2}{2} + V_1 + V_0$$

At node c

$$\frac{V_c - V_0}{2} + \frac{V_c}{2} + \frac{V_c - V_b}{1} = -4 V_c$$

$$2 V_b + V_0 = 4 V_c$$

$$V_c = \frac{V_b}{2} + \frac{V_0}{4}$$

At node b

$$\frac{V_b - V_a}{1} + \frac{V_b - V_c}{1} + \frac{V_b - V_1}{2} = 5 V_b$$

$$2V_a + 2V_c + V = 5 V_b$$

At node a $\frac{V_a - V_2}{2} + \frac{V_a - V_b}{1} = 3 V_a$

$$2V_b + V_2 = 3 V_a$$

$$V_c = \frac{V_b}{2} + \frac{V_0}{1}$$

25. (B) Conversion time of successive approximation is independent of analog output.

26. (C) 27. (B) 28. (A) 29. (A) 30. (B)

31. (A)

$$R_f = 20 \Omega,$$

$$R_L = 980 \Omega$$

$$V_m = 50\sqrt{2} = 70.7 \text{ V}$$

Mean load current

$$I_{dc} = \frac{2 I_m}{\pi}$$

$$I_m = \frac{V_m}{R_f + R_L} = \frac{70.7}{(20 + 980)}$$

$$= 70.7 \text{ m amp.}$$

32. (A) 33. (B) 34. (B) 35. (D) 36. (D)

37. (B)

38. (A)

$$V_Z = 15 V_{diode}$$

$$V_Z = 15 \text{ V at } I_{ZT}$$

$$I_{ZK} = 0.25 \text{ mA}$$

$$I_{ZT} = 17 \text{ mA}$$

$$Z_{ZT} = 14 \Omega$$

For I_{ZK} :

$$V_{out} = V_Z$$

$$= 15 \text{ V} - \Delta I_Z Z_{ZT}$$

$$= 15 \text{ V} - (I_{ZT} - I_{ZK}) Z_{ZT}$$

$$= 15 \text{ V} - (16.75 \text{ mA}) (14 \Omega)$$

$$= 15 \text{ V} - 0.235 \text{ V}$$

$$= 14.76 \text{ V}$$

For Zener maximum current,

The power dissipation is 1W

$$I_{ZM} = \frac{P_{D(max)}}{V_Z} = \frac{1 \text{ W}}{15 \text{ V}}$$

$$= 66.7 \text{ mA}$$

For I_{ZM} : $V_{out} = V_Z = 15 \text{ V} + \Delta I_Z Z_{ZT}$

$$= 15 \text{ V} + (I_{ZM} - I_{ZT}) Z_{ZT}$$

$$= 15 \text{ V} + (49.7 \text{ mA}) (14 \Omega)$$

$$= 15.7 \text{ V}$$

R also can be calculated

$$R = \frac{V_{IN} - V_Z}{I_{ZM}} = \frac{24 \text{ V} - 15.7 \text{ V}}{66.7 \text{ mA}}$$

$$= 124 \Omega$$

39. (B) 40. (D)

41. (B)

$$V_S = I_D \cdot R_S$$

$$= (2 \times 10^{-3}) \times 1000 = 2 \text{ V}$$

Now,

$$V_D = V_{DD} - I_D R_D$$

$$= 10 - (2 \times 10^{-3}) (2 \times 10^3)$$

$$= 10 - 4 = 6 \text{ V}$$

$$V_{DS} = V_D - V_S$$

$$= 6 - 2 = 4 \text{ V}$$

Now gate to source voltage

$$V_{GS} = -V_S$$

$$= 2 \text{ V}$$

42. (B)

$$I_D = k (V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$$

$$= \frac{10 \text{ mA}}{(8 \text{ V} - 2 \text{ V})^2} = \frac{10 \text{ mA}}{36 \text{ V}^2}$$

$$= \frac{5 \text{ mA}}{18 \text{ V}^2}$$

$$I_D = \frac{5}{18} (4 \text{ V} - 2 \text{ V})^2$$

$$= \frac{5}{18} \times 4 \text{ mA} = \frac{10}{9} \text{ mA}$$

$$= 1.11 \text{ mA}$$

43. (A)

44. (B)

45. (C)

46. (B)

47. (C)

48. (D)

49. (A)

50. (B)

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

$$= (25 \text{ mA}) \left(1 - \frac{5 \text{ V}}{15 \text{ V}} \right)^2$$

$$= 25 \text{ mA} (1 - 0.333)^2$$

$$= 11.1 \text{ mA}$$

$$R_S = \left(\frac{V_{GS}}{I_D} \right)$$

$$= \frac{5 \text{ V}}{11.1 \text{ mA}} = 450 \Omega$$

51. (A) 52. (D) 53. (A) 54. (C) 55. (B)
 56. (A) 57. (B) 58. (C) 59. (D) 60. (D)
 61. (C) 62. (B) 63. (C)

64. (B) $A_{Cl} = 29,$
 $B = \frac{1}{29}$
 $= \frac{R_3}{R_f} = \frac{R_f}{R_3} = 29$
 $R_f = 29 R_3$
 $= 29 (10 \text{ k}\Omega)$
 $= 290 \text{ k}\Omega.$

65. (B) frequency $= \frac{1}{2\pi \sqrt{6} RC}$
 $R_1 = R_2 = R_3 = R$
 $C_1 = C_2 = C_3 = C$
 frequency $= \frac{1}{2\pi \sqrt{6} \times (10 \text{ k}\Omega) (0.001 \mu\text{F})}$
 $= 6.5 \text{ kHz}.$

66. (C) 67. (A) 68. (B)

69. (A) The Channels have closed-loop voltage gain of $A_{V_1}(\text{CL}) = -\frac{100 \text{ k}\Omega}{20 \text{ k}\Omega} = -5$

$$A_{V_2}(\text{CL}) = \frac{-100 \text{ k}\Omega}{10 \text{ k}\Omega} = -10$$

$$A_{V_3}(\text{CL}) = \frac{-100 \text{ k}\Omega}{50 \text{ k}\Omega} = -2$$

The $V_{out} = (-50) (100 \text{ m V}_{pp})$
 $+ (-10) (200 \text{ m V}_{pp}) + (-2) (300 \text{ m V}_{pp})$
 $= -3.1 \text{ V}_{pp}.$

70. (B) $A_{V(\text{CL})} = \frac{3.9 \text{ k}\Omega}{100 \Omega} + 1$
 $= 40$
 $f_2(\text{CL}) = \frac{1 \text{ MHz}}{40}$
 $= 25 \text{ kHz}.$

71. (B) $V_{out} = \text{dB}$
 equivalent of 40 is 32 dB
 $= [20 \text{ dB} + 6 \text{ dB} + 6 \text{ dB}]$
 $= 32 \text{ dB}]$

Since $A_{V(\text{CL})}$ breaks at 25 kHz it is down 20 dB at 250 kHz.

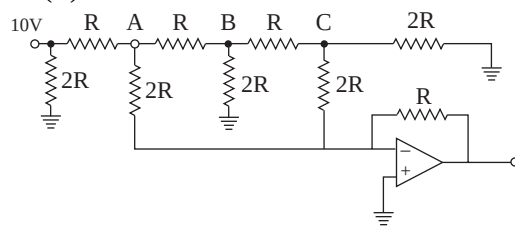
$$A_{V(\text{CL})} = 12 \text{ dB at } 250 \text{ kHz}$$

12 dB equivalent to 4

So $V_{out} = 4 (50 \text{ m V}_{pp})$
 $= 200 \text{ m V}_{pp}.$

72. (B) 73. (D) 74. (A) 75. (B) 76. (B)
 77. (C)

78. (B)



Given $R = 10 \text{ k}\Omega$
 Since $V_+ = 0$
 So in Op-Amp $V_+ = V_-$
 So $V_- = 0$

At node A,

$$\frac{V_A - V_R}{R} + \frac{V_A - V_B}{R} + \frac{V_A - V_-}{2R} = 0$$

$$5 V_A - 2 V_B - 2 V_R = 0$$

At node B

$$\frac{V_B - V_A}{R} + \frac{V_B - V_C}{R} + \frac{V_B}{2R} = 0$$

$$5 V_B - 2 V_A - 2 V_C = 0$$

At node C

$$\frac{V_C - V_B}{R} + \frac{V_C - V_-}{2R} + \frac{V_C}{2R} = 0$$

$$4 V_C - 2 V_B = 0$$

Solve

$$V_A = 5 \text{ V},$$

$$V_B = 2.5 \text{ V}$$

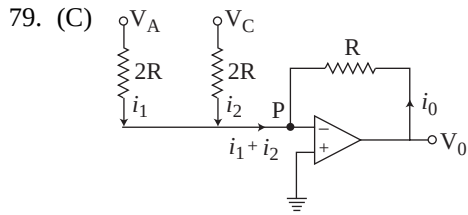
$$V_C = 1.25 \text{ V}$$

$$i = \frac{V_C}{2R}$$

$$= \frac{1.25}{2 \times 10 \times 10^3}$$

$$= 0.0625 \text{ mA}$$

$$= 62.5 \mu\text{A}$$



$$i_1 = \frac{V_A}{2R} = \frac{5}{20} \text{ mA}$$

$$i_2 = \frac{V_C}{2R} = \frac{1.25}{20} \text{ mA}$$

$$i_0 = \frac{V_0}{10} \text{ mA}$$

apply KCL at node P

$$i_1 + i_2 + i_0 = 0$$

$$\frac{V_0}{10} = -[i_1 + i_2]$$

$$= -\frac{5}{20} - \frac{2.5}{20}$$

$$V_0 = -3.125 \text{ V}$$

80. (D) If $V_{in} > V_{DAC}$, output 1 otherwise 0.
Hence if $V_{DAC} \leq 6.2 \text{ V}$, output 1 and counter counts

If $V_{DAC} > 6.2 \text{ V}$, output 0, counter stop.

$$\begin{aligned} V_{DAC} &= \sum_{n=0}^3 2^{n-1} b_n \\ &= 2^{-1} b_0 + 2^0 b_1 + 2^1 b_2 + 2^2 b_3 > 6.21 \\ &= 4b_3 + 2b_2 + b_1 + \frac{1}{2} b_0 > 6.21 \end{aligned}$$

Which gives now 1101
= 13

81. (B) Error Voltage

$$\begin{aligned} V_{DAC} - V_{1A} &= 6.5 - 6.2 \\ &= 0.3 \text{ volt} \end{aligned}$$

82. (C)

83. (D) $np = (n_i)^2$ (constant)

By doping acceptor impurities with a concentration of N_A per cm^3

$$p \approx N_A$$

$$N_A \cdot n = n_i^2$$

or,
$$n = \frac{n_i^2}{N_A}$$

84. (D)

85. (C)

$$V_1 \approx V_2$$

$$\begin{aligned} V_2 &= 1 \times \left(\frac{1}{1+1} \right) = \frac{1}{2} \\ &= 0.5 \text{ V} \end{aligned}$$

From KCL at V_1

$$\frac{V_0 - V_1}{2 \text{ k}\Omega} = \frac{V_1 - 1}{1 \text{ k}\Omega} = 0$$

$$\Rightarrow V_0 - V_1 = 2V_1 - 2$$

$$V_0 = 3V_1 - 2$$

$$\begin{aligned} V_0 &= 3 \times 0.5 - 2 \\ &= 1.5 - 2 = -0.5 \end{aligned}$$

86. (D) This is a circuit diagram of log amplifier

$$V_0 = -V_T \ln V_i + C$$

$$V_{01} = -V_T \ln V_{i1} + C$$

$$= -V_T \ln 2 + C$$

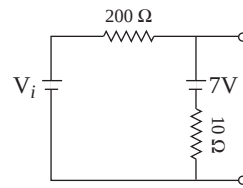
$$V_{02} = -V_T \ln V_{i2} + C$$

$$= -V_T \ln 4 + C$$

$$V_{01} - V_{02} = -V_T \ln 2 + V_T \ln 4$$

$$= V_T \ln 2.$$

87. (C)



$$V_i = 10 \text{ V}$$

$$i = \frac{10 - 7}{210} = \frac{3}{210} \text{ amp}$$

$$V_{01} = 7 + 10i$$

$$= 7 + 10 \times \frac{3}{210}$$

$$= 7.14 \text{ volt}$$

when

$$V_i = 16 \text{ V}$$

$$i = \left(\frac{9}{210} \right) \text{ amp}$$

$$V_{02} = 7 + 10 \left(\frac{9}{210} \right)$$

$$= 7.43 \text{ V}$$

88. (D)

89. (A) MUX

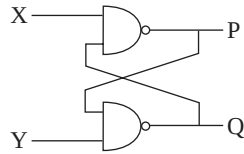
A (S_1)	B (S_0)	Y_1
0	0	0
0	1	1
1	0	1
1	1	0

For MUX 2

Y (S ₁)	C (S ₀)	Y
0	0	0
1	1	0
1	0	1
0	1	1

$$\begin{aligned}
 Y &= \overline{Y_1} C + Y_1 \overline{C} \\
 &= (\overline{AB} + A \overline{B}) C + (\overline{A} B + A \overline{B}) \overline{C} \\
 Y &= ABC + \overline{A} \overline{B} C + \overline{A} B \overline{C} + A \overline{B} \overline{C}
 \end{aligned}$$

90. (C)



$$\begin{aligned}
 P_{t+1} &= \overline{X} \overline{Q}_t \\
 &= \overline{X} + \overline{Q}_t \\
 Q_{t+1} &= \overline{Y} \cdot P_t \\
 &= \overline{Y} + \overline{P}_t
 \end{aligned}$$

X	Y	P _t	Q _t	P _{t+1}	Q _{t+1}
0	1	d	d	1	0
0	0	1	0	1	1
1	1	1	1	0	1
Not Stable				1	0

↓
because it is
continuously change
its state

 91. (B) $np = n_i^2$
Let N_d be the doping concentration in n -type material

$$n \cong N_d$$

$$\text{So, } p = \frac{n_i^2}{N_d}$$

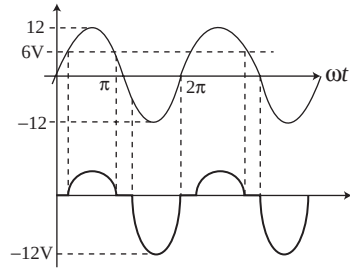
$$\text{So, } p \propto \frac{1}{N_d}$$

92. (B) 93. (D)

$$\begin{aligned}
 94. (B) \quad \overline{F} &= q(\overline{V} \times \overline{B}) \\
 &= -e(VB) \hat{a}_x \times \hat{a}_z \\
 &= -eVB \hat{a}_y
 \end{aligned}$$

$$\begin{aligned}
 95. (D) \quad \text{Ratio} &= \frac{qn \mu_n}{qn_i (\mu_n + \mu_p)} \\
 &= \frac{n}{n_i \left[1 + \frac{\mu_p}{\mu_n} \right]} \\
 &= \frac{4 \cdot 2 \times 10^8}{1 \cdot 5 \times 10^4 (1 + 0 \cdot 4)} \\
 &= 20,000
 \end{aligned}$$

96. (B)


 During $0 < \omega t < \pi/6$

Diode is off and not conducting

$$V_R = 0$$

 From $\frac{\pi}{6} < \omega t < \pi$

$$V_Z = 6V$$

$$V_0 = 12 \sin \omega t - V_Z$$

 $\pi < \omega t < 2\pi$ means diode in conduct.

$$\begin{aligned}
 97. (B) \quad C &= \frac{\epsilon_r \epsilon_0 A}{d} \\
 \frac{C}{A} &= \frac{\epsilon_r \epsilon_0}{d} \\
 &= 10 \mu F
 \end{aligned}$$

98. (B) 99. (D)

 100. (D) Initially $S_0 = 0$,
 $C_0 = 1$

After one clock pulse

$$S = Q_A \cdot Q_B = 0$$

$$C_1 = Q_A \cdot Q_B = 1$$

After second clock pulse

$$\begin{aligned}
 S_2 &= [Q_A \times Q_B] + S_A \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 C_2 &= Q_A Q_B \\
 &= 1
 \end{aligned}$$

C _K	Q _A	Q _B
1	1	1
2	1	1

101. (B)	C_K	D_3	D_2	D_1	D_0
	1	0	0	0	0
	2	0	0	0	1
	3	0	0	1	0
	4	0	0	1	1
	5	0	0	0	0
	6	0	0	0	1
	7	0	0	1	0
	8	0	0	1	1
	9	1	0	0	0
	10	1	0	0	1
	11	1	0	1	0
	12	1	0	1	1
	13	1	0	0	0
	14	1	0	0	1
	15	1	0	1	0

Means C_{k-5} to C_{k-8} count will be constant and output will rise.

102. (A)

103. (B)

$$\frac{\mu_e}{\mu_n} = 3$$

$$\frac{\sigma_A}{\sigma_B} = \frac{N_A \mu_p}{N_D \mu_e}$$

$$\frac{\mu_p}{\mu_e} = \frac{1}{3}$$

104. (C) In CB configuration

$$I_E = I_C + I_{CBO}$$

$$= \alpha I_E + I_{CBO}$$

$$I_C \approx I_E$$

$$I_E = \frac{V_{BE}}{2\alpha}$$

105. (B)

$$R_{th} = 5 || 5$$

$$= 2.5$$

$$V_{th} = 10 \text{ V} - 2.5 \text{ V}$$

$$= 7.5 \text{ V}$$

106. (C)

107. (A) At 1000 Hz, frequency $Z_C \rightarrow \infty$
Hence high pass filter

$$\omega_C = \frac{1}{RC}$$

$$= 10^3 \text{ rad/sec}$$

108. (B) 109. (C) 110. (B) 111. (A)

112. (C) The maximum power transfer theorem

will give $P_{max} = \frac{E^2}{4R}$

when $R = R_L$

$$P_{max} = \frac{10^2}{4 \times 100}$$

$$= 0.25 \text{ W}$$

113. (D)

114. (B)

$$R_i = \frac{V_i}{I_i}$$

$$R_i = 10 \text{ k}\Omega$$

115. (A)

$$V_{BE} = 0.7 \quad [\text{Forward bias}]$$

$$V_{BC} = -V_{CB}$$

$$= -0.2 \text{ V}$$

[Reverse bias]

Hence transistor operate in normal active mode.

116. (B)

117. (A) Second order low pass filter.

118. (D) $2^7 = 128$
For 100 increment 7 bits required.

119. (D)

120. (A)

$$E = h\nu = \frac{hc}{\lambda}$$

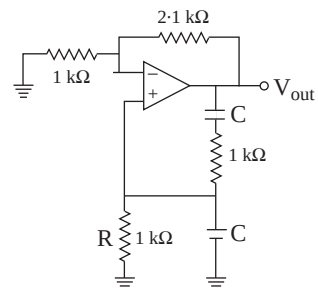
$$\frac{E_2}{E_1} = \frac{\lambda_1}{\lambda_2}$$

$$E_2 = \frac{1.1}{0.87} \times 1.12 \text{ eV}$$

$$= 1.41609$$

121. (D) 122. (B) 123. (B)

124. (A)



$$\frac{V_0 - V_1}{X_0 + R} = \frac{V_1}{X_C} + \frac{V_1}{R}$$

$$V_0 = (X_0 + R) \left[\frac{1}{X_C} + \frac{1}{R} \right] V_1 + V_1$$

$$\begin{aligned}\frac{V_0}{V_1} &= \left(\frac{X_C + R}{X_C R} \right)^2 + 1 \\ &= \text{feedback gain} \\ X_C &= \frac{-j}{\omega C} \\ R &= 1 \times 10^3 \\ \frac{V_0}{V_1} &= \frac{-\frac{1}{\omega^2 C^2} + R^2 - \frac{2jR}{\omega C}}{-\frac{R}{j\omega C}} + 1 \\ &= j\omega C \left(\frac{-\frac{1}{\omega^2 C^2} + R^2 - \frac{2jR}{\omega C}}{R} \right) - 1 \\ &= \frac{-j\omega C \left(R^2 - \frac{1}{\omega^2 C^2} \right) - 3R}{R}\end{aligned}$$

For oscillation, imaginary part is equal to 0

$$\begin{aligned}\text{So, } R^2 - \frac{1}{\omega^2 C^2} &= 0 \\ C &= \frac{1}{\omega R} \\ &= \frac{1}{2\pi} \mu\text{F}.\end{aligned}$$

125. (A) 126. (A) 127. (B) 128. (B) 129. (B)
130. (B) 131. (B) 132. (C) 133. (A) 134. (A)
135. (D) 136. (B) 137. (C) 138. (C) 139. (B)
140. (B) 141. (B) 142. (C) 143. (C) 144. (D)
145. (D) 146. (C) 147. (C)

$$\begin{aligned}148. \text{ (B) } I_{10k\Omega} &= \frac{12V}{10k\Omega} = 1.2 \text{ mA} \\ I_{5k\Omega} &= \frac{10V}{5k\Omega} = 2 \text{ mA} \\ I_{2k\Omega} &= 1.2 \text{ mA} + 2 \text{ mA} \\ &= 3.2 \text{ mA} \\ V_{out} &= -\frac{(3.2) \times 2}{\text{mA} \cdot k\Omega} \\ &= -64 \text{ V}\end{aligned}$$

$$\begin{aligned}149. \text{ (B) } 10_{10} &= 1010_2 \\ &[\text{Switches } D_3 \text{ and } D_1 \text{ are closed}] \\ I_3 &= \frac{5V}{12.5k\Omega} = 0.4 \text{ mA} \\ I_1 &= \frac{5V}{50k\Omega} = 0.1 \text{ mA} \\ V_{out} &= -[0.4 \text{ mA} + 0.1 \text{ mA}] \times 20k\Omega \\ &= -10 \text{ V}\end{aligned}$$

150. (B) The exact weights in volts are

$$\begin{array}{cccc|cccc} D_1 & C_1 & B_1 & A_1 & D_0 & C_1 & B_1 & A_1 \\ 8.0 & 4.0 & 2.0 & 1.0 & 0.8 & 0.4 & 0.2 & 0.1 \end{array}$$

For 0101 and 1000

$$\begin{aligned}V_{out} &= C_1 + A_1 + D_0 \\ &= 4V + 1V + 0.8 \\ &= 5.8 \text{ V}\end{aligned}$$

Easier way

$$\begin{aligned}101011000 &= 58_{10} \\ \text{step size} &= 0.1\end{aligned}$$

$$\begin{aligned}\text{So } V_{out} &= 58 \times \text{step size} \\ &= 5.8 \text{ V}\end{aligned}$$

151. (C) An eight stage counter has 2^8 – count levels.

At a clock rate of 2 MHz, the count period—

$$\begin{aligned}T &= \frac{1}{f} \\ &= \frac{1}{2 \times 10^6} = 0.5 \mu\text{s}\end{aligned}$$

Time period for one conversion

$$\begin{aligned}&= 0.5 \times 256 \\ &= 128 \mu\text{s}\end{aligned}$$

152. (A) 153. (B) 154. (A) 155. (D) 156. (B)
157. (C) 158. (C) 159. (C) 160. (B) 161. (C)
162. (A) 163. (C) 164. (C) 165. (B) 166. (A)
167. (D) 168. (C) 169. (D) 170. (B) 171. (B)
172. (D) 173. (C) 174. (A) 175. (A) 176. (A)
177. (C) 178. (D) 179. (A) 180. (C) 181. (C)
182. (A) 183. (B) 184. (C) 185. (C) 186. (A)
187. (C) 188. (A) 189. (B) 190. (B) 191. (C)
192. (B) 193. (A) 194. (B) 195. (A) 196. (C)
197. (C) 198. (A) 199. (C) 200. (C) 201. (C)
202. (C)

203. (B) Shorting 5 V source :

$$R_{TH} = \frac{2}{3} + 2 = \frac{8}{3} || 1$$

$$1 = \frac{8}{11} \Omega$$

$$\text{I mesh, } V_{TH} = 5 \times \frac{2}{3} = \frac{10}{3} \text{ V, } \frac{2}{3} \Omega$$

$$\begin{aligned}\text{II mesh, } \frac{10}{3} \times \left(\frac{2}{3} + 2 \right) || 1 \\ &= \frac{10}{11} \text{ V}\end{aligned}$$

204. (B)

205. (A) Efficiency of Rectifier

$$\eta = \frac{\text{Power of D.C. output (P}_{dc})}{\text{Power of A.C. input (P}_{ac})}$$

For a half-wave rectifier

$$P_{dc} = \left(\frac{I_m}{\pi}\right)^2 R_L$$

$$P_{ac} = \left(\frac{I_m}{2}\right)^2 (r_f + R_L)$$

$$\begin{aligned}\eta_{1/2} &= \frac{I_m^2}{\pi^2} \cdot \frac{R_L \cdot 4}{I_m^2 (r_f + R_L)} \\ &= \frac{4R_L}{\pi^2 (r_f + R_L)}\end{aligned}$$

For a full wave rectifier

$$P_{dc} = \left(\frac{2I_m}{\pi}\right)^2 R_L$$

$$P_{ac} = \left(\frac{I_m}{\sqrt{2}}\right)^2 (r_f + R_L)$$

$$\begin{aligned}\eta &= \frac{4I_m^2 R_L \cdot 2}{\pi^2 I_m^2 (r_f + R_L)} \\ &= \frac{8R_L}{\pi^2 (r_f + R_L)}\end{aligned}$$

$$\begin{aligned}\therefore \frac{\eta}{\eta_{1/2}} &= \frac{8R_L}{\pi^2 (r_f + R_L)} \times \frac{\pi^2 (r_f + R_L)}{4R_L} \\ &= 2\end{aligned}$$

η (full wave)

$$= 2 \eta_{1/2} \text{ (half wave)}$$

206. (A) 207. (A) 208. (C)

209. (C) For full wave rectification

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

$$\therefore I_{dc} = \frac{2I_m}{\pi}$$

$$\begin{aligned}\text{Ripple factor} &= \frac{I_{ac}}{I_{dc}} \\ &= \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}} \\ &= \sqrt{\frac{I_{rms}^2}{I_{dc}^2} - 1}\end{aligned}$$

$$\text{Here } \frac{I_{rms}}{I_{dc}} = \frac{I_m}{\sqrt{2}} \times \frac{\pi}{2I_m} = \frac{\pi}{2\sqrt{2}}$$

$$\begin{aligned}\text{So Ripple factor} &= \sqrt{\frac{\pi^2}{4 \times 2} - 1} \\ &= \sqrt{\frac{\pi^2}{8} - 1} \\ &= \sqrt{1.2346 - 1} \\ &= \sqrt{0.2346} \\ &= 0.48\end{aligned}$$

210. (A)

211. (D) Given

$$\beta = 49,$$

$$I_{CBO} = 1 \mu A$$

$$I_B = 10 \mu A$$

$$\alpha = \frac{\beta}{1 + \beta}$$

$$= \frac{49}{50} = 0.98$$

$$I_{CEO} = \frac{1}{1 - \alpha} \cdot I_{CBO}$$

$$= \frac{1}{0.02} \mu A$$

$$= 50 \mu A$$

$$I_C = \beta I_B + I_{CEO}$$

$$= 49 \times 10 + 50$$

$$= 540 \mu A$$

212. (C) Collector current,

$$I_C = \beta I_B,$$

where I_B is given by,

$$I_B R_B + V_{BE} = 2$$

$$I_B R_B = 2 - V_{BE}$$

$$= 2 - 0.6$$

$$= 1.4$$

$$R_B = \frac{1.4}{I_C / \beta}$$

$$= \frac{1.4 \beta}{I_C}$$

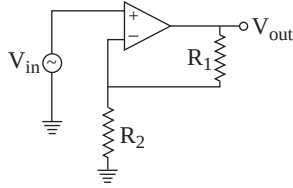
$$= \frac{1.4 \times 100}{2 \times 10^{-3}}$$

$$= 0.7 \times 10^5$$

$$= 70 \text{ k}\Omega$$

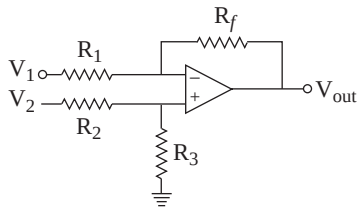
213. (B) 214. (C) 215. (D) 216. (B)

217. (C)



In our problem, $R_1 = 2R$
and $R_2 = R$

218. (D) The circuit given in the problem is the Differential Amplifier type



which has the output V_0 expressed as

$$V_0 = \frac{1 + \frac{R_f}{R_1}}{1 + \frac{R_2}{R_3}} V_2 - \frac{R_f}{R_1} V_1$$

In this problem

$$R_1 = R_2 = R_3 = R$$

$$\therefore 1 + \frac{R_f}{R_1} = 1 + \frac{3R}{R} = 4$$

$$1 + \frac{R_2}{R_3} = 1 + \frac{R}{R} = 2$$

$$\frac{R_f}{R_1} = \frac{3R}{R}$$

$$= 3$$

$$\therefore V_0 = \frac{4}{2} V_2 - 3V_1$$

$$= 2V_2 - 3V_1$$

219. (D) 220. (A) 221. (C)

$$222. (C) \quad A_V = \frac{R_S}{R_S + \frac{1}{g_m}}$$

$$\text{If} \quad \frac{1}{A_V} = 1 + \frac{g_m}{R_S}$$

$$\frac{10}{9} = 1 + \frac{1}{9}$$

$$= \frac{1}{g_m R_S}$$

$$\text{which gives us} \quad g_m R_S = 9$$

$$\text{or,} \quad g_m = \frac{9}{R_S}$$

$$= 3 \text{ mA/V}$$

223. (A) 224. (C) 225. (A) 226. (A) 227. (C)
228. (A) 229. (B) 230. (D) 231. (C) 232. (A)
233. (C) 234. (A) 235. (B) 236. (B) 237. (B)
238. (B) 239. (B) 240. (A) 241. (A) 242. (A)
243. (B) 244. (B) 245. (A) 246. (B) 247. (C)
248. (C) 249. (D) 250. (C) 251. (C) 252. (B)
253. (B) 254. (A) 255. (D) 256. (C) 257. (D)
258. (B) 259. (B) 260. (C) 261. (C) 262. (B)
263. (D) 264. (B) 265. (B) 266. (B) 267. (C)
268. (B) 269. (A) 270. (C) 271. (A) 272. (D)
273. (D) 274. (A) 275. (C) 276. (A) 277. (C)
278. (A) 279. (A) 280. (D) 281. (B) 282. (B)
283. (D) 284. (B) 285. (C) 286. (C) 287. (D)
288. (A) 289. (D) 290. (C) 291. (C) 292. (C)
293. (C) 294. (B) 295. (D) 296. (A) 297. (B)
298. (C) 299. (B) 300. (C) 301. (B) 302. (C)
303. (C) 304. (C) 305. (C) 306. (B) 307. (A)
308. (C) 309. (B) 310. (D) 311. (C) 312. (B)
313. (A) 314. (C) 315. (B)

